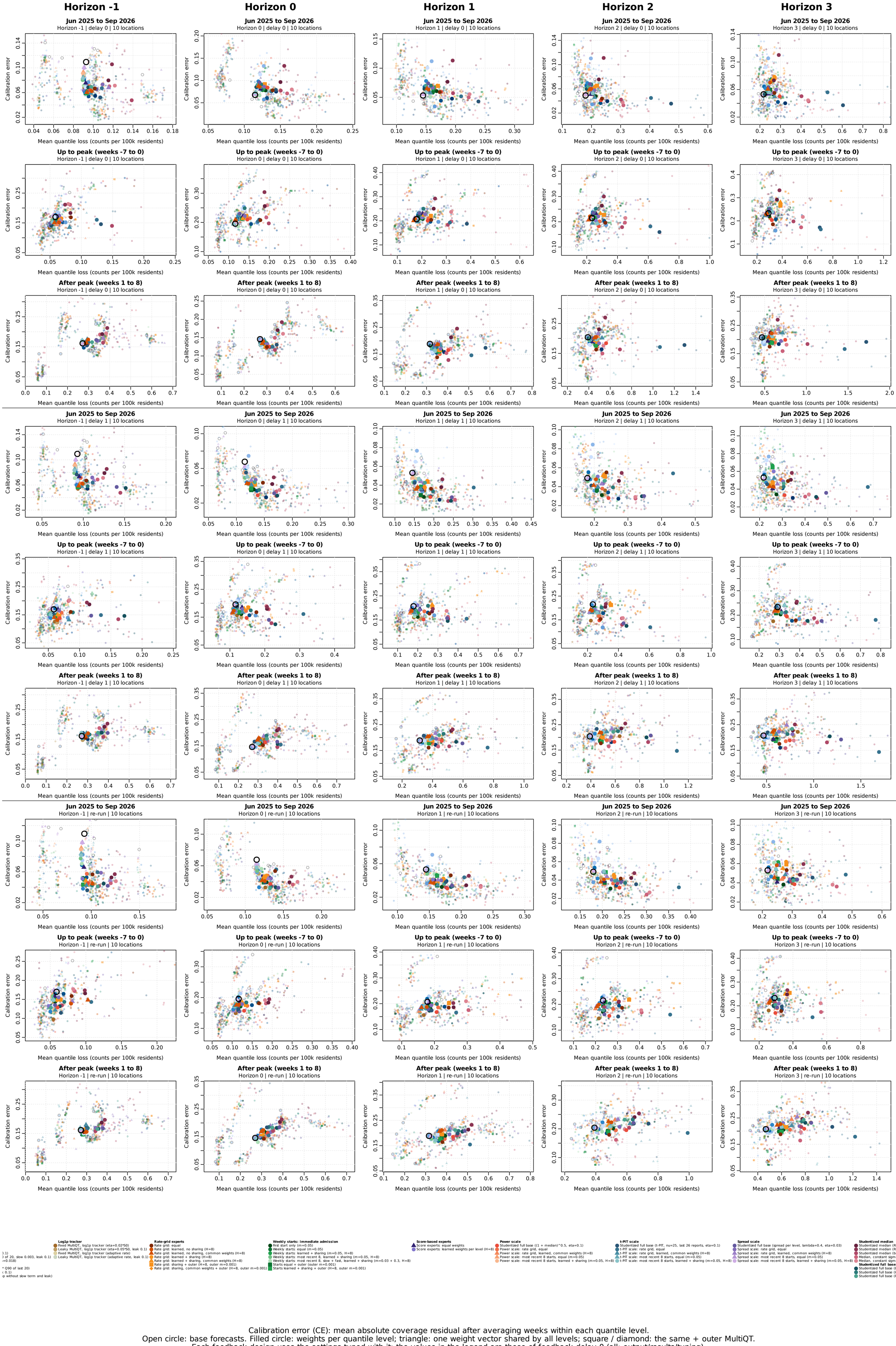
covid | the 10 largest locations without Florida | evaluation: Jun 2025 to Sep 2026 | raw-count updates, loss in counts per 100k residents

Small: single locations; large: means over locations; axes: 1st to 99th percentile of both.



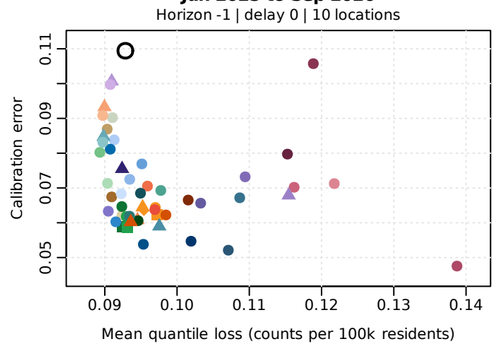
Calibration error (CE): mean absolute coverage residual after averaging weeks within each quantile level.
Open circle: base forecasts. Filled circle: weights per quantile level; triangle: one weight vector shared by all levels; square / diamond: the same + outer MultiQT.
Each feedback design uses the settings tuned with it; the values in the legend are those of feedback delay 0 (all: output/results/tuning).

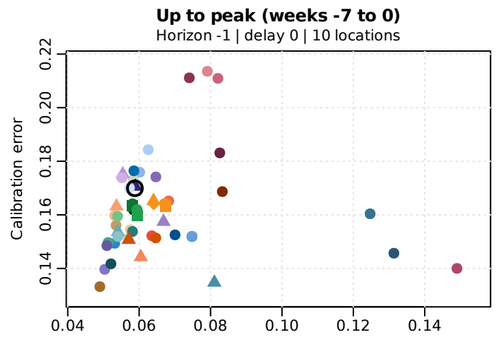
covid | the 10 largest locations without Florida | evaluation: Jun 2025 to Sep 2026 | raw-count updates, loss in counts per 100k residents

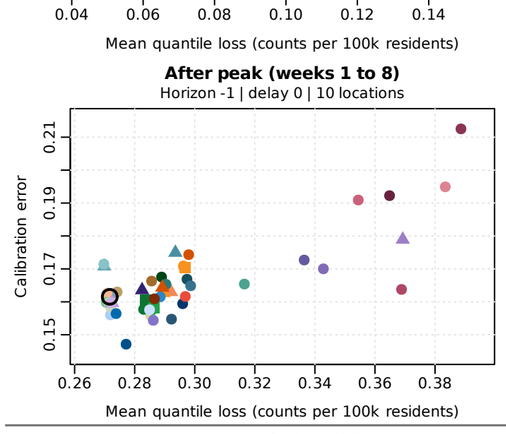
Means over locations only; axes fitted to the means.

Feedback delay 0

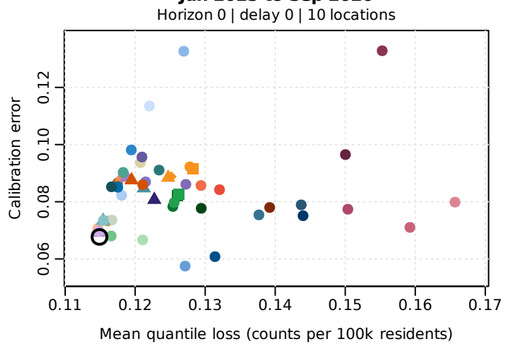
Horizon -1

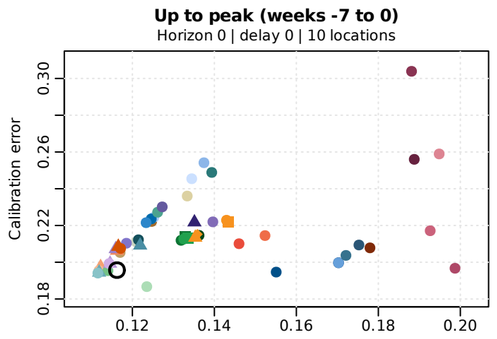


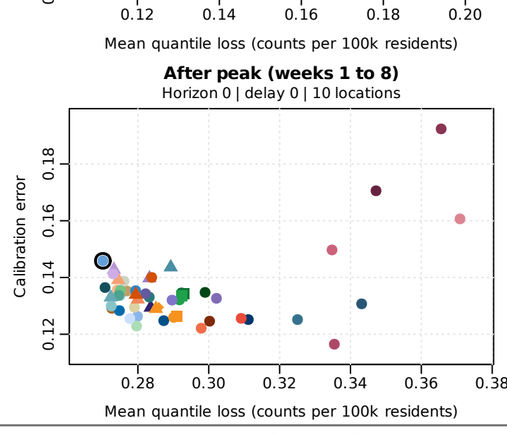




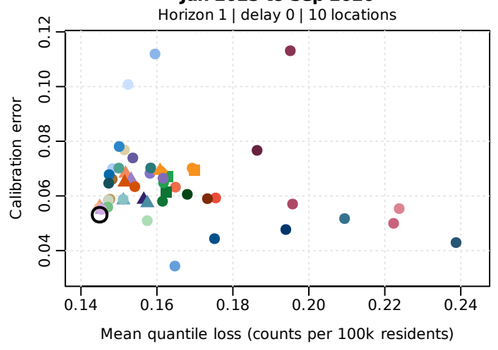
Horizon 0

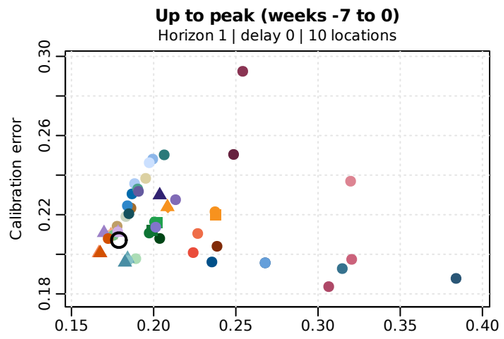


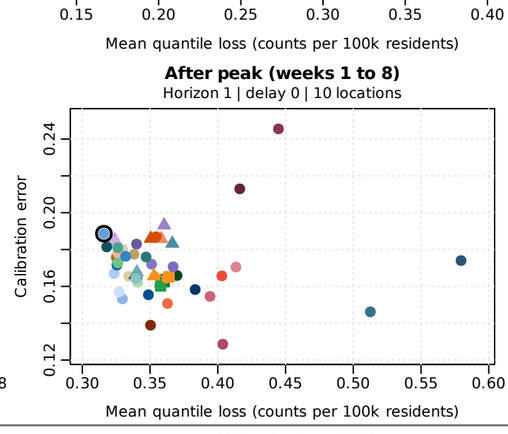




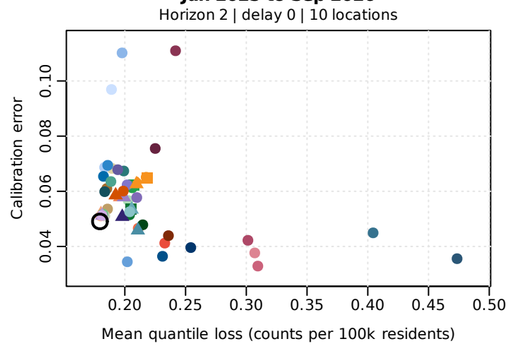
Horizon 1

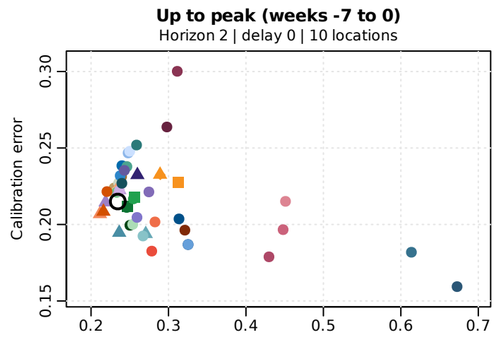


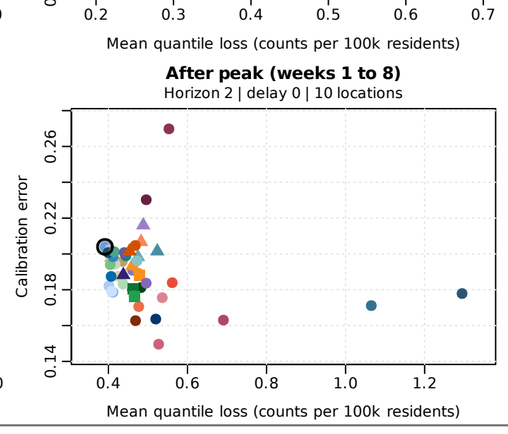




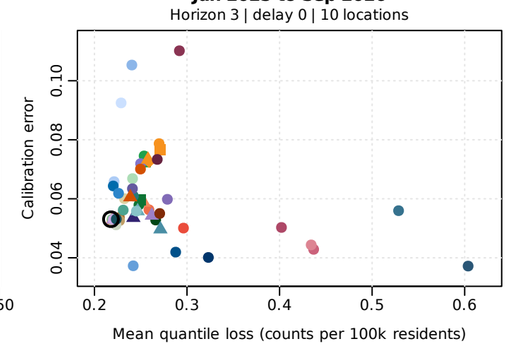
Horizon 2

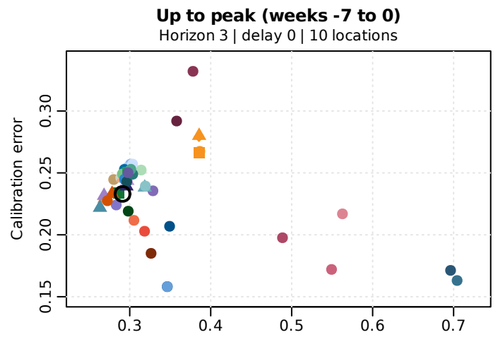


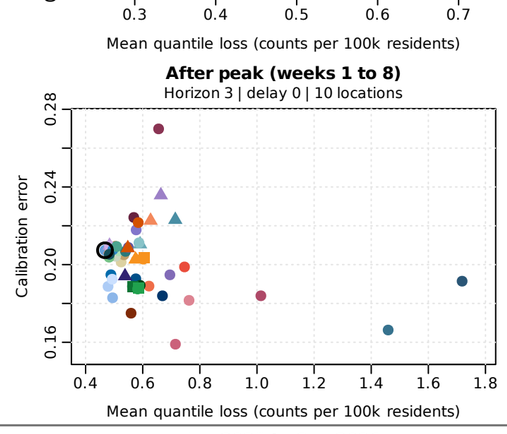




Horizon 3

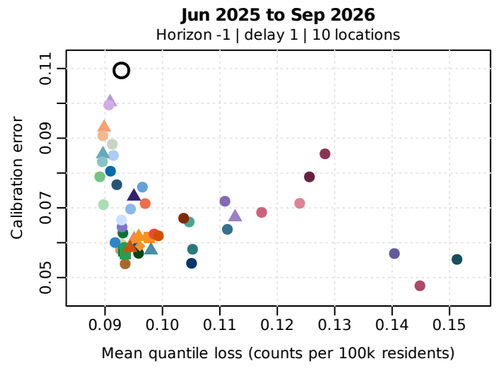


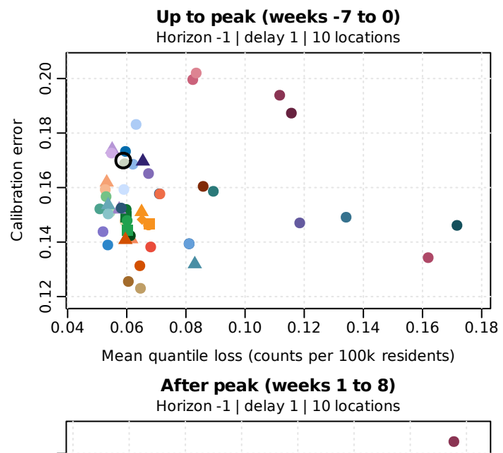


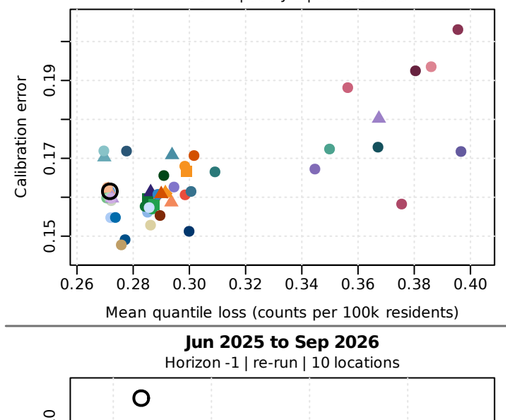


Feedback delay 1

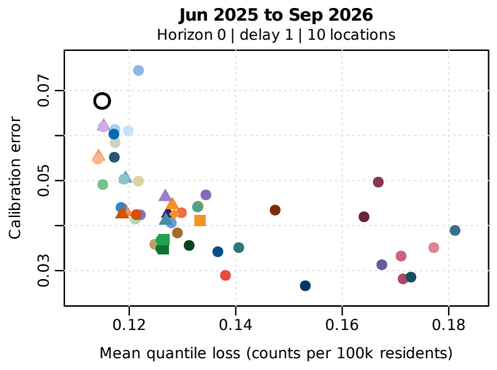
Horizon -1

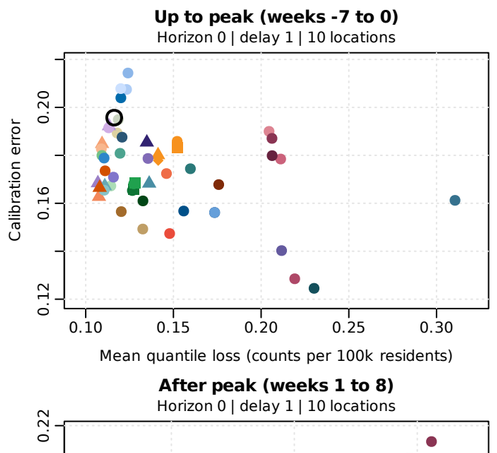


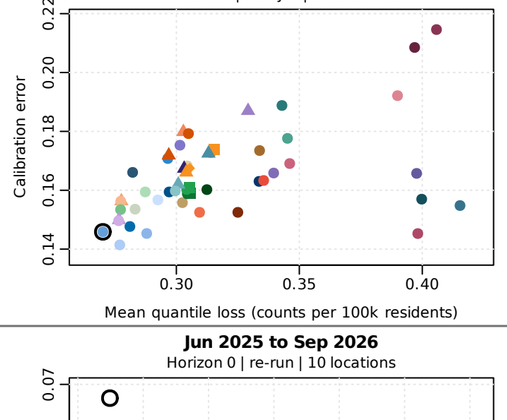




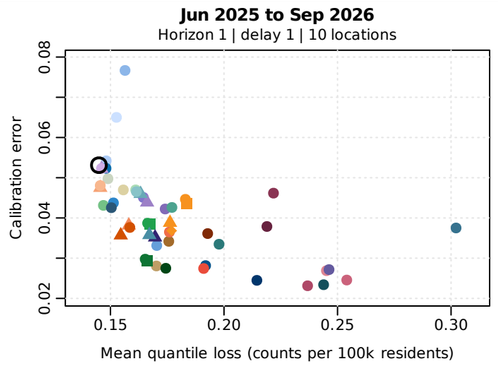
Horizon 0

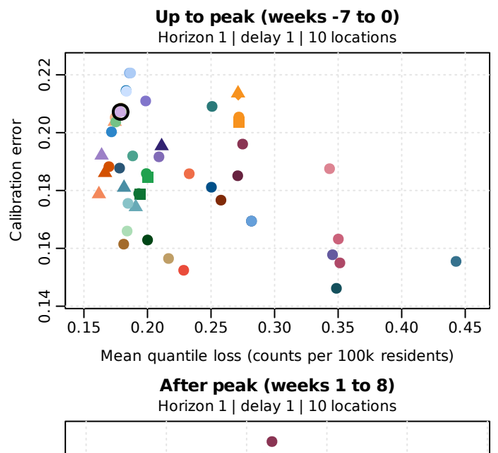


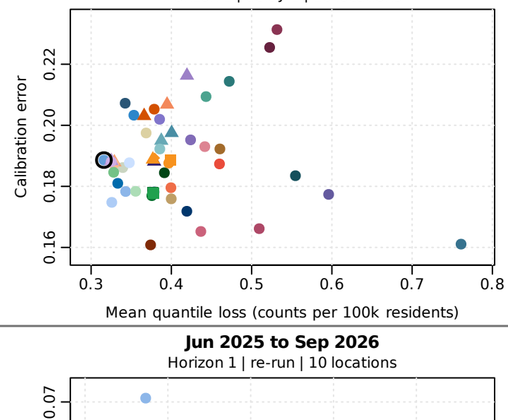




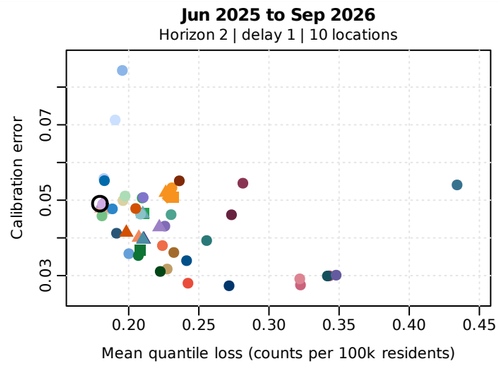
Horizon 1

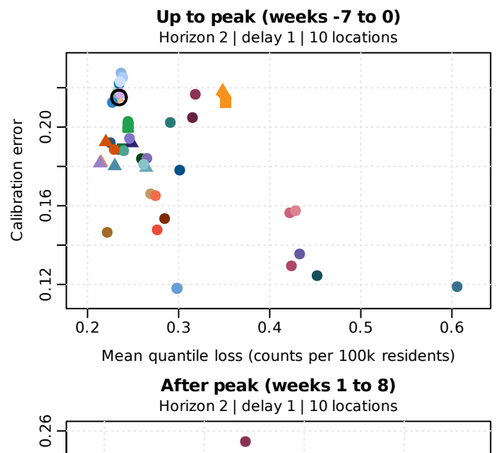


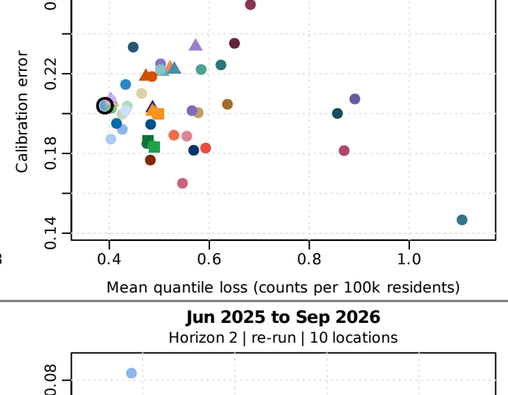




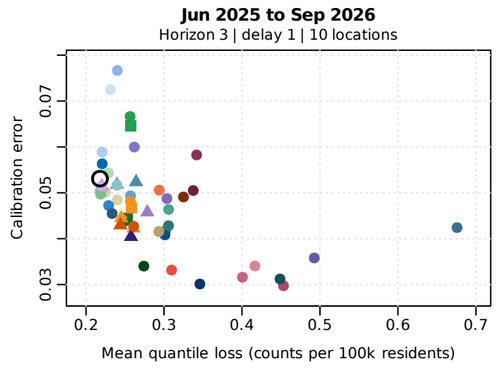
Horizon 2

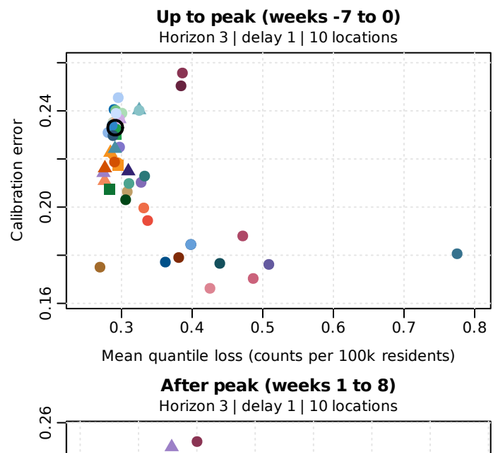


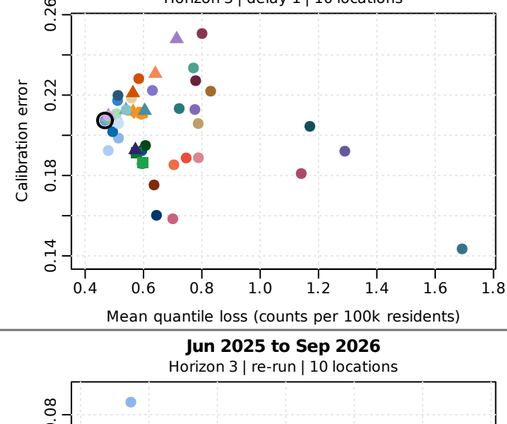




Horizon 3

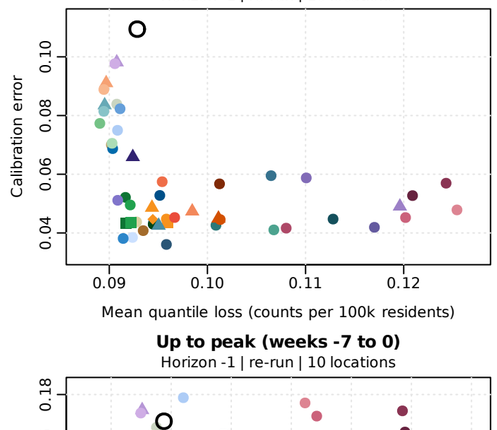


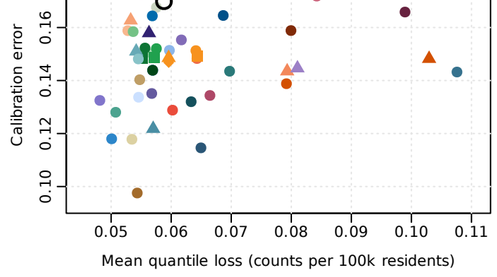


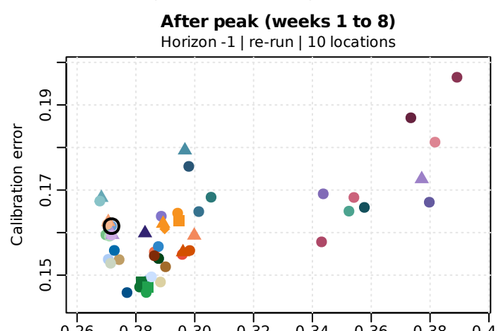


Feedback re-run

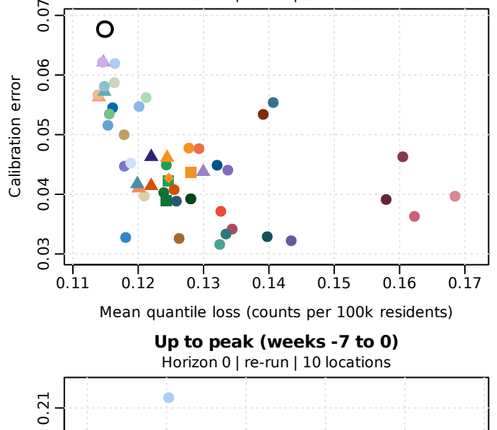
Horizon -1

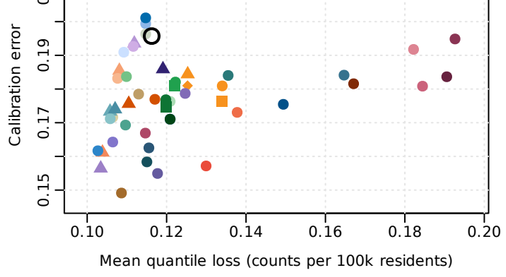


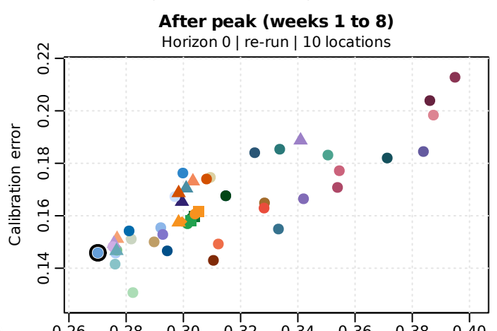




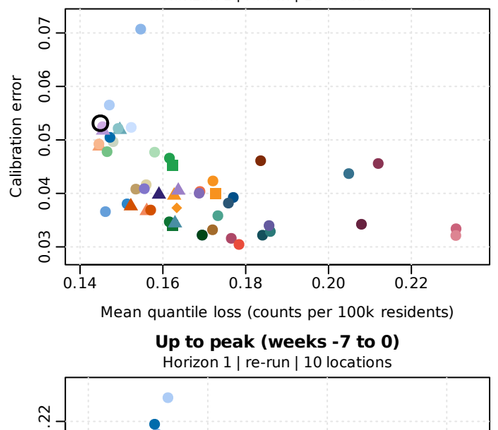
Horizon 0

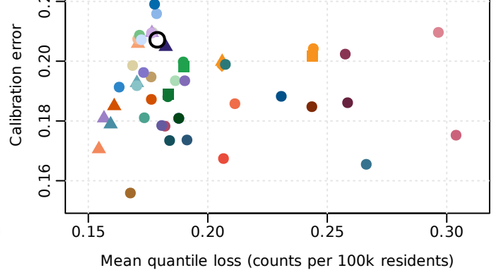


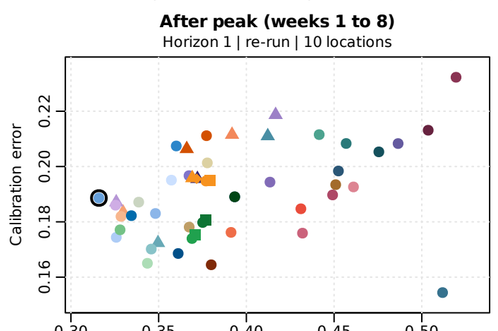




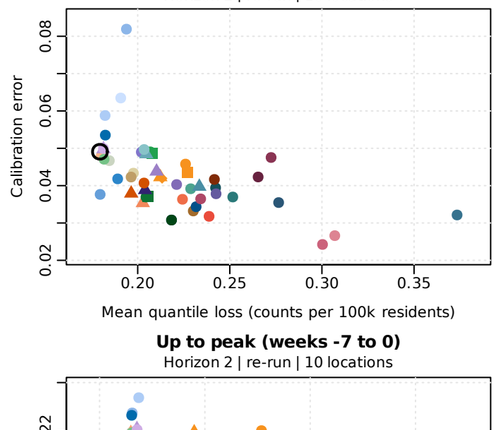
Horizon 1

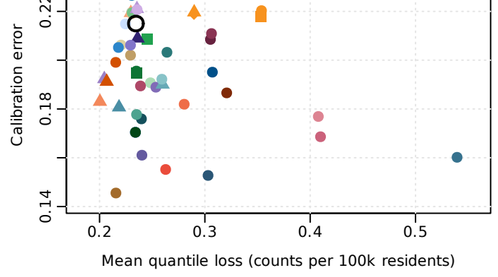


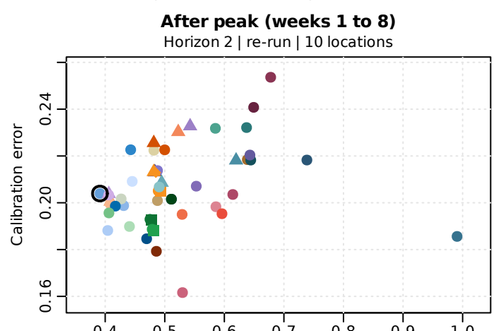




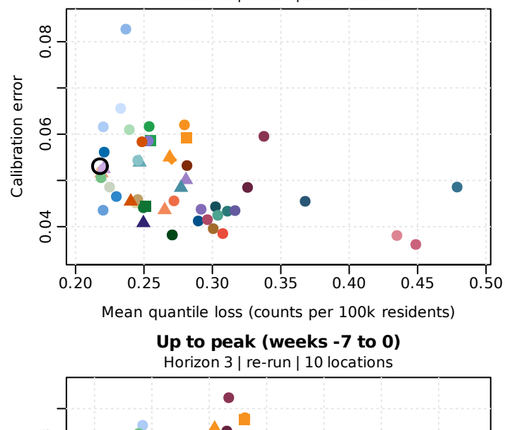
Horizon 2

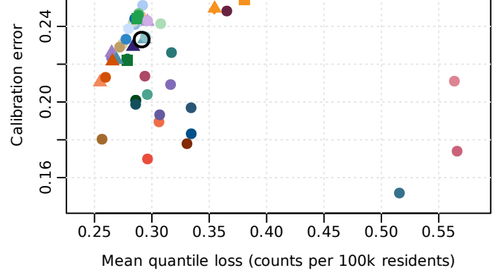


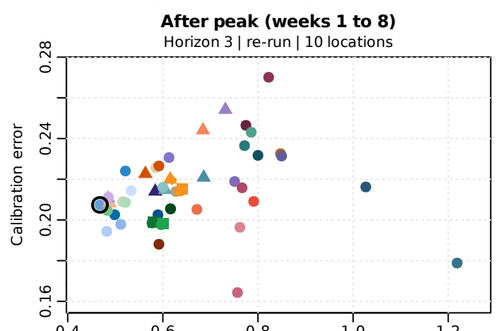


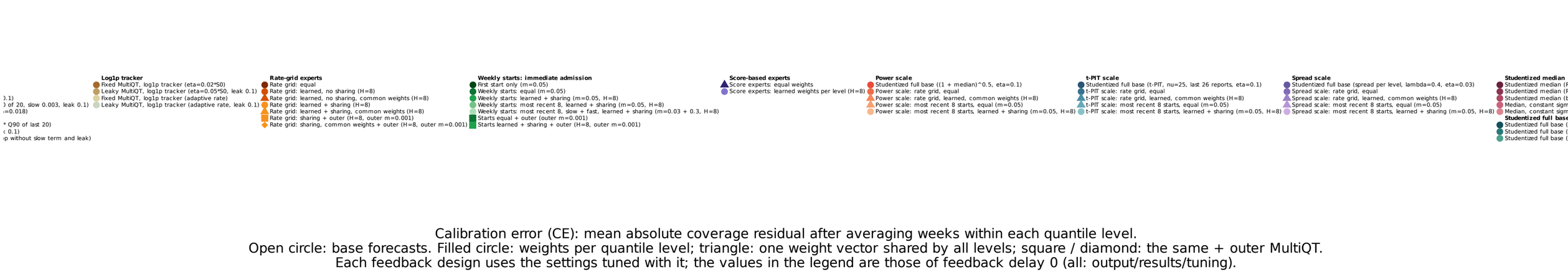


Horizon 3









Calibration error (CE): mean absolute coverage residual after averaging weeks within each quantile level.
Open circle: base forecasts. Filled circle: weights per quantile level; triangle: one weight vector shared by all levels; square / diamond: the same + outer MultiQT.
Each feedback design uses the settings tuned with it; the values in the legend are those of feedback delay 0 (all: output/results/tuning).

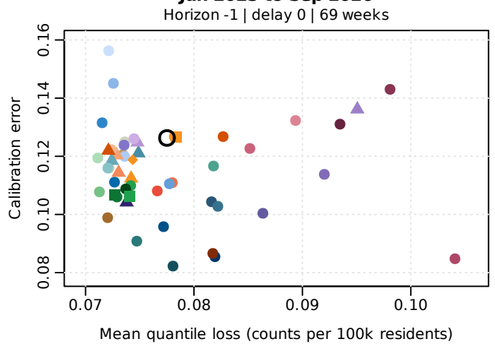
covid | United States (US) | evaluation: Jun 2025 to Sep 2026 | raw-count updates, loss in counts per 100k residents

Rows: feedback design (delay d: learn once, d weeks after the first report, from the report current then; re-run: re-run every week on the reports known then) and window; columns: horizon.

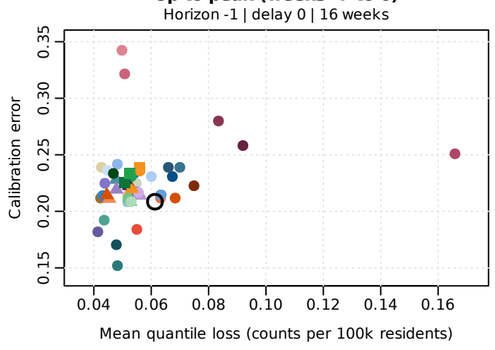
Horizon -1

Feedback delay 0

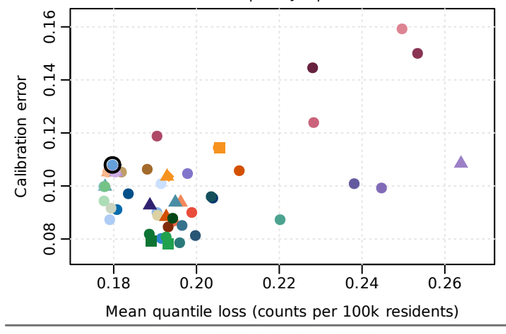
Jun 2025 to Sep 2026
Horizon -1 | delay 0 | 69 weeks



Up to peak (weeks -7 to 0)
Horizon -1 | delay 0 | 16 weeks

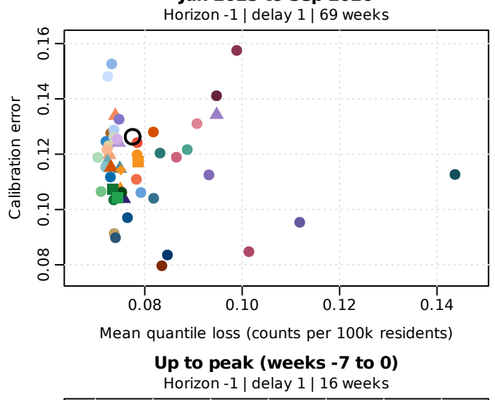


After peak (weeks 1 to 8)
Horizon -1 | delay 0 | 16 weeks

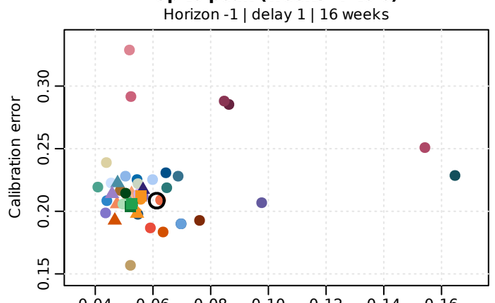


Feedback delay 1

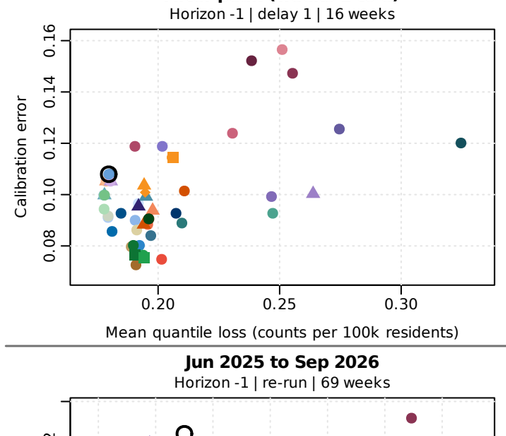
Jun 2025 to Sep 2026
Horizon -1 | delay 1 | 69 weeks



Up to peak (weeks -7 to 0)
Horizon -1 | delay 1 | 16 weeks

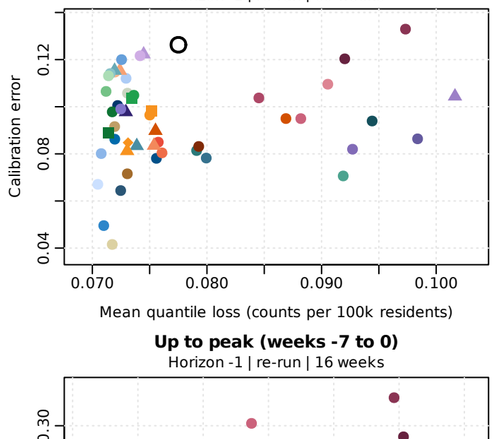


After peak (weeks 1 to 8)
Horizon -1 | delay 1 | 16 weeks

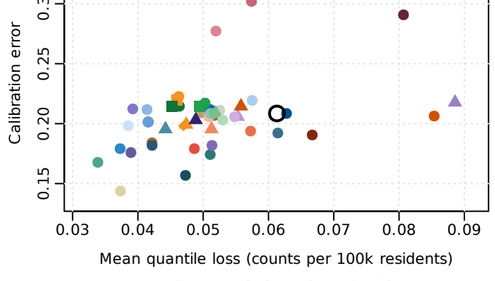


Feedback re-run

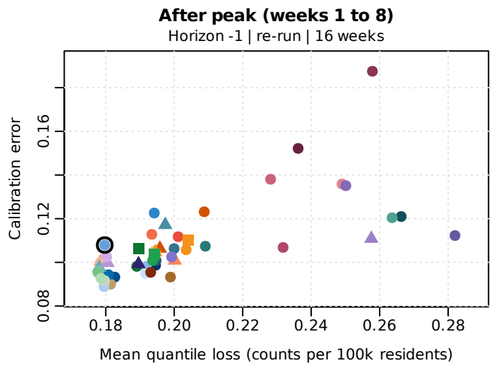
Jun 2025 to Sep 2026
Horizon -1 | re-run | 69 weeks



Up to peak (weeks -7 to 0)
Horizon -1 | re-run | 16 weeks



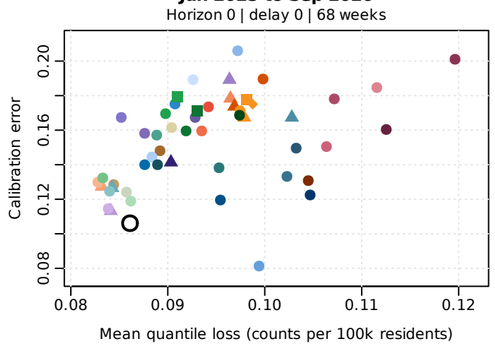
After peak (weeks 1 to 8)
Horizon -1 | re-run | 16 weeks



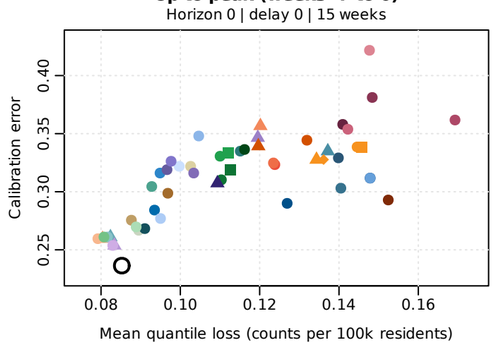
Horizon 0

Feedback delay 0

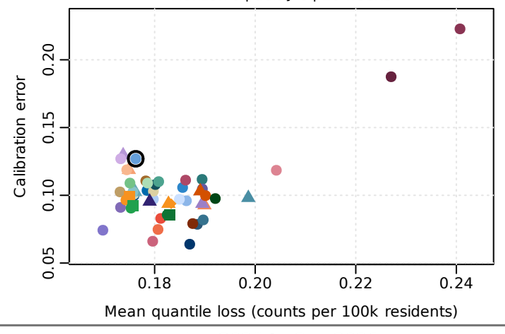
Jun 2025 to Sep 2026
Horizon 0 | delay 0 | 68 weeks



Up to peak (weeks -7 to 0)
Horizon 0 | delay 0 | 15 weeks

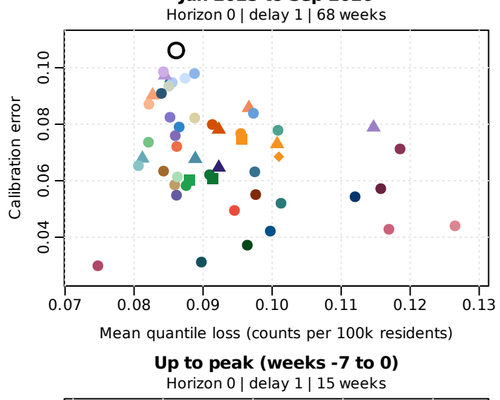


After peak (weeks 1 to 8)
Horizon 0 | delay 0 | 16 weeks

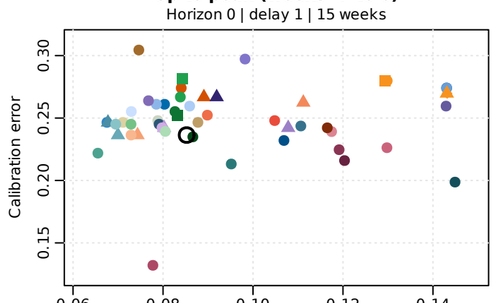


Feedback delay 1

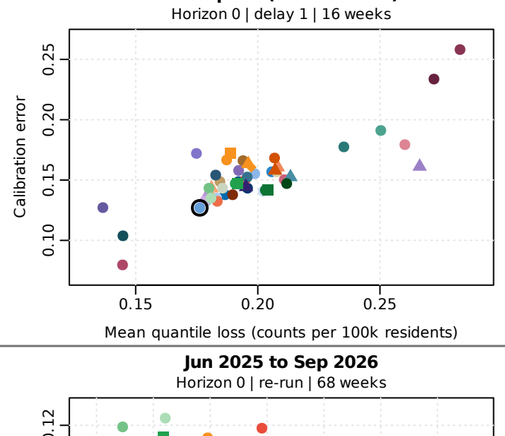
Jun 2025 to Sep 2026
Horizon 0 | delay 1 | 68 weeks



Up to peak (weeks -7 to 0)
Horizon 0 | delay 1 | 15 weeks

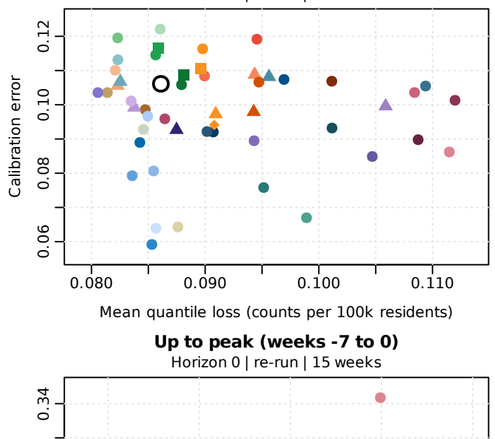


After peak (weeks 1 to 8)
Horizon 0 | delay 1 | 16 weeks

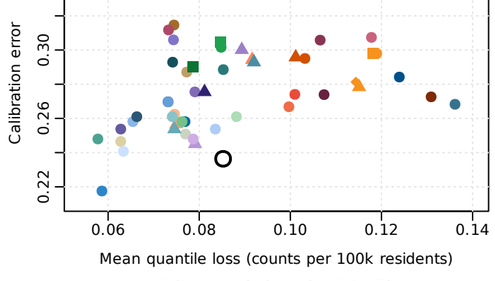


Feedback re-run

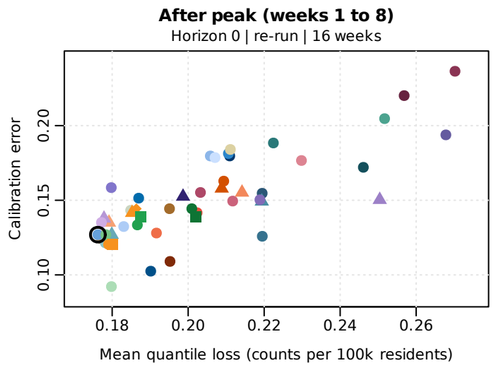
Jun 2025 to Sep 2026
Horizon 0 | re-run | 68 weeks



Up to peak (weeks -7 to 0)
Horizon 0 | re-run | 15 weeks



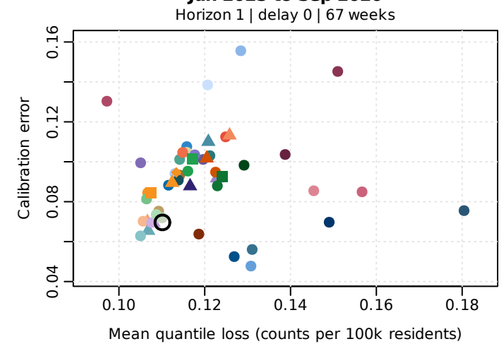
After peak (weeks 1 to 8)
Horizon 0 | re-run | 16 weeks



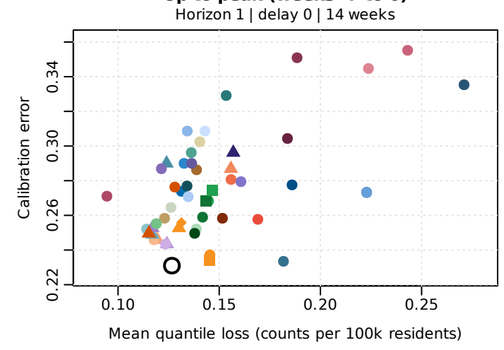
Horizon 1

Feedback delay 0

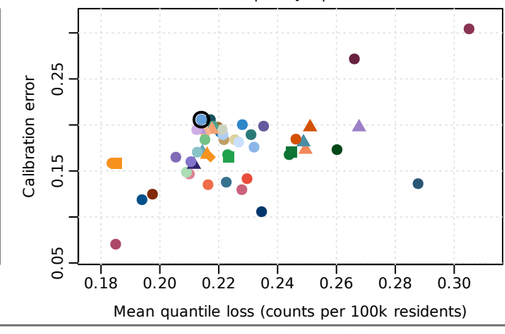
Jun 2025 to Sep 2026
Horizon 1 | delay 0 | 67 weeks



Up to peak (weeks -7 to 0)
Horizon 1 | delay 0 | 14 weeks

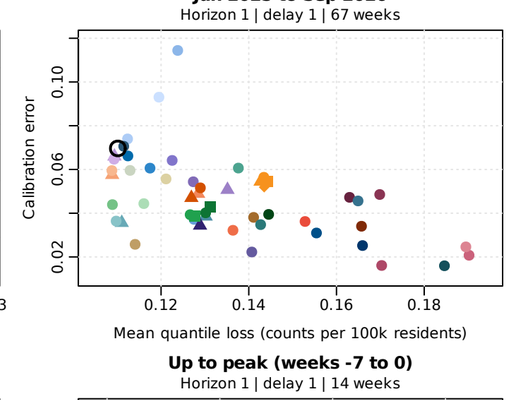


After peak (weeks 1 to 8)
Horizon 1 | delay 0 | 16 weeks

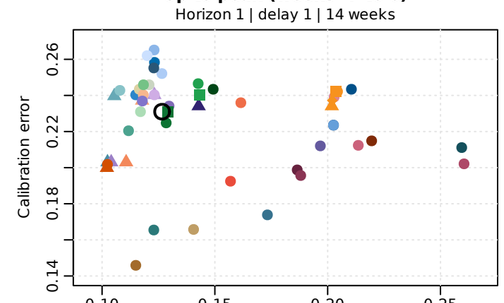


Feedback delay 1

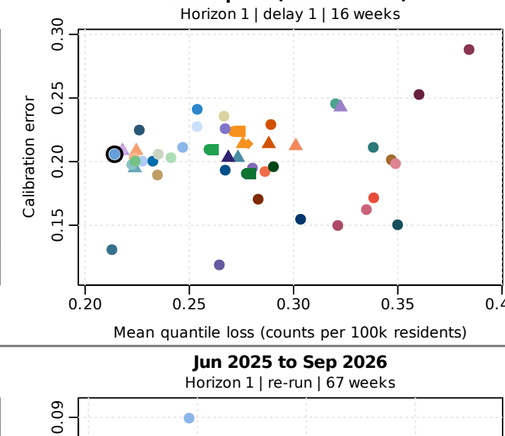
Jun 2025 to Sep 2026
Horizon 1 | delay 1 | 67 weeks



Up to peak (weeks -7 to 0)
Horizon 1 | delay 1 | 14 weeks

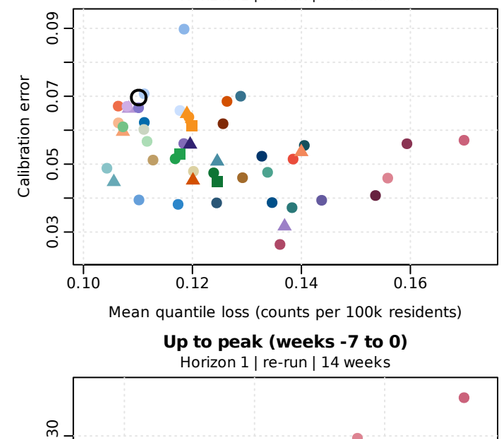


After peak (weeks 1 to 8)
Horizon 1 | delay 1 | 16 weeks

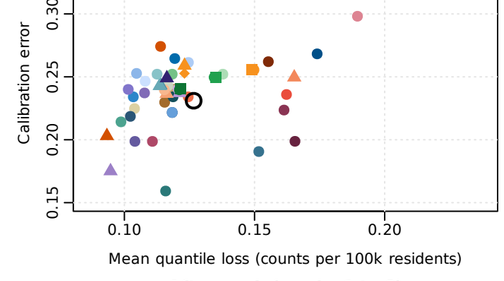


Feedback re-run

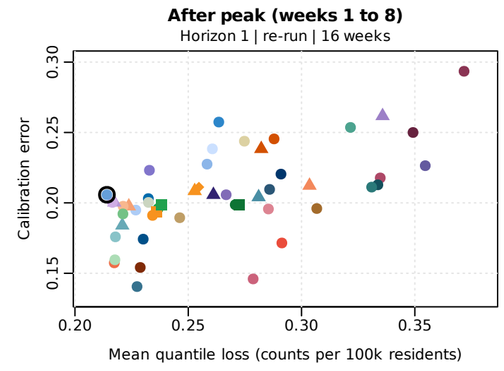
Jun 2025 to Sep 2026
Horizon 1 | re-run | 67 weeks



Up to peak (weeks -7 to 0)
Horizon 1 | re-run | 14 weeks



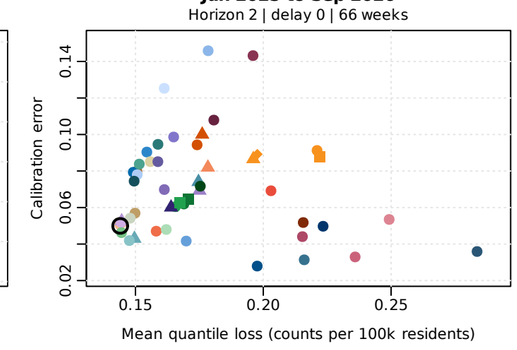
After peak (weeks 1 to 8)
Horizon 1 | re-run | 16 weeks



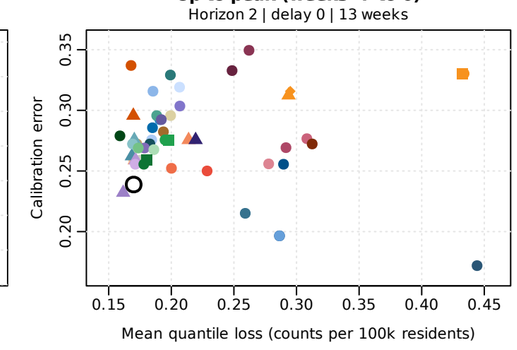
Horizon 2

Feedback delay 0

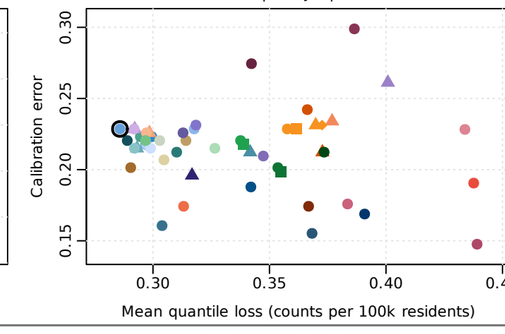
Jun 2025 to Sep 2026
Horizon 2 | delay 0 | 66 weeks



Up to peak (weeks -7 to 0)
Horizon 2 | delay 0 | 13 weeks

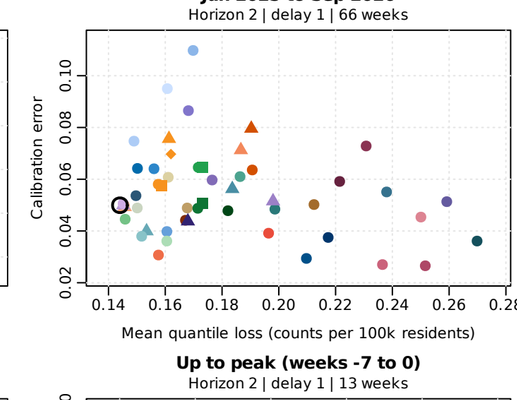


After peak (weeks 1 to 8)
Horizon 2 | delay 0 | 16 weeks

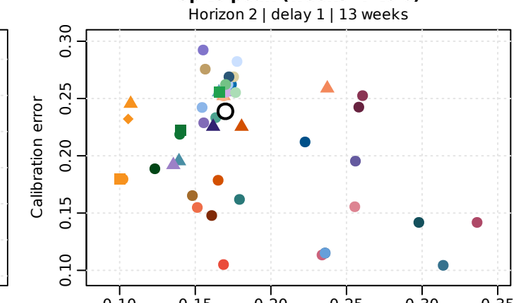


Feedback delay 1

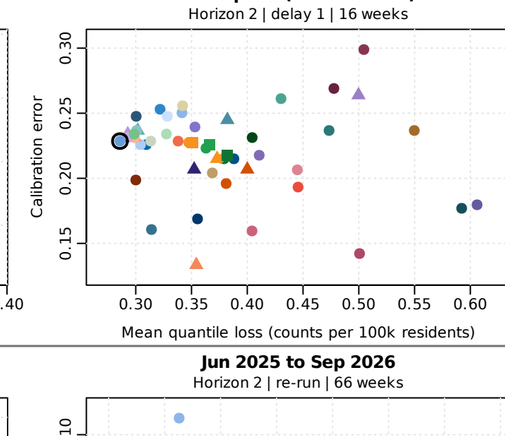
Jun 2025 to Sep 2026
Horizon 2 | delay 1 | 66 weeks



Up to peak (weeks -7 to 0)
Horizon 2 | delay 1 | 13 weeks

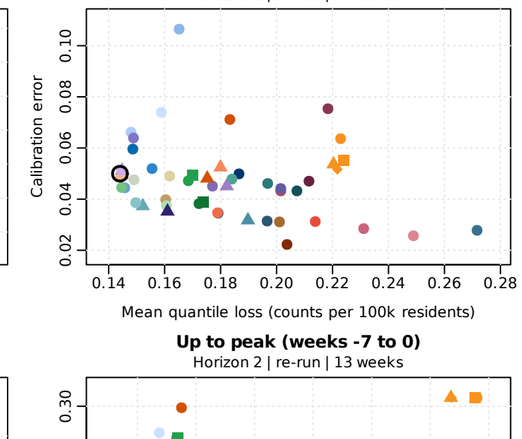


After peak (weeks 1 to 8)
Horizon 2 | delay 1 | 16 weeks

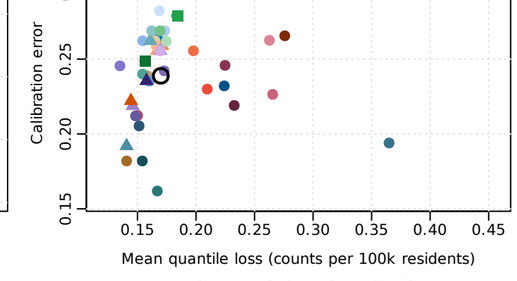


Feedback re-run

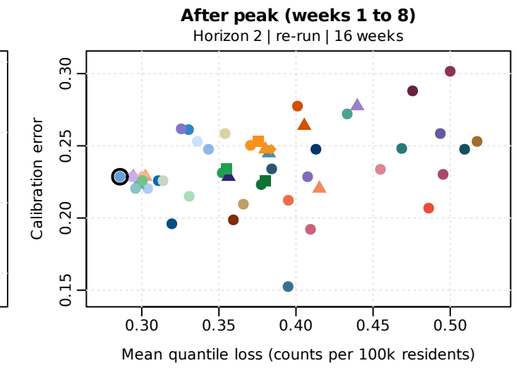
Jun 2025 to Sep 2026
Horizon 2 | re-run | 66 weeks



Up to peak (weeks -7 to 0)
Horizon 2 | re-run | 13 weeks



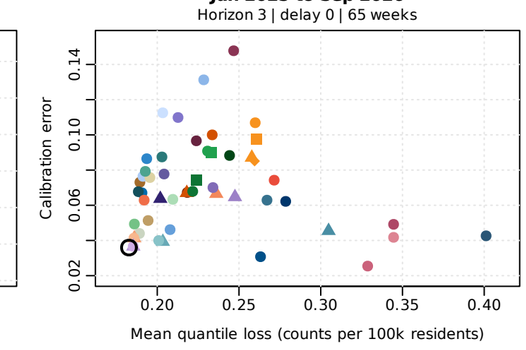
After peak (weeks 1 to 8)
Horizon 2 | re-run | 16 weeks



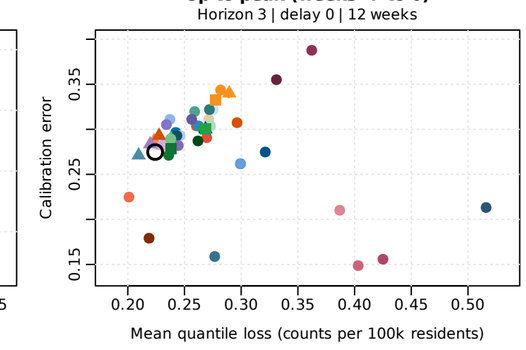
Horizon 3

Feedback delay 0

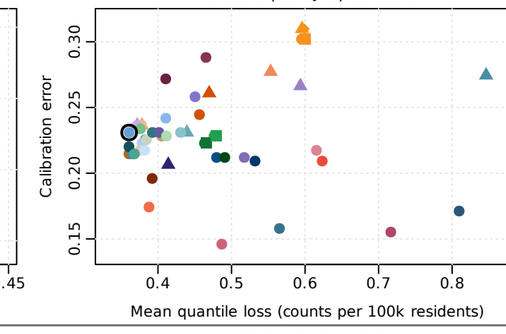
Jun 2025 to Sep 2026
Horizon 3 | delay 0 | 65 weeks



Up to peak (weeks -7 to 0)
Horizon 3 | delay 0 | 12 weeks

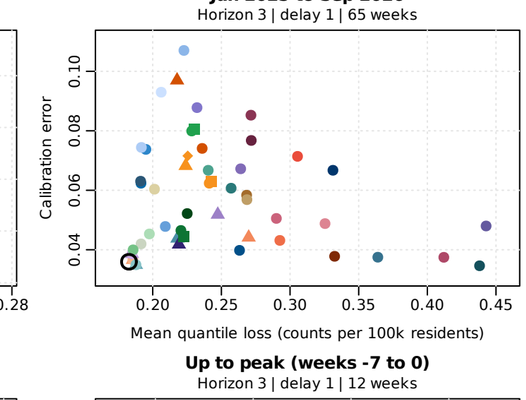


After peak (weeks 1 to 8)
Horizon 3 | delay 0 | 16 weeks

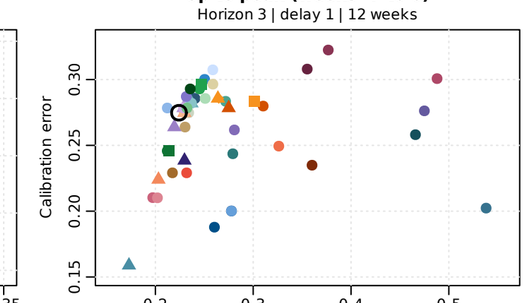


Feedback delay 1

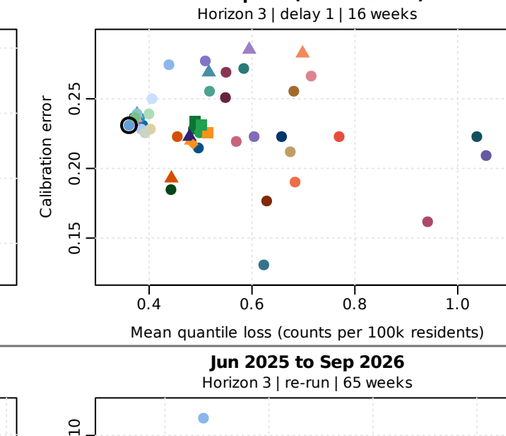
Jun 2025 to Sep 2026
Horizon 3 | delay 1 | 65 weeks



Up to peak (weeks -7 to 0)
Horizon 3 | delay 1 | 12 weeks

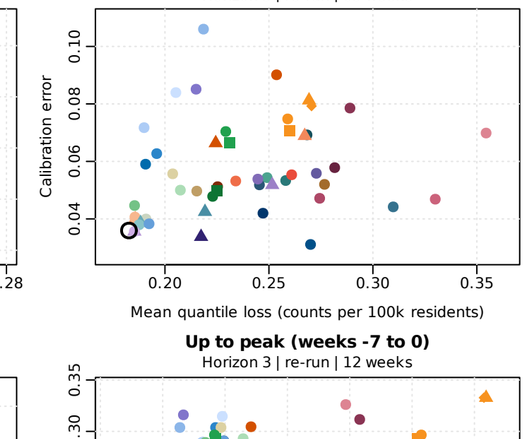


After peak (weeks 1 to 8)
Horizon 3 | delay 1 | 16 weeks

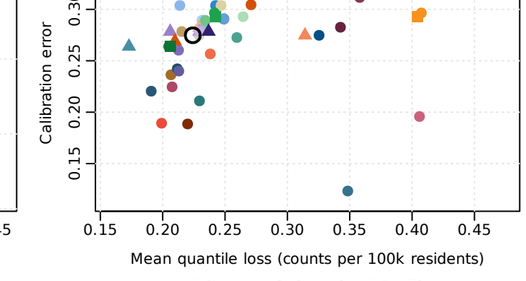


Feedback re-run

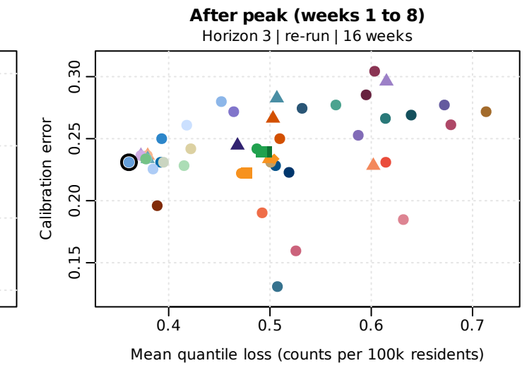
Jun 2025 to Sep 2026
Horizon 3 | re-run | 65 weeks

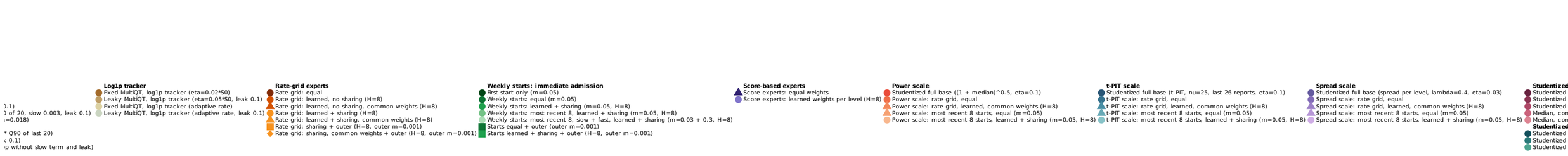


Up to peak (weeks -7 to 0)
Horizon 3 | re-run | 12 weeks



After peak (weeks 1 to 8)
Horizon 3 | re-run | 16 weeks





Calibration error (CE): mean absolute coverage residual after averaging weeks within each quantile level.
Open circle: base forecasts. Filled circle: weights per quantile level; triangle: one weight vector shared by all levels; square / diamond: the same + outer MultiQT.
Each feedback design uses the settings tuned with it; the values in the legend are those of feedback delay 0 (all: output/results/tuning).

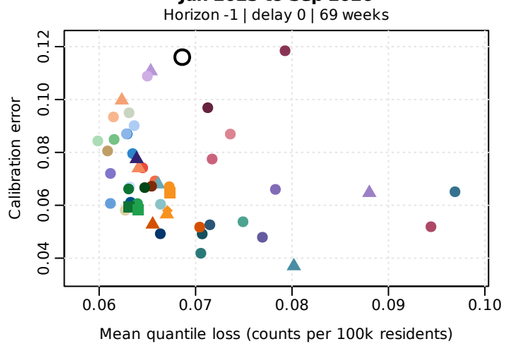
covid | California (06) | evaluation: Jun 2025 to Sep 2026 | raw-count updates, loss in counts per 100k residents

Rows: feedback design (delay d: learn once, d weeks after the first report, from the report current then; re-run: re-run every week on the reports known then) and window; columns: horizon.

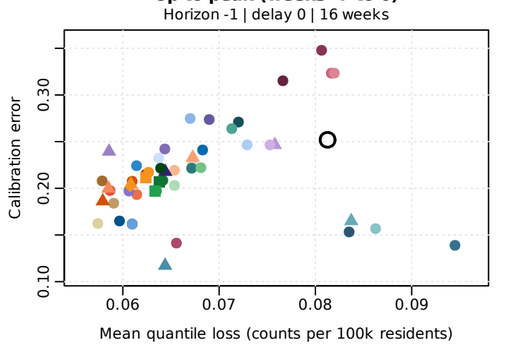
Horizon -1

Feedback delay 0

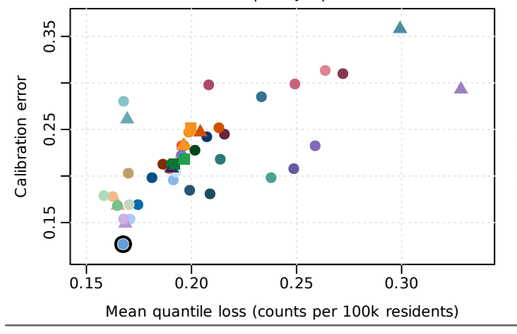
Jun 2025 to Sep 2026
Horizon -1 | delay 0 | 69 weeks



Up to peak (weeks -7 to 0)
Horizon -1 | delay 0 | 16 weeks

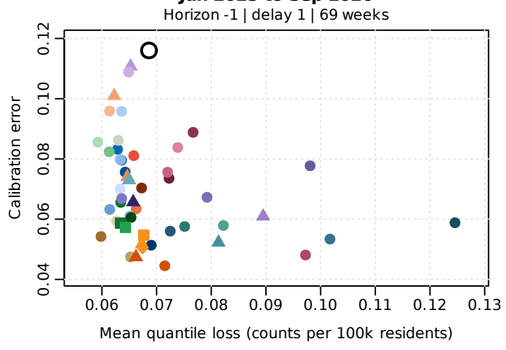


After peak (weeks 1 to 8)
Horizon -1 | delay 0 | 9 weeks

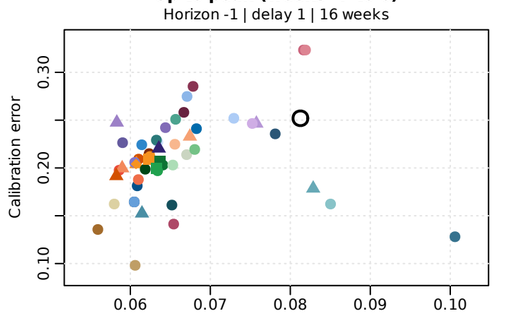


Feedback delay 1

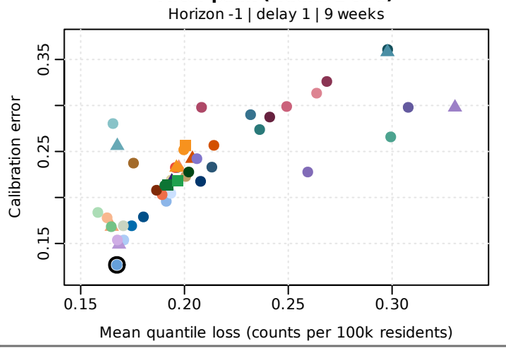
Jun 2025 to Sep 2026
Horizon -1 | delay 1 | 69 weeks



Up to peak (weeks -7 to 0)
Horizon -1 | delay 1 | 16 weeks

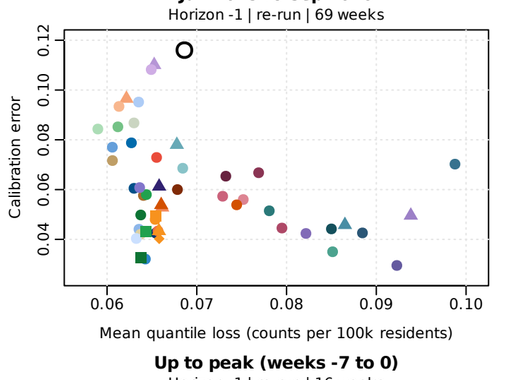


After peak (weeks 1 to 8)
Horizon -1 | delay 1 | 9 weeks

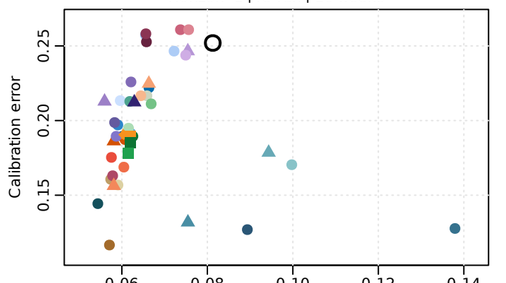


Feedback re-run

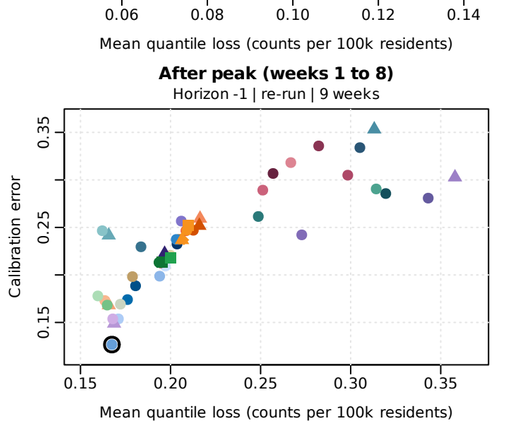
Jun 2025 to Sep 2026
Horizon -1 | re-run | 69 weeks



Up to peak (weeks -7 to 0)
Horizon -1 | re-run | 16 weeks



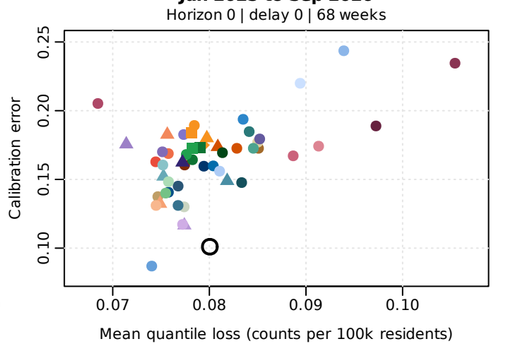
After peak (weeks 1 to 8)
Horizon -1 | re-run | 9 weeks



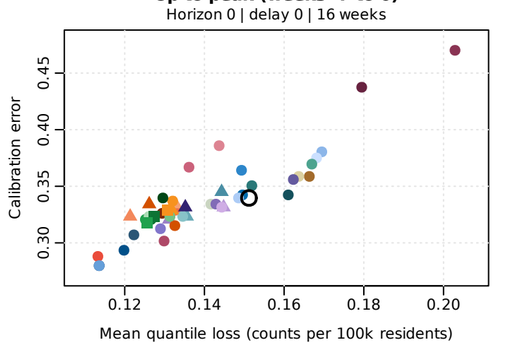
Horizon 0

Feedback delay 0

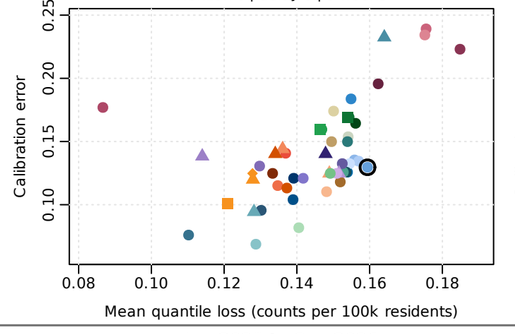
Jun 2025 to Sep 2026
Horizon 0 | delay 0 | 68 weeks



Up to peak (weeks -7 to 0)
Horizon 0 | delay 0 | 16 weeks

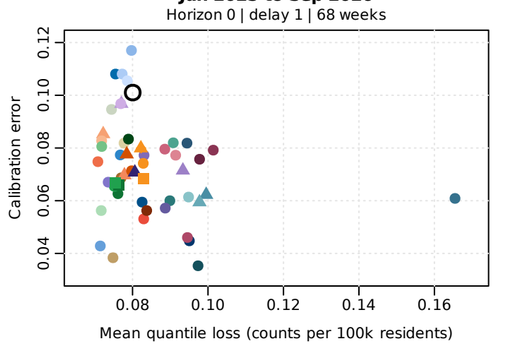


After peak (weeks 1 to 8)
Horizon 0 | delay 0 | 9 weeks

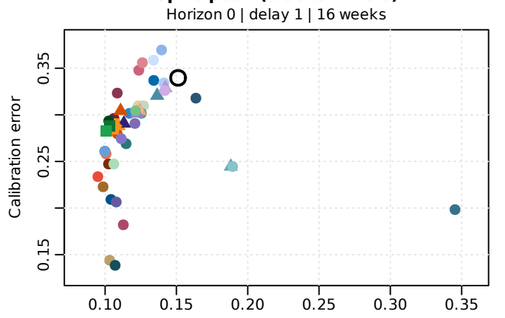


Feedback delay 1

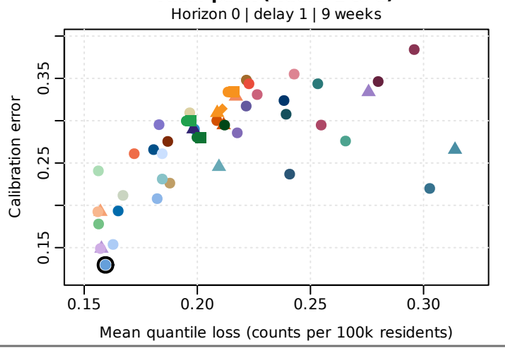
Jun 2025 to Sep 2026
Horizon 0 | delay 1 | 68 weeks



Up to peak (weeks -7 to 0)
Horizon 0 | delay 1 | 16 weeks

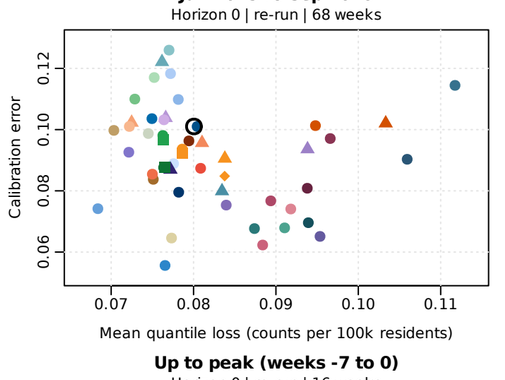


After peak (weeks 1 to 8)
Horizon 0 | delay 1 | 9 weeks

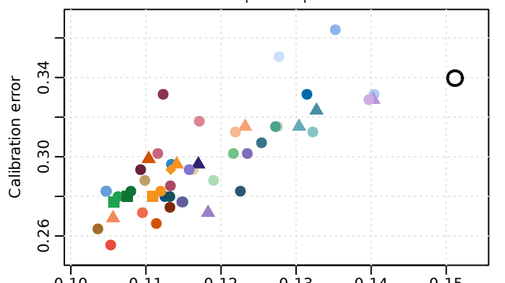


Feedback re-run

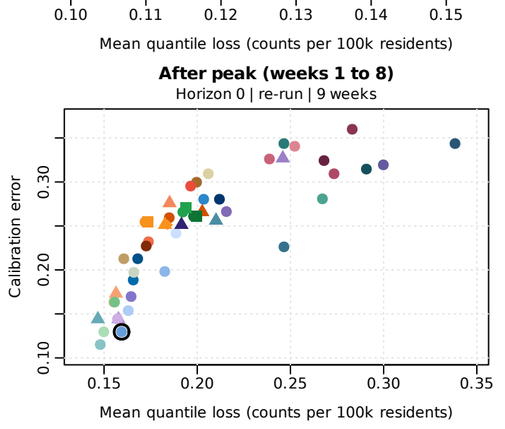
Jun 2025 to Sep 2026
Horizon 0 | re-run | 68 weeks



Up to peak (weeks -7 to 0)
Horizon 0 | re-run | 16 weeks



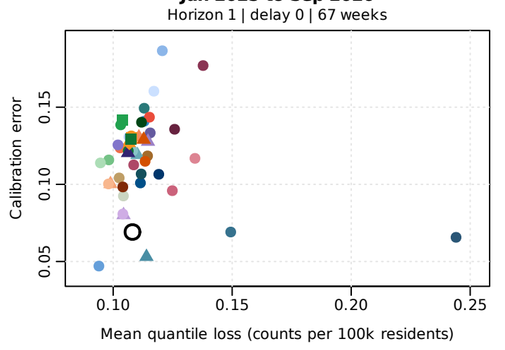
After peak (weeks 1 to 8)
Horizon 0 | re-run | 9 weeks



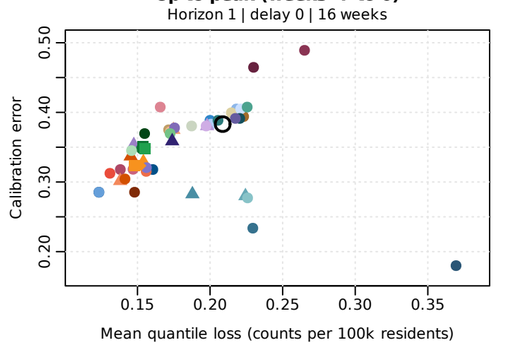
Horizon 1

Feedback delay 0

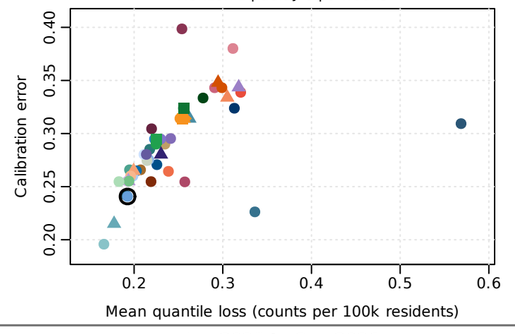
Jun 2025 to Sep 2026
Horizon 1 | delay 0 | 67 weeks



Up to peak (weeks -7 to 0)
Horizon 1 | delay 0 | 16 weeks

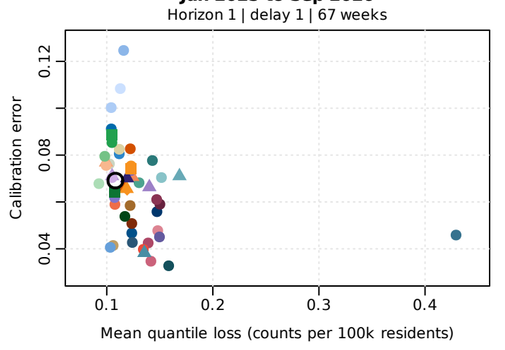


After peak (weeks 1 to 8)
Horizon 1 | delay 0 | 9 weeks

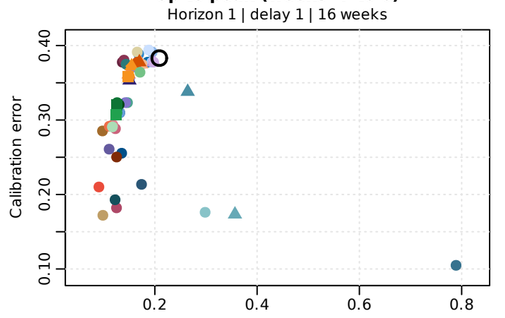


Feedback delay 1

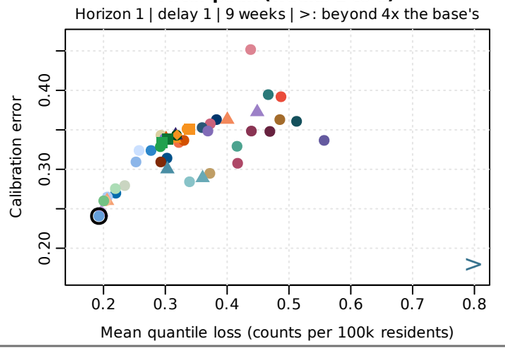
Jun 2025 to Sep 2026
Horizon 1 | delay 1 | 67 weeks



Up to peak (weeks -7 to 0)
Horizon 1 | delay 1 | 16 weeks

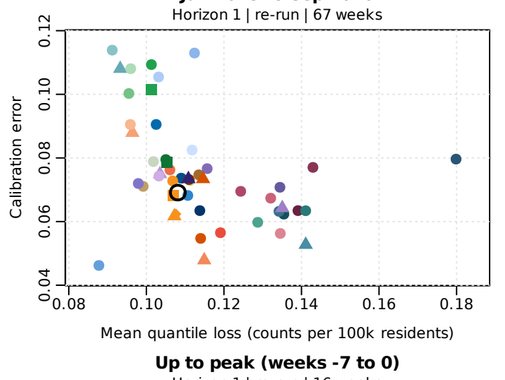


After peak (weeks 1 to 8)
Horizon 1 | delay 1 | 9 weeks | >: beyond 4x the base's

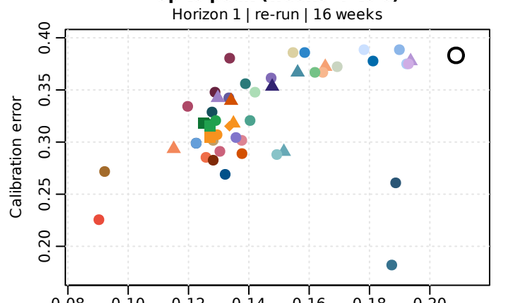


Feedback re-run

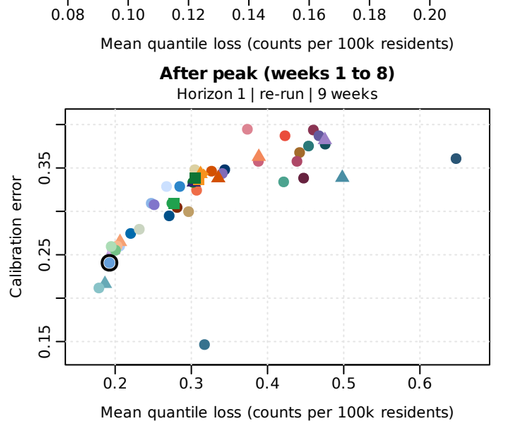
Jun 2025 to Sep 2026
Horizon 1 | re-run | 67 weeks



Up to peak (weeks -7 to 0)
Horizon 1 | re-run | 16 weeks



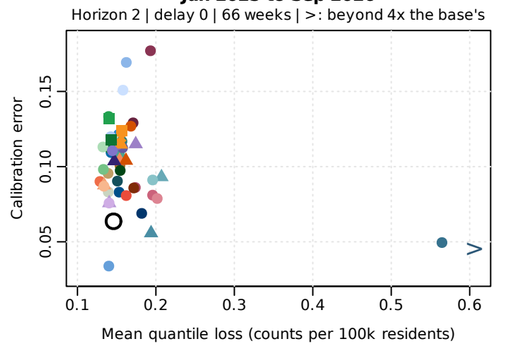
After peak (weeks 1 to 8)
Horizon 1 | re-run | 9 weeks



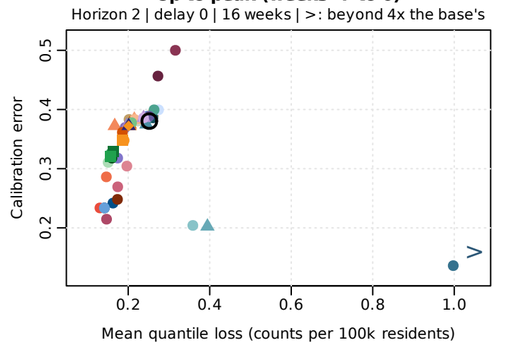
Horizon 2

Feedback delay 0

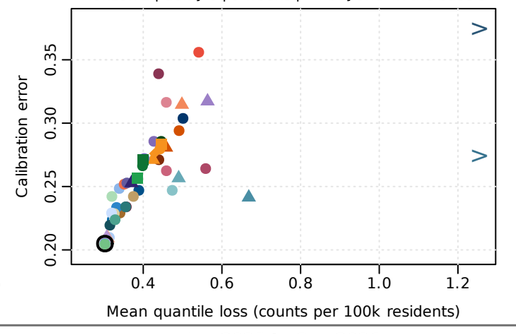
Jun 2025 to Sep 2026
Horizon 2 | delay 0 | 66 weeks | >: beyond 4x the base's



Up to peak (weeks -7 to 0)
Horizon 2 | delay 0 | 16 weeks | >: beyond 4x the base's

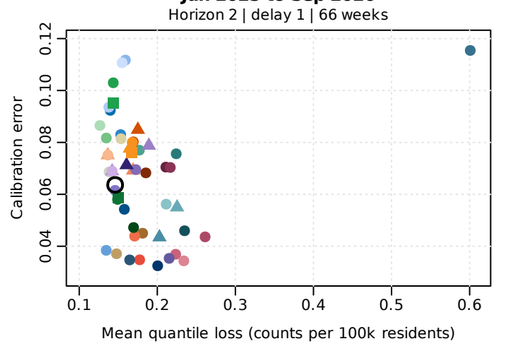


After peak (weeks 1 to 8)
Horizon 2 | delay 0 | 9 weeks | >: beyond 4x the base's

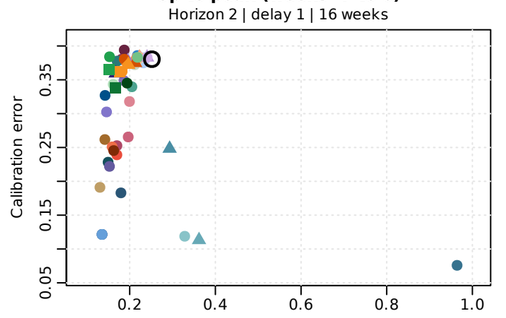


Feedback delay 1

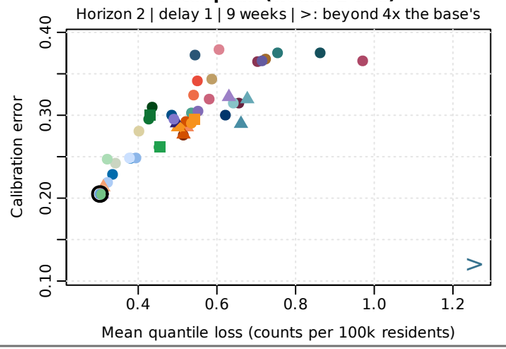
Jun 2025 to Sep 2026
Horizon 2 | delay 1 | 66 weeks



Up to peak (weeks -7 to 0)
Horizon 2 | delay 1 | 16 weeks

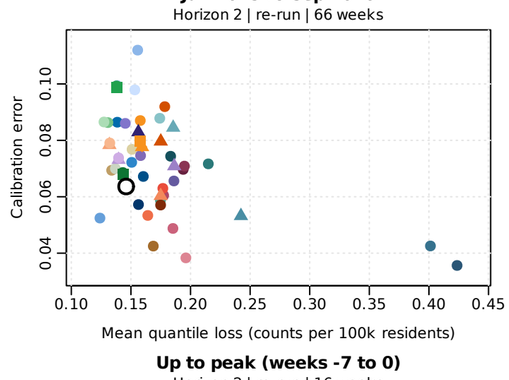


After peak (weeks 1 to 8)
Horizon 2 | delay 1 | 9 weeks | >: beyond 4x the base's

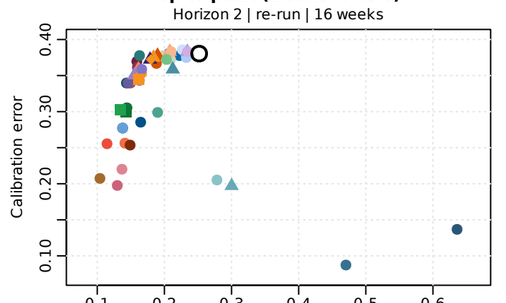


Feedback re-run

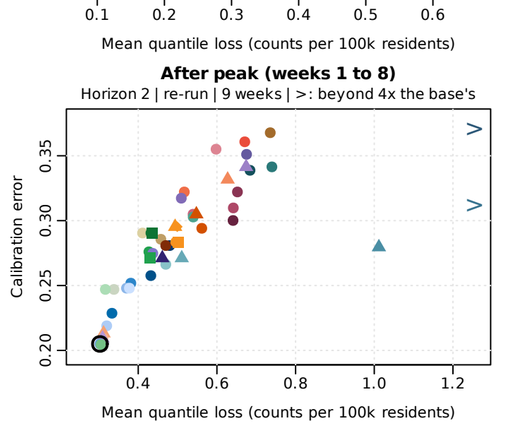
Jun 2025 to Sep 2026
Horizon 2 | re-run | 66 weeks



Up to peak (weeks -7 to 0)
Horizon 2 | re-run | 16 weeks



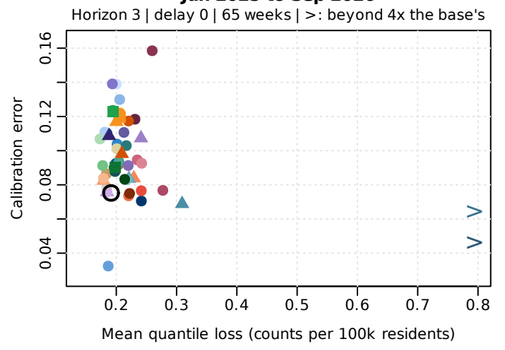
After peak (weeks 1 to 8)
Horizon 2 | re-run | 9 weeks | >: beyond 4x the base's



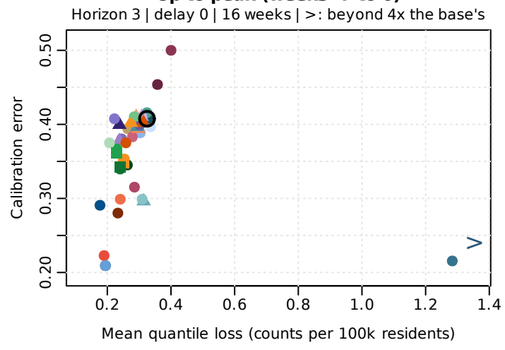
Horizon 3

Feedback delay 0

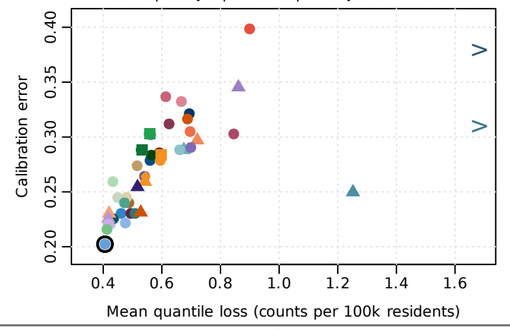
Jun 2025 to Sep 2026
Horizon 3 | delay 0 | 65 weeks | >: beyond 4x the base's



Up to peak (weeks -7 to 0)
Horizon 3 | delay 0 | 16 weeks | >: beyond 4x the base's

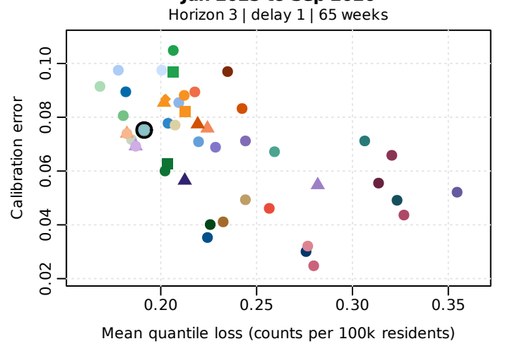


After peak (weeks 1 to 8)
Horizon 3 | delay 0 | 9 weeks | >: beyond 4x the base's

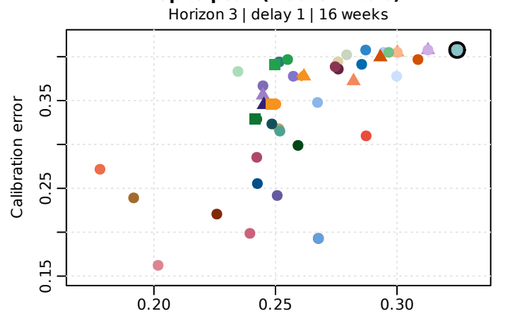


Feedback delay 1

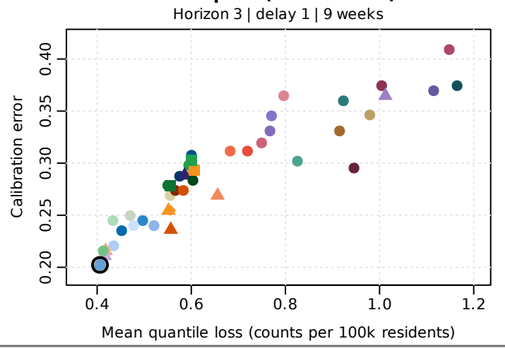
Jun 2025 to Sep 2026
Horizon 3 | delay 1 | 65 weeks



Up to peak (weeks -7 to 0)
Horizon 3 | delay 1 | 16 weeks

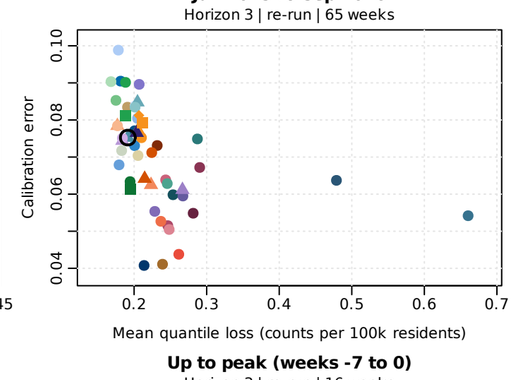


After peak (weeks 1 to 8)
Horizon 3 | delay 1 | 9 weeks

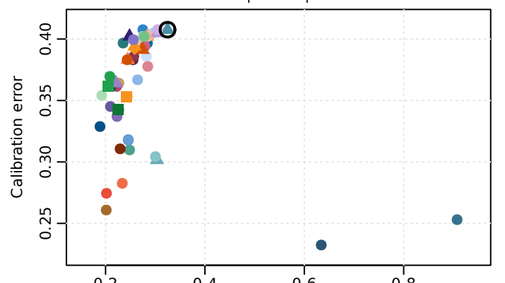


Feedback re-run

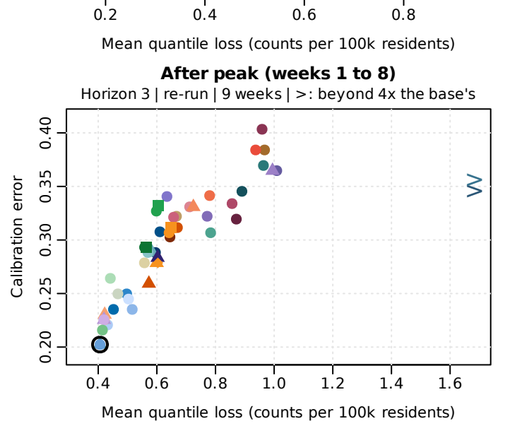
Jun 2025 to Sep 2026
Horizon 3 | re-run | 65 weeks

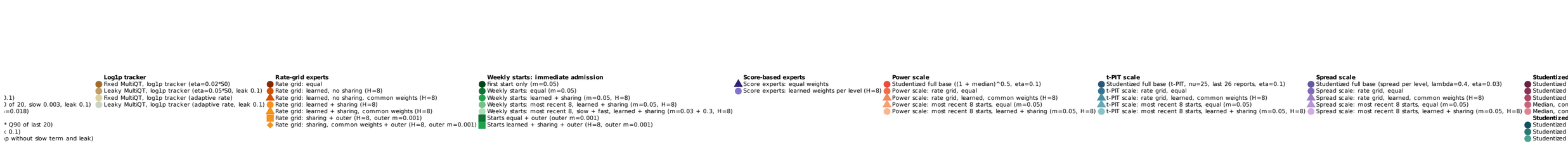


Up to peak (weeks -7 to 0)
Horizon 3 | re-run | 16 weeks



After peak (weeks 1 to 8)
Horizon 3 | re-run | 9 weeks | >: beyond 4x the base's





Calibration error (CE): mean absolute coverage residual after averaging weeks within each quantile level.
Open circle: base forecasts. Filled circle: weights per quantile level; triangle: one weight vector shared by all levels; square / diamond: the same + outer MultiQT.
Each feedback design uses the settings tuned with it; the values in the legend are those of feedback delay 0 (all: output/results/tuning).

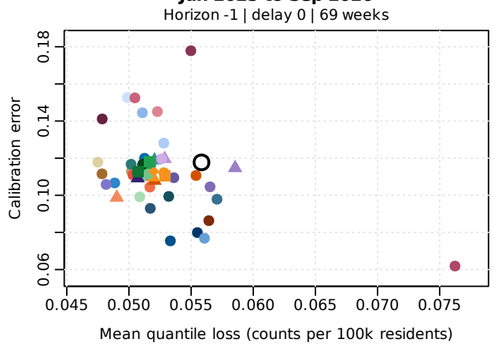
covid | Texas (48) | evaluation: Jun 2025 to Sep 2026 | raw-count updates, loss in counts per 100k residents

Rows: feedback design (delay d: learn once, d weeks after the first report, from the report current then; re-run: re-run every week on the reports known then) and window; columns: horizon.

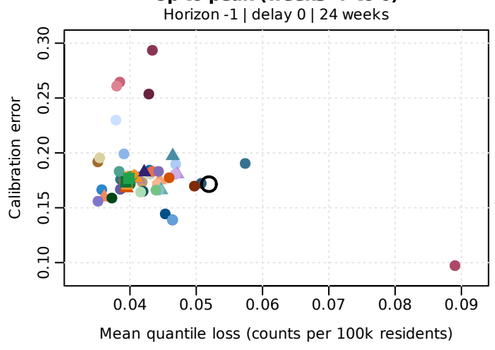
Horizon -1

Feedback delay 0

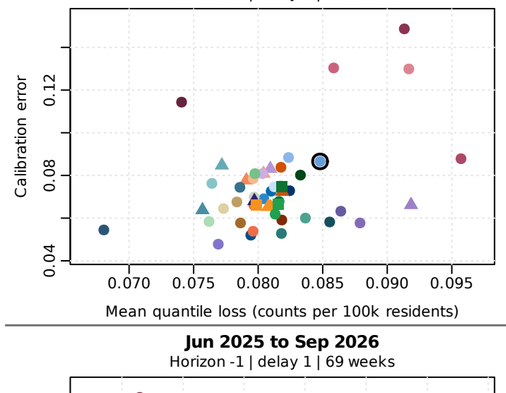
Jun 2025 to Sep 2026
Horizon -1 | delay 0 | 69 weeks



Up to peak (weeks -7 to 0)
Horizon -1 | delay 0 | 24 weeks

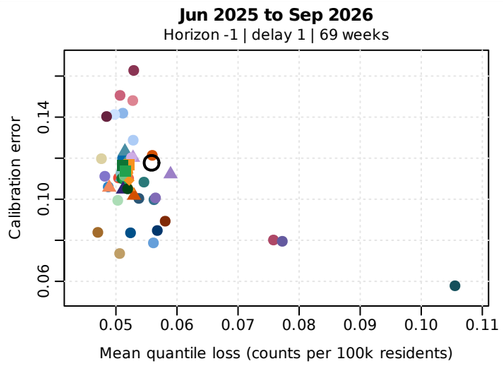


After peak (weeks 1 to 8)
Horizon -1 | delay 0 | 19 weeks

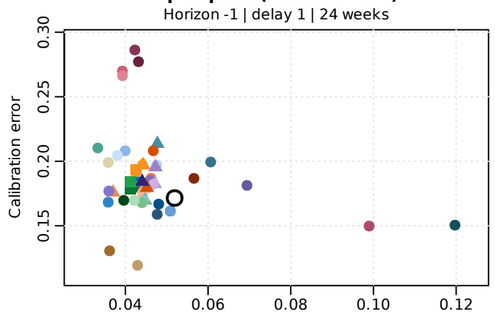


Feedback delay 1

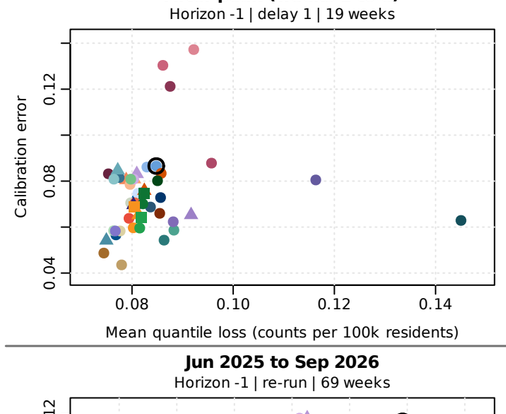
Jun 2025 to Sep 2026
Horizon -1 | delay 1 | 69 weeks



Up to peak (weeks -7 to 0)
Horizon -1 | delay 1 | 24 weeks

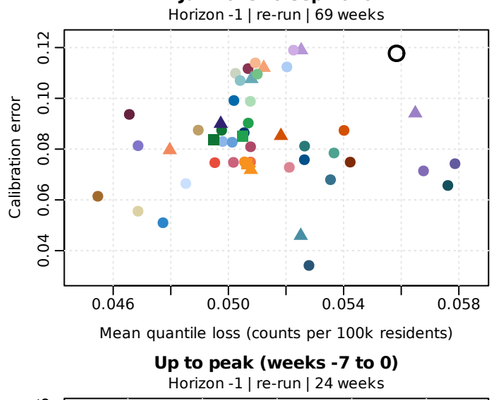


After peak (weeks 1 to 8)
Horizon -1 | delay 1 | 19 weeks

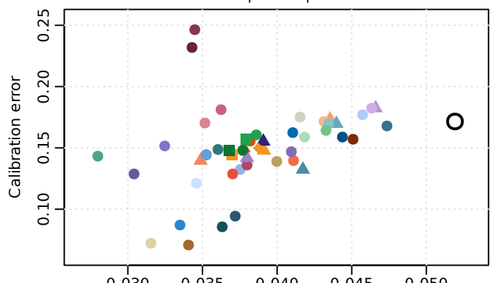


Feedback re-run

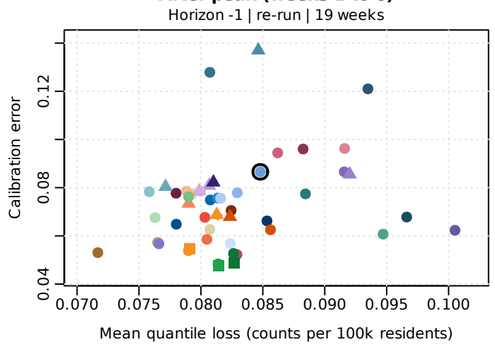
Jun 2025 to Sep 2026
Horizon -1 | re-run | 69 weeks



Up to peak (weeks -7 to 0)
Horizon -1 | re-run | 24 weeks



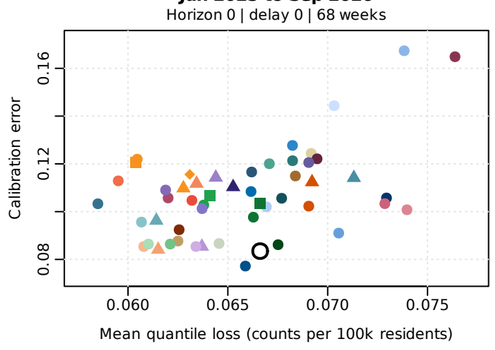
After peak (weeks 1 to 8)
Horizon -1 | re-run | 19 weeks



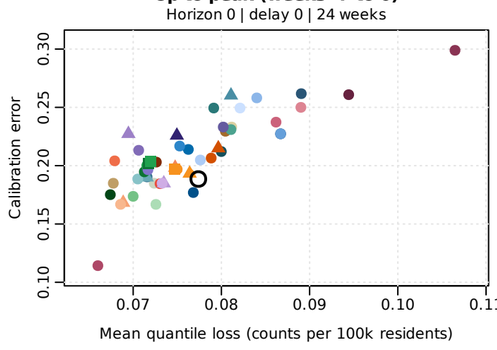
Horizon 0

Feedback delay 0

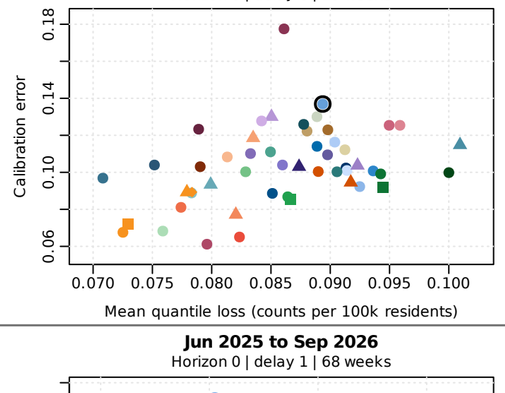
Jun 2025 to Sep 2026
Horizon 0 | delay 0 | 68 weeks



Up to peak (weeks -7 to 0)
Horizon 0 | delay 0 | 24 weeks

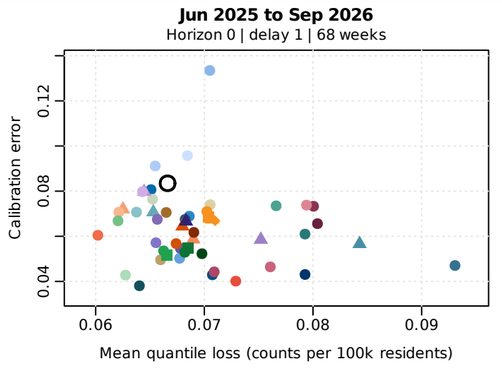


After peak (weeks 1 to 8)
Horizon 0 | delay 0 | 19 weeks

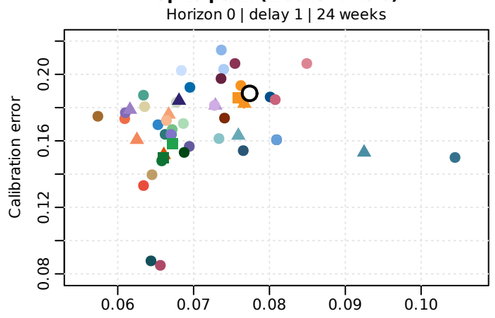


Feedback delay 1

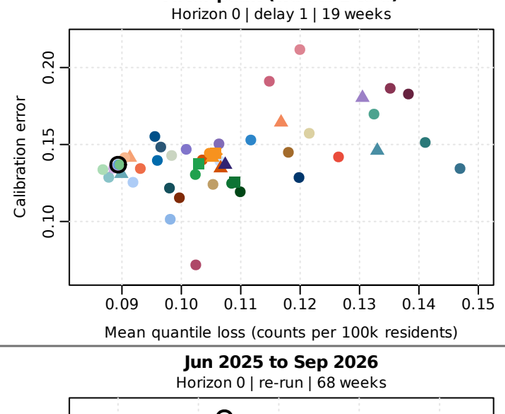
Jun 2025 to Sep 2026
Horizon 0 | delay 1 | 68 weeks



Up to peak (weeks -7 to 0)
Horizon 0 | delay 1 | 24 weeks

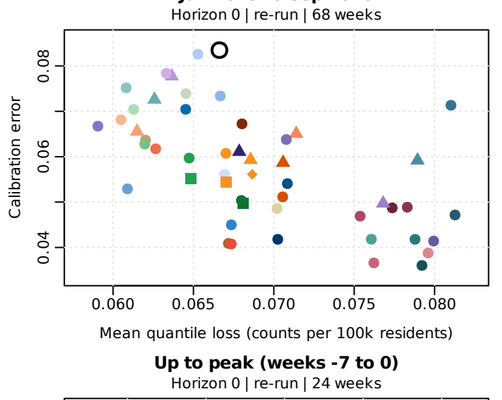


After peak (weeks 1 to 8)
Horizon 0 | delay 1 | 19 weeks

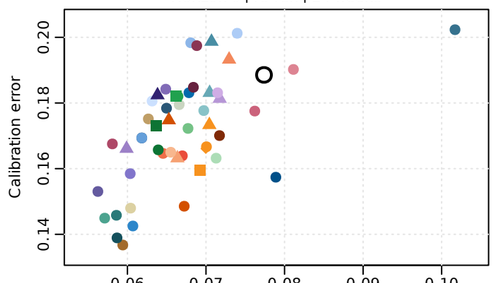


Feedback re-run

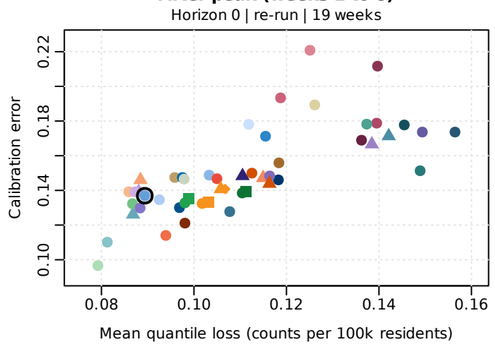
Jun 2025 to Sep 2026
Horizon 0 | re-run | 68 weeks



Up to peak (weeks -7 to 0)
Horizon 0 | re-run | 24 weeks



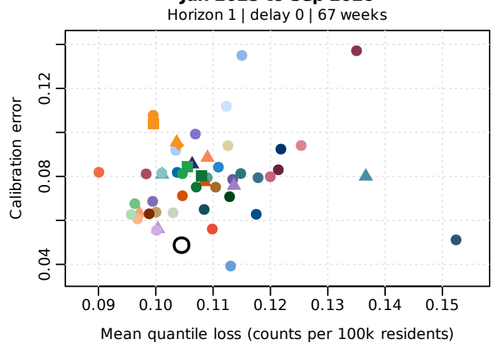
After peak (weeks 1 to 8)
Horizon 0 | re-run | 19 weeks



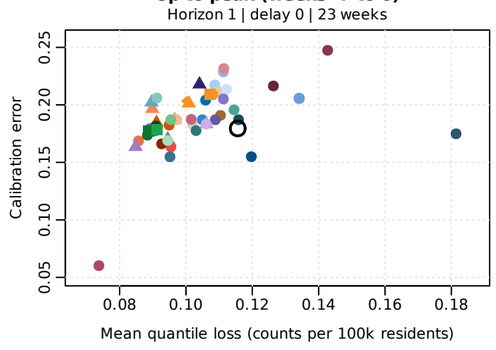
Horizon 1

Feedback delay 0

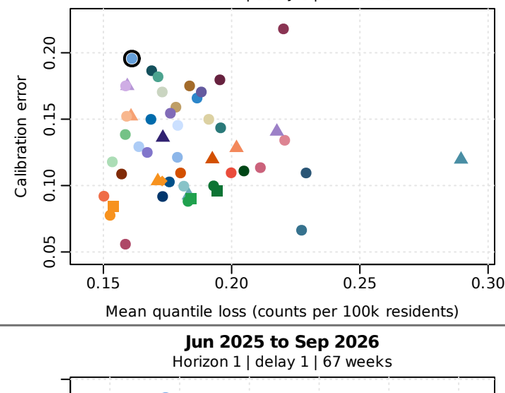
Jun 2025 to Sep 2026
Horizon 1 | delay 0 | 67 weeks



Up to peak (weeks -7 to 0)
Horizon 1 | delay 0 | 23 weeks

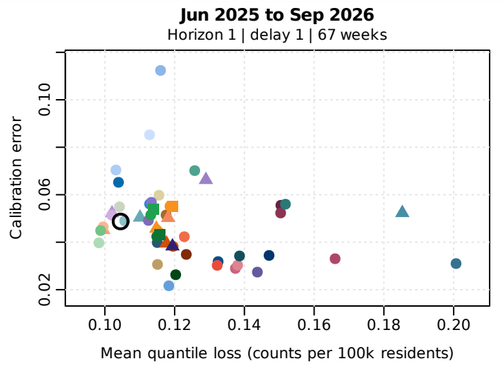


After peak (weeks 1 to 8)
Horizon 1 | delay 0 | 19 weeks

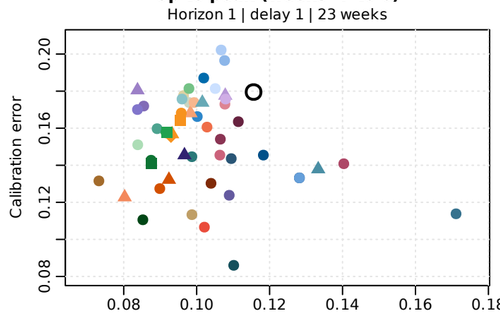


Feedback delay 1

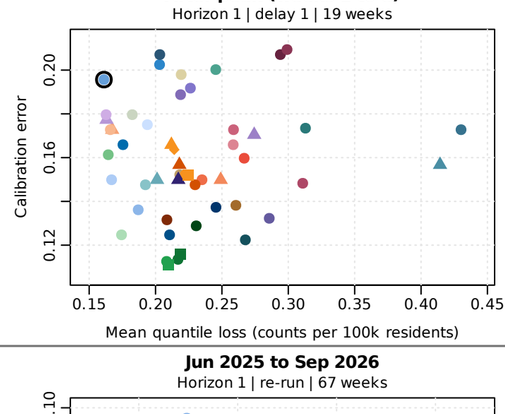
Jun 2025 to Sep 2026
Horizon 1 | delay 1 | 67 weeks



Up to peak (weeks -7 to 0)
Horizon 1 | delay 1 | 23 weeks

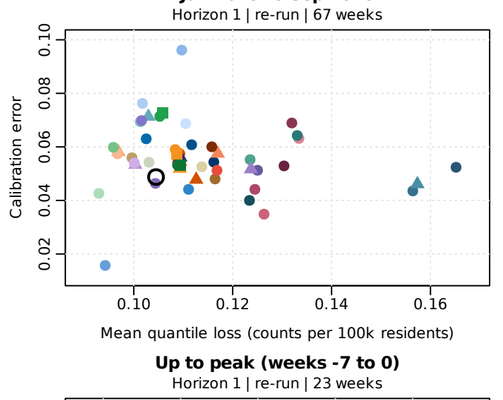


After peak (weeks 1 to 8)
Horizon 1 | delay 1 | 19 weeks

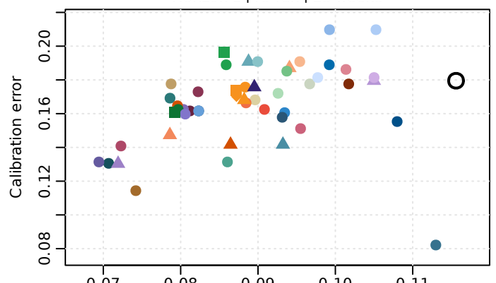


Feedback re-run

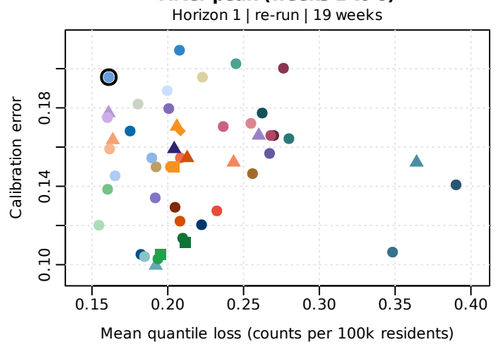
Jun 2025 to Sep 2026
Horizon 1 | re-run | 67 weeks



Up to peak (weeks -7 to 0)
Horizon 1 | re-run | 23 weeks



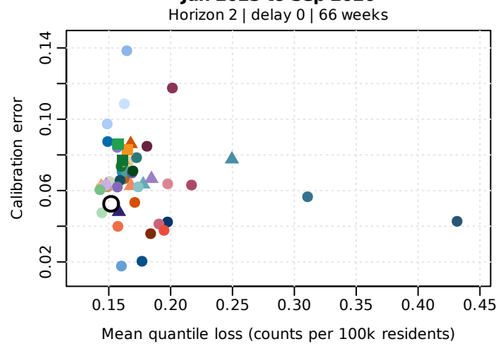
After peak (weeks 1 to 8)
Horizon 1 | re-run | 19 weeks



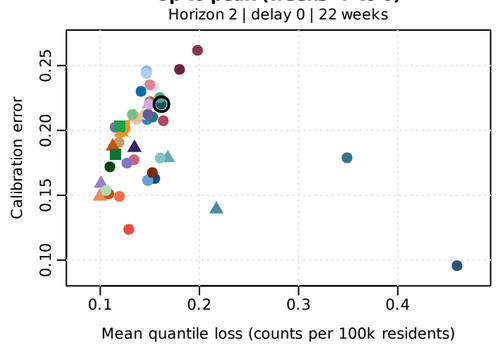
Horizon 2

Feedback delay 0

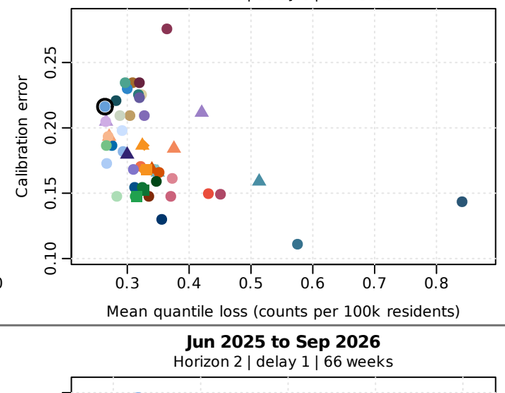
Jun 2025 to Sep 2026
Horizon 2 | delay 0 | 66 weeks



Up to peak (weeks -7 to 0)
Horizon 2 | delay 0 | 22 weeks

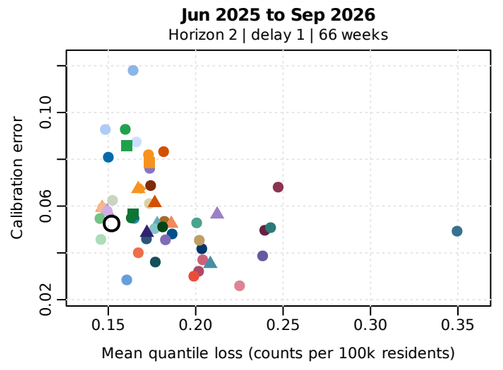


After peak (weeks 1 to 8)
Horizon 2 | delay 0 | 19 weeks

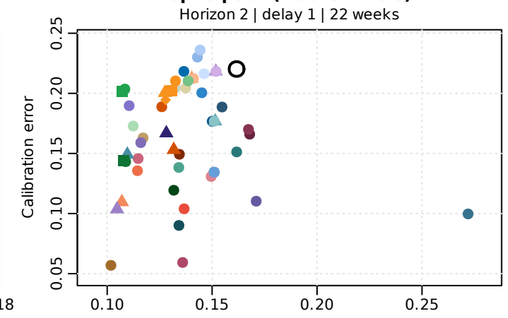


Feedback delay 1

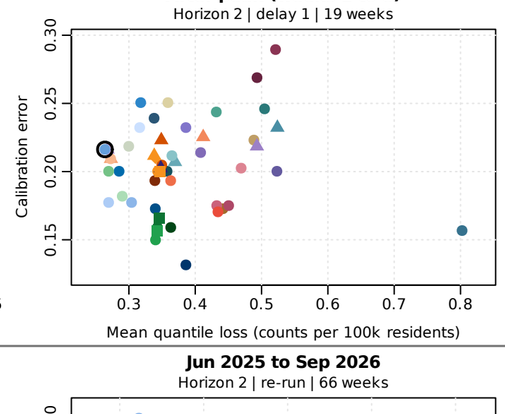
Jun 2025 to Sep 2026
Horizon 2 | delay 1 | 66 weeks



Up to peak (weeks -7 to 0)
Horizon 2 | delay 1 | 22 weeks

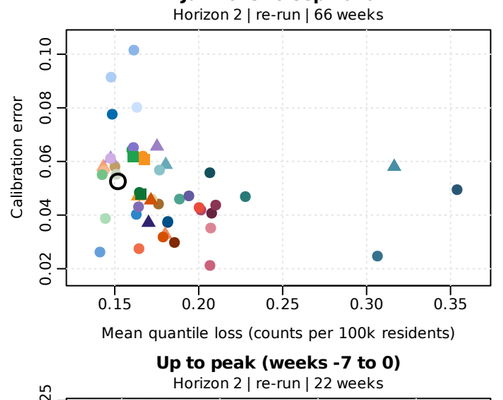


After peak (weeks 1 to 8)
Horizon 2 | delay 1 | 19 weeks

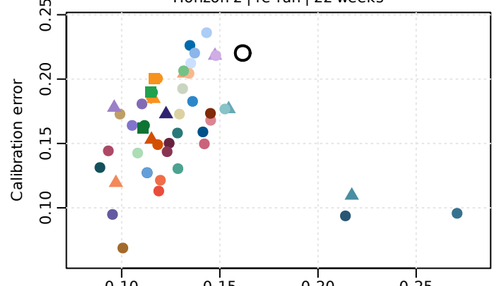


Feedback re-run

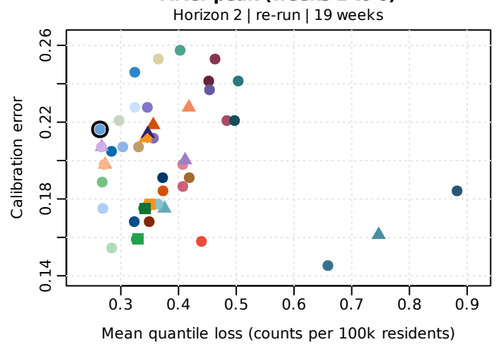
Jun 2025 to Sep 2026
Horizon 2 | re-run | 66 weeks



Up to peak (weeks -7 to 0)
Horizon 2 | re-run | 22 weeks



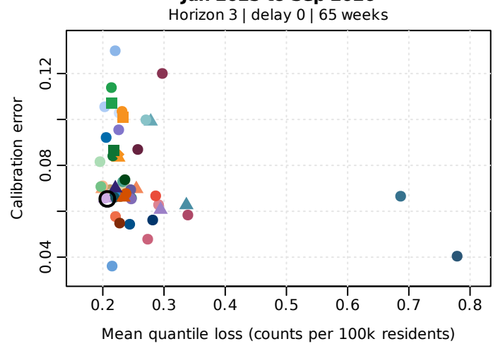
After peak (weeks 1 to 8)
Horizon 2 | re-run | 19 weeks



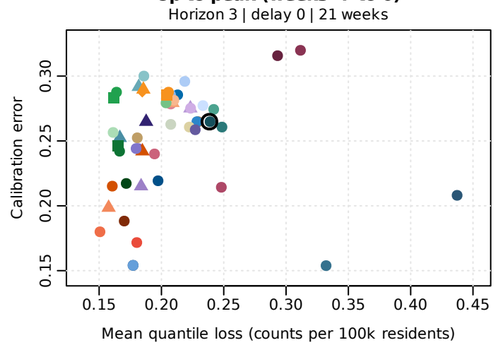
Horizon 3

Feedback delay 0

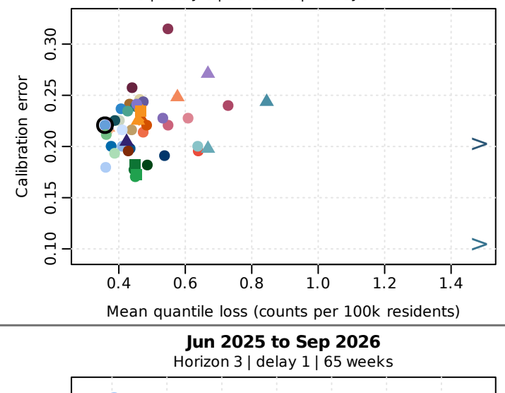
Jun 2025 to Sep 2026
Horizon 3 | delay 0 | 65 weeks



Up to peak (weeks -7 to 0)
Horizon 3 | delay 0 | 21 weeks

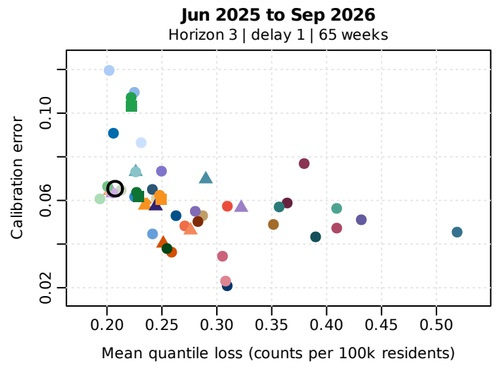


After peak (weeks 1 to 8)
Horizon 3 | delay 0 | 19 weeks | >: beyond 4x the base's

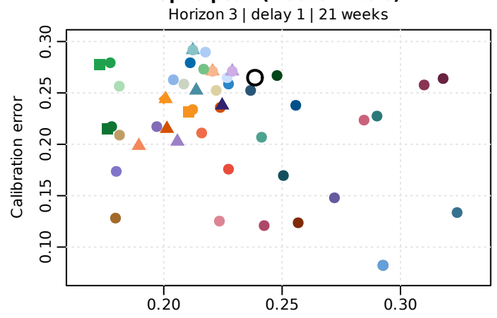


Feedback delay 1

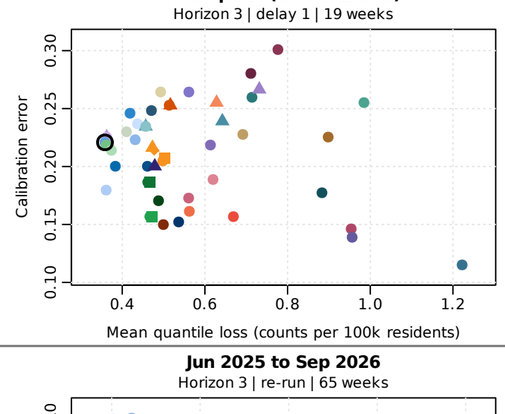
Jun 2025 to Sep 2026
Horizon 3 | delay 1 | 65 weeks



Up to peak (weeks -7 to 0)
Horizon 3 | delay 1 | 21 weeks

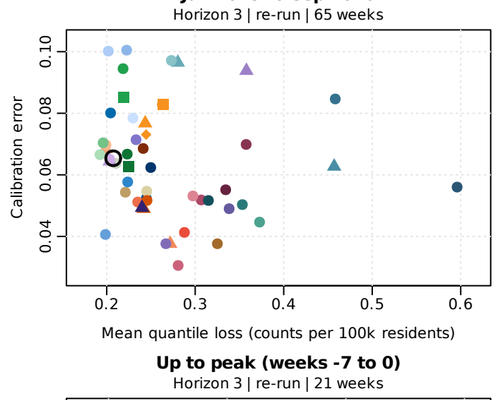


After peak (weeks 1 to 8)
Horizon 3 | delay 1 | 19 weeks

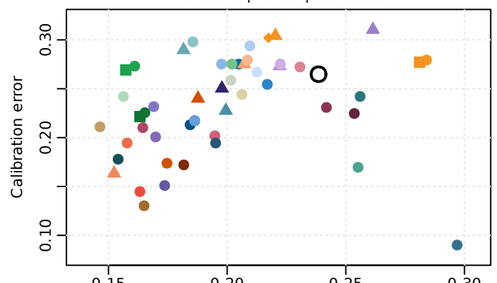


Feedback re-run

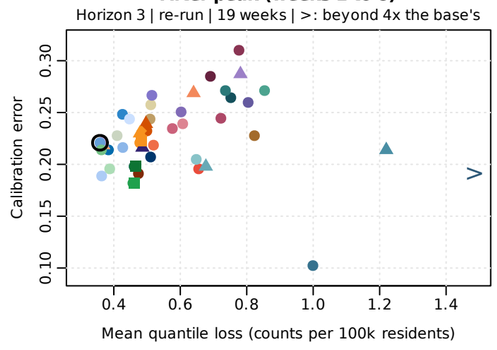
Jun 2025 to Sep 2026
Horizon 3 | re-run | 65 weeks

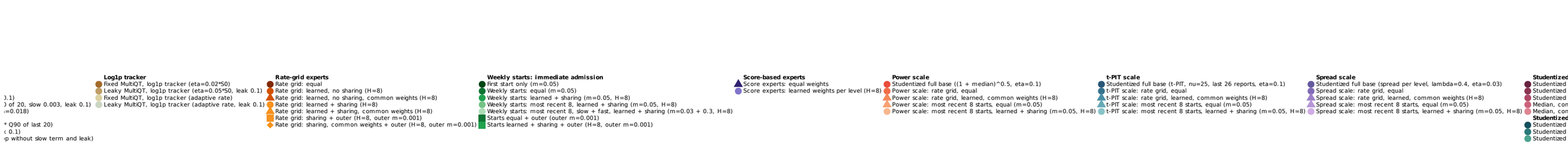


Up to peak (weeks -7 to 0)
Horizon 3 | re-run | 21 weeks



After peak (weeks 1 to 8)
Horizon 3 | re-run | 19 weeks | >: beyond 4x the base's





Calibration error (CE): mean absolute coverage residual after averaging weeks within each quantile level.
Open circle: base forecasts. Filled circle: weights per quantile level; triangle: one weight vector shared by all levels; square / diamond: the same + outer MultiQT.
Each feedback design uses the settings tuned with it; the values in the legend are those of feedback delay 0 (all: output/results/tuning).

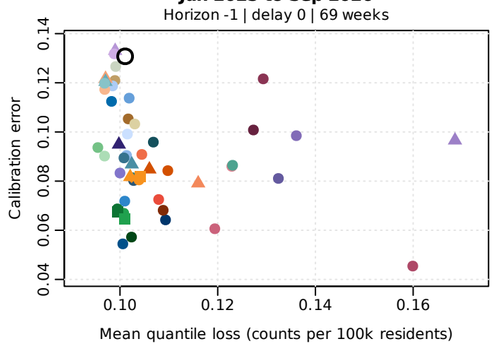
covid | New York (36) | evaluation: Jun 2025 to Sep 2026 | raw-count updates, loss in counts per 100k residents

Rows: feedback design (delay d: learn once, d weeks after the first report, from the report current then; re-run: re-run every week on the reports known then) and window; columns: horizon.

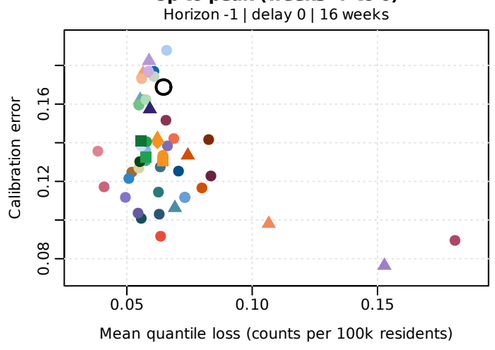
Horizon -1

Feedback delay 0

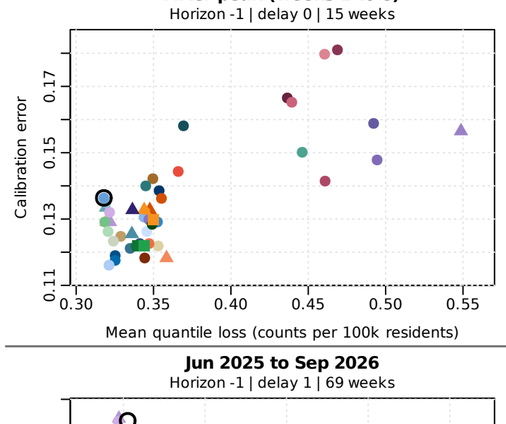
Jun 2025 to Sep 2026
Horizon -1 | delay 0 | 69 weeks



Up to peak (weeks -7 to 0)
Horizon -1 | delay 0 | 16 weeks

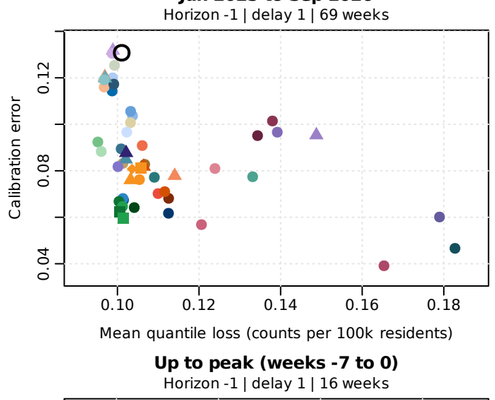


After peak (weeks 1 to 8)
Horizon -1 | delay 0 | 15 weeks

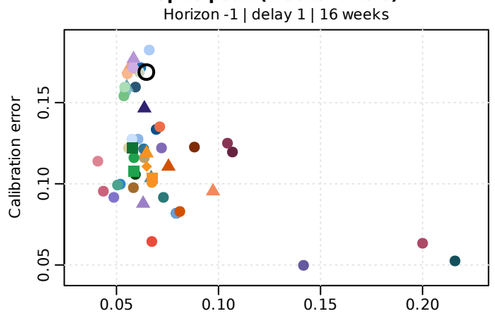


Feedback delay 1

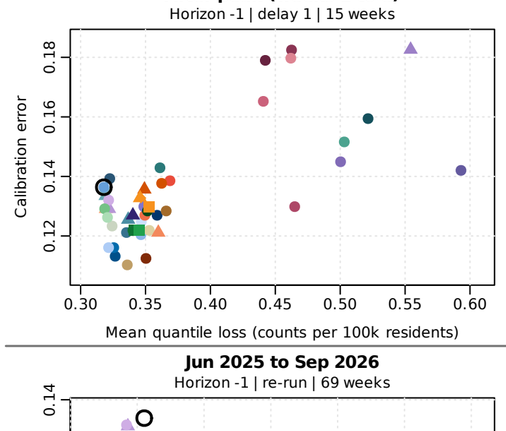
Jun 2025 to Sep 2026
Horizon -1 | delay 1 | 69 weeks



Up to peak (weeks -7 to 0)
Horizon -1 | delay 1 | 16 weeks

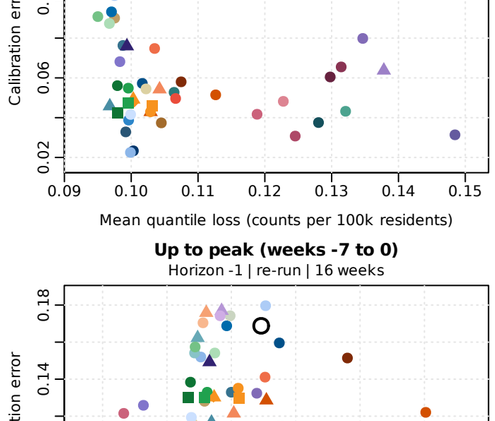


After peak (weeks 1 to 8)
Horizon -1 | delay 1 | 15 weeks

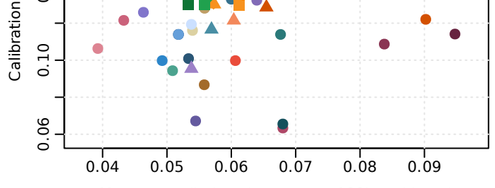


Feedback re-run

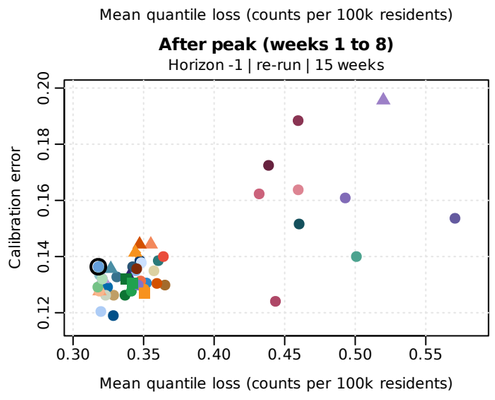
Jun 2025 to Sep 2026
Horizon -1 | re-run | 69 weeks



Up to peak (weeks -7 to 0)
Horizon -1 | re-run | 16 weeks



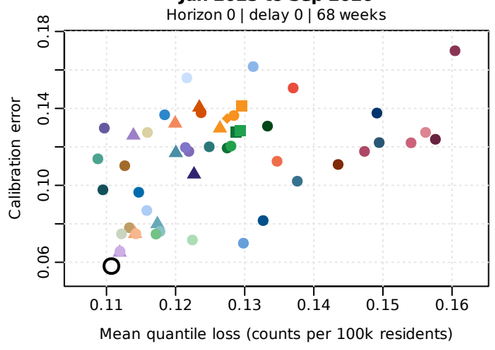
After peak (weeks 1 to 8)
Horizon -1 | re-run | 15 weeks



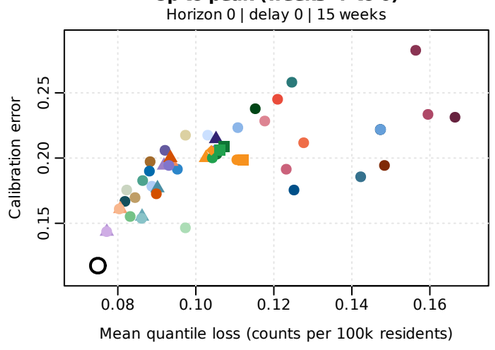
Horizon 0

Feedback delay 0

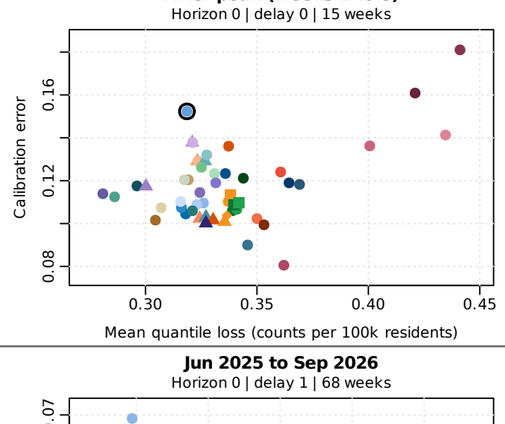
Jun 2025 to Sep 2026
Horizon 0 | delay 0 | 68 weeks



Up to peak (weeks -7 to 0)
Horizon 0 | delay 0 | 15 weeks

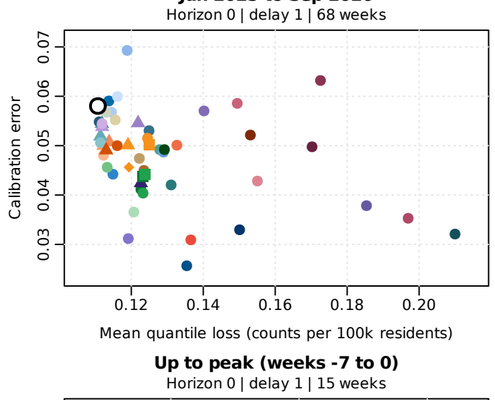


After peak (weeks 1 to 8)
Horizon 0 | delay 0 | 15 weeks

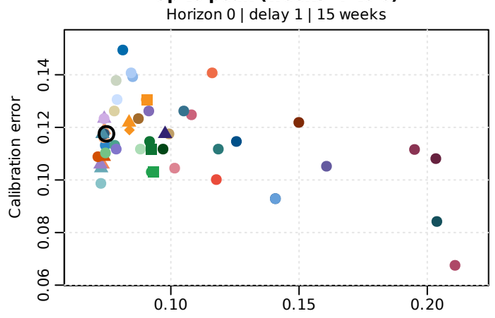


Feedback delay 1

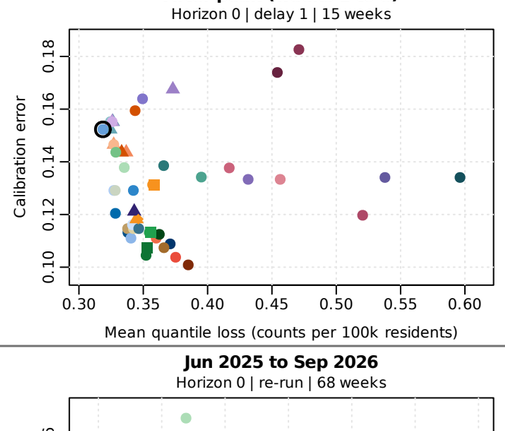
Jun 2025 to Sep 2026
Horizon 0 | delay 1 | 68 weeks



Up to peak (weeks -7 to 0)
Horizon 0 | delay 1 | 15 weeks

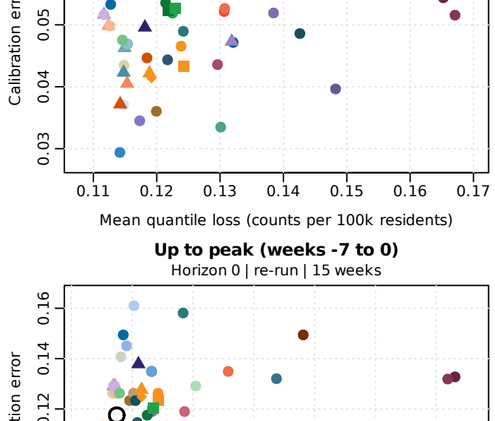


After peak (weeks 1 to 8)
Horizon 0 | delay 1 | 15 weeks

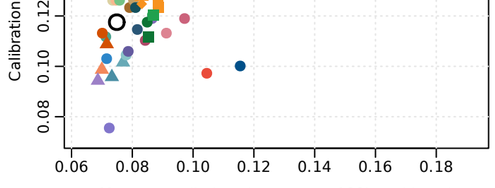


Feedback re-run

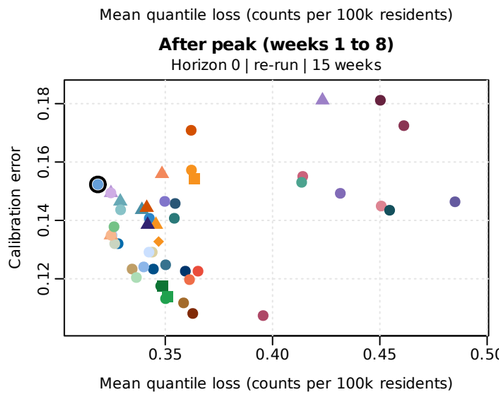
Jun 2025 to Sep 2026
Horizon 0 | re-run | 68 weeks



Up to peak (weeks -7 to 0)
Horizon 0 | re-run | 15 weeks



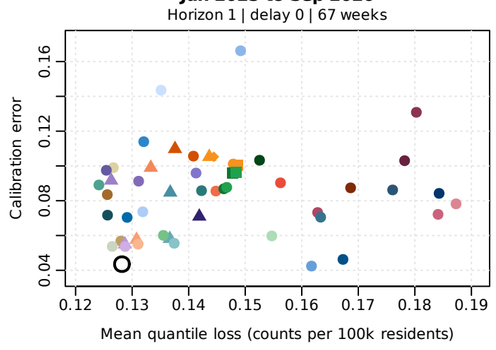
After peak (weeks 1 to 8)
Horizon 0 | re-run | 15 weeks



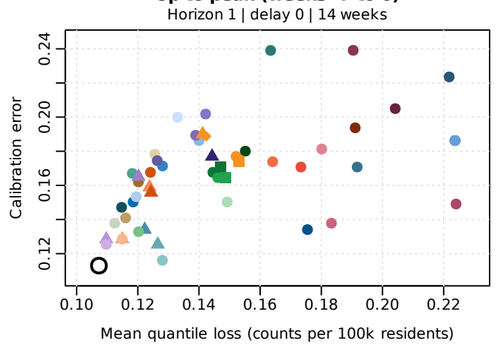
Horizon 1

Feedback delay 0

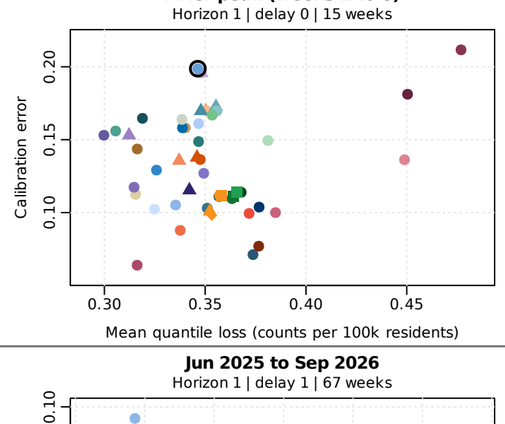
Jun 2025 to Sep 2026
Horizon 1 | delay 0 | 67 weeks



Up to peak (weeks -7 to 0)
Horizon 1 | delay 0 | 14 weeks

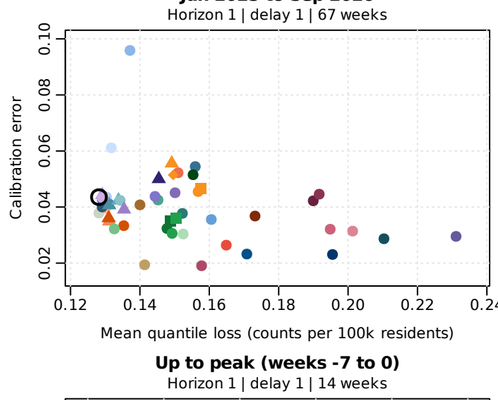


After peak (weeks 1 to 8)
Horizon 1 | delay 0 | 15 weeks

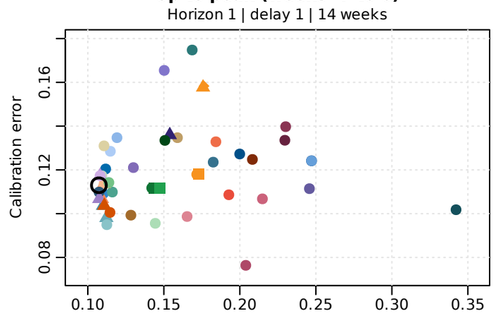


Feedback delay 1

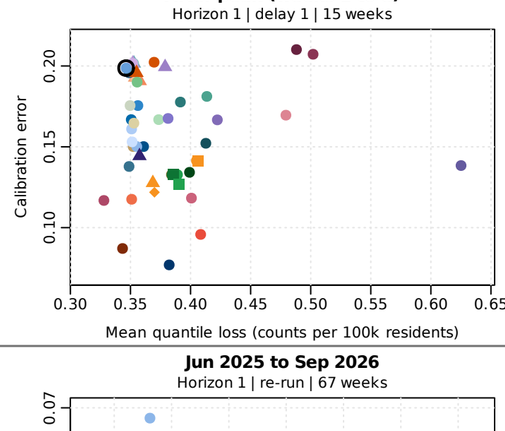
Jun 2025 to Sep 2026
Horizon 1 | delay 1 | 67 weeks



Up to peak (weeks -7 to 0)
Horizon 1 | delay 1 | 14 weeks

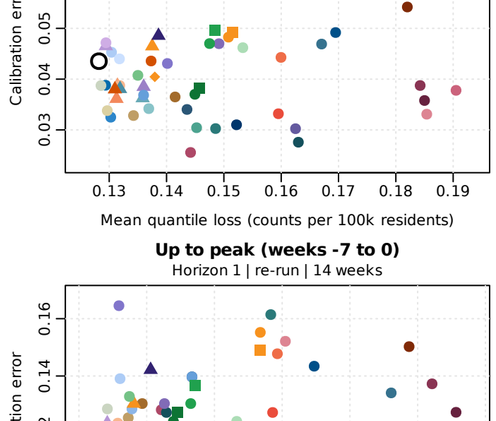


After peak (weeks 1 to 8)
Horizon 1 | delay 1 | 15 weeks

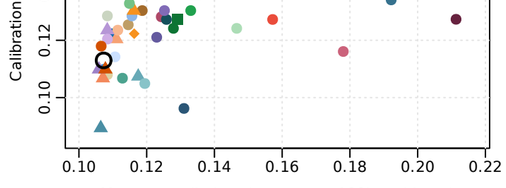


Feedback re-run

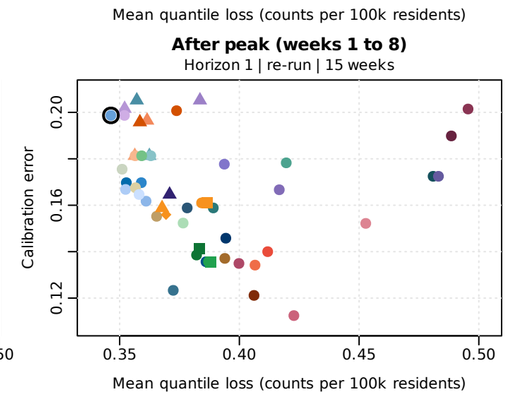
Jun 2025 to Sep 2026
Horizon 1 | re-run | 67 weeks



Up to peak (weeks -7 to 0)
Horizon 1 | re-run | 14 weeks



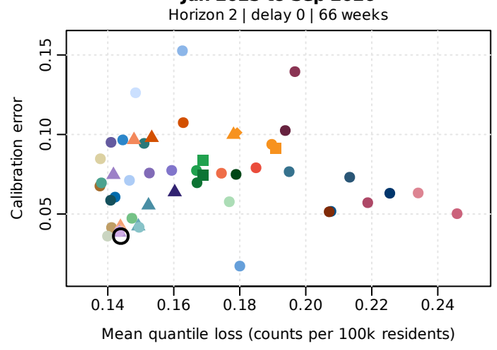
After peak (weeks 1 to 8)
Horizon 1 | re-run | 15 weeks



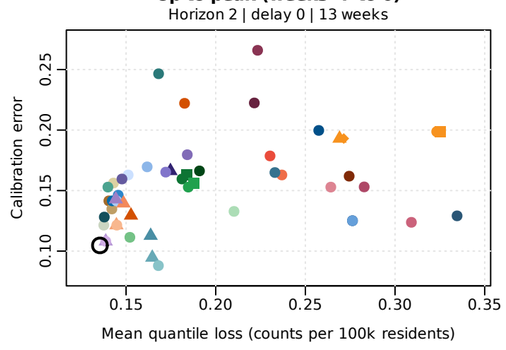
Horizon 2

Feedback delay 0

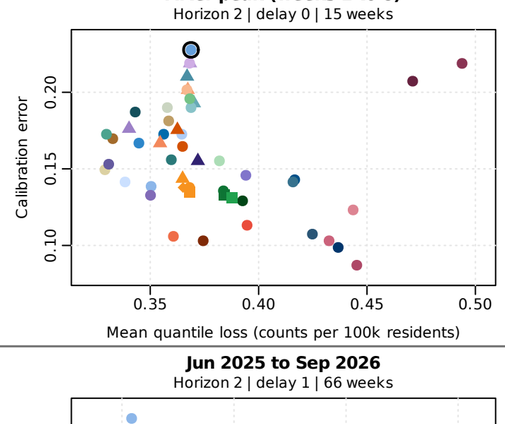
Jun 2025 to Sep 2026
Horizon 2 | delay 0 | 66 weeks



Up to peak (weeks -7 to 0)
Horizon 2 | delay 0 | 13 weeks

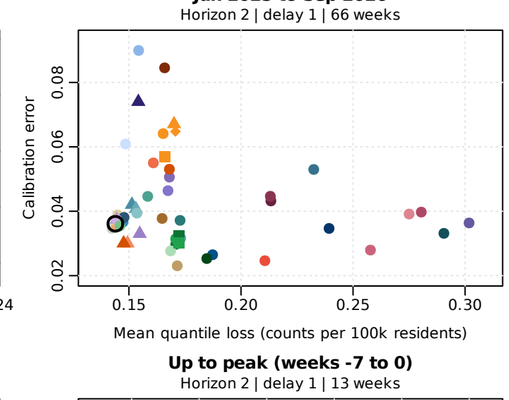


After peak (weeks 1 to 8)
Horizon 2 | delay 0 | 15 weeks

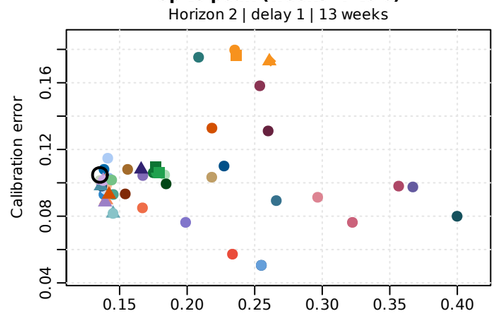


Feedback delay 1

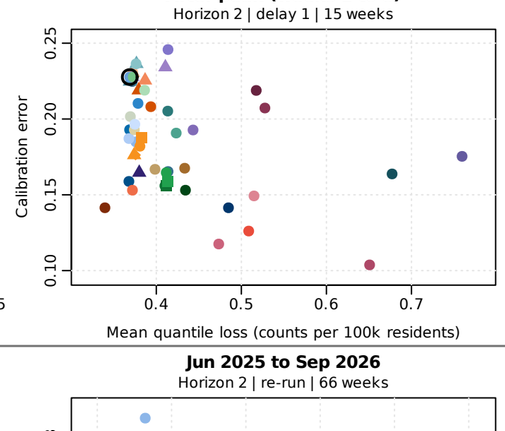
Jun 2025 to Sep 2026
Horizon 2 | delay 1 | 66 weeks



Up to peak (weeks -7 to 0)
Horizon 2 | delay 1 | 13 weeks

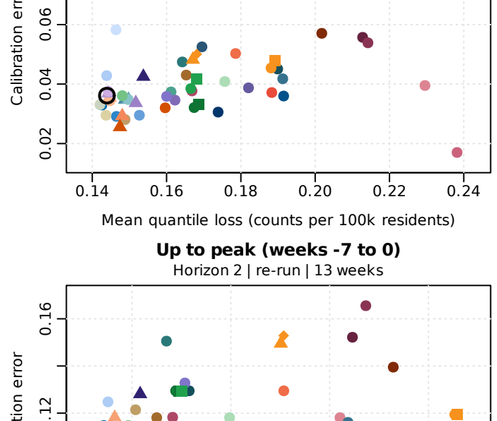


After peak (weeks 1 to 8)
Horizon 2 | delay 1 | 15 weeks

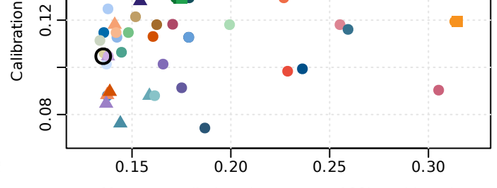


Feedback re-run

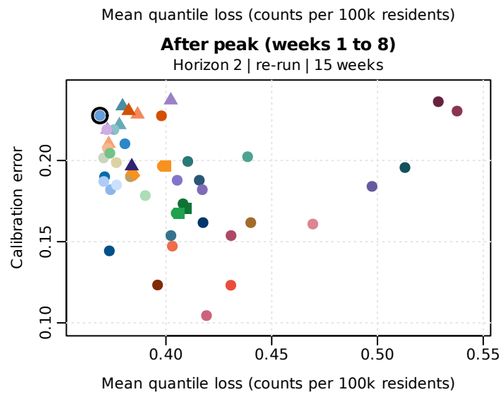
Jun 2025 to Sep 2026
Horizon 2 | re-run | 66 weeks



Up to peak (weeks -7 to 0)
Horizon 2 | re-run | 13 weeks



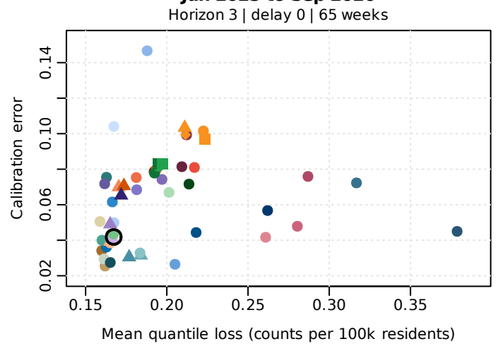
After peak (weeks 1 to 8)
Horizon 2 | re-run | 15 weeks



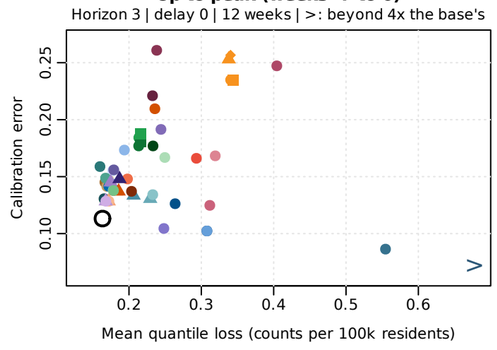
Horizon 3

Feedback delay 0

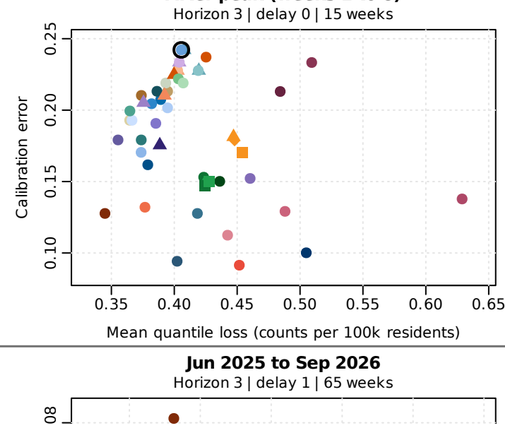
Jun 2025 to Sep 2026
Horizon 3 | delay 0 | 65 weeks



Up to peak (weeks -7 to 0)
Horizon 3 | delay 0 | 12 weeks | >: beyond 4x the base's

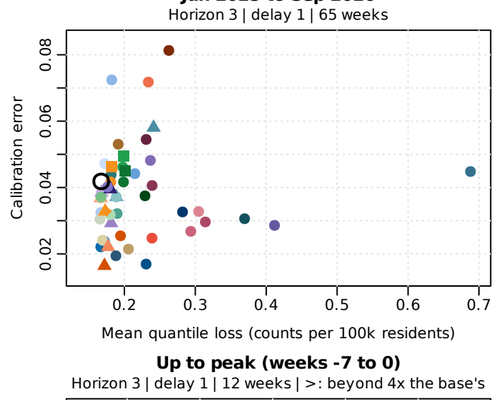


After peak (weeks 1 to 8)
Horizon 3 | delay 0 | 15 weeks

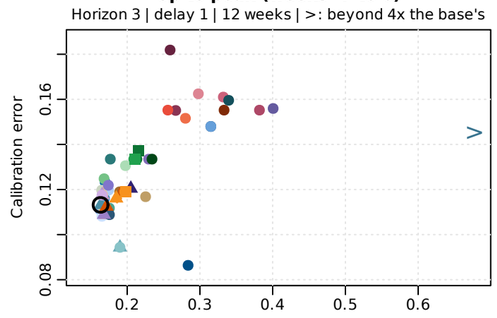


Feedback delay 1

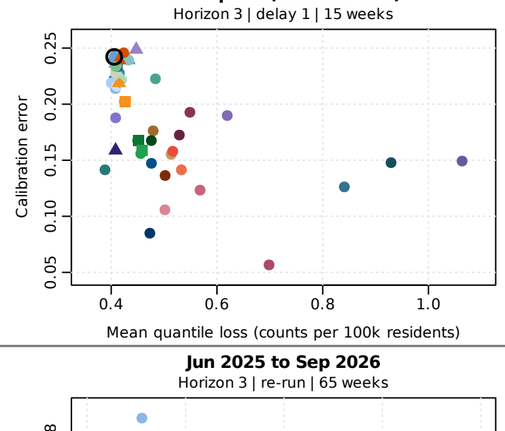
Jun 2025 to Sep 2026
Horizon 3 | delay 1 | 65 weeks



Up to peak (weeks -7 to 0)
Horizon 3 | delay 1 | 12 weeks | >: beyond 4x the base's

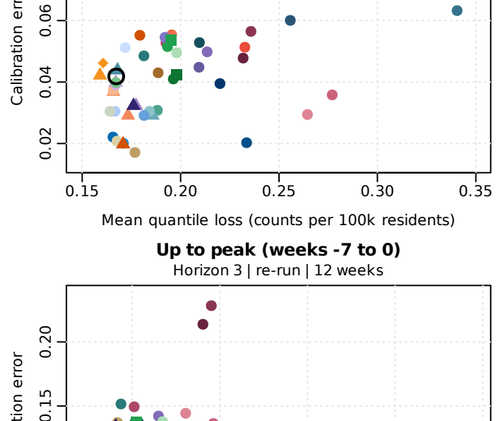


After peak (weeks 1 to 8)
Horizon 3 | delay 1 | 15 weeks

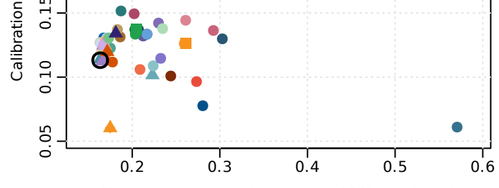


Feedback re-run

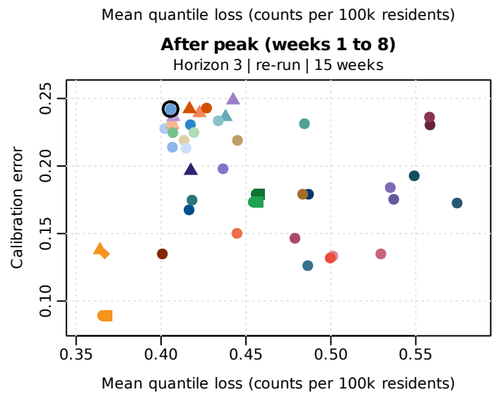
Jun 2025 to Sep 2026
Horizon 3 | re-run | 65 weeks

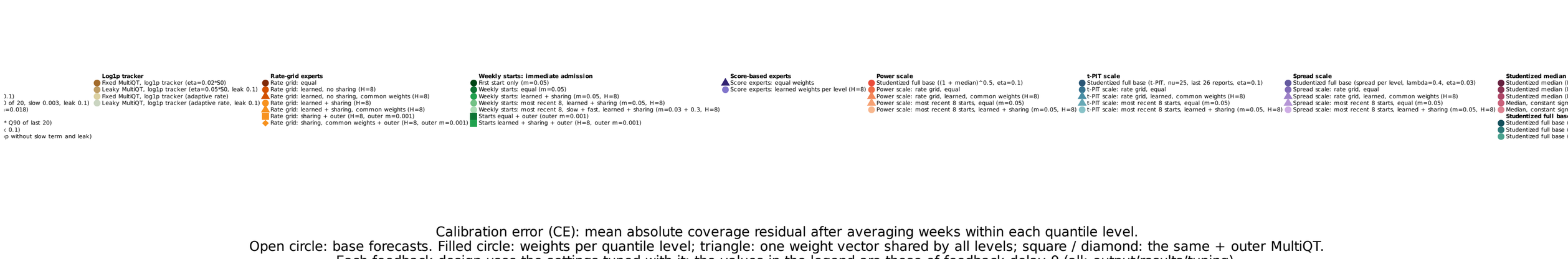


Up to peak (weeks -7 to 0)
Horizon 3 | re-run | 12 weeks



After peak (weeks 1 to 8)
Horizon 3 | re-run | 15 weeks

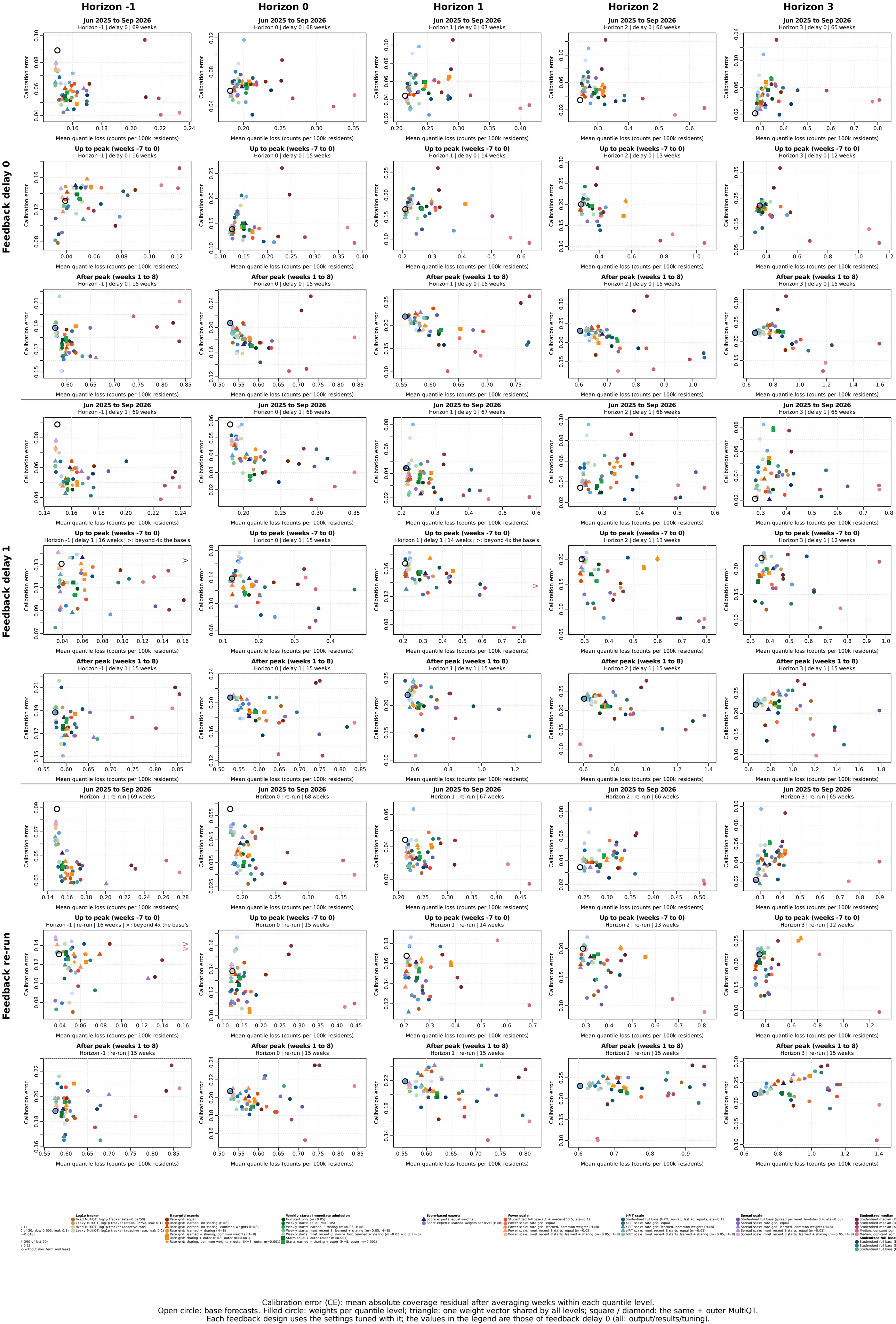




Calibration error (CE): mean absolute coverage residual after averaging weeks within each quantile level.
Open circle: base forecasts. Filled circle: weights per quantile level; triangle: one weight vector shared by all levels; square / diamond: the same + outer MultiQT.
Each feedback design uses the settings tuned with it; the values in the legend are those of feedback delay 0 (all: output/results/tuning).

covid | Pennsylvania (42) | evaluation: Jun 2025 to Sep 2026 | raw-count updates, loss in counts per 100k residents

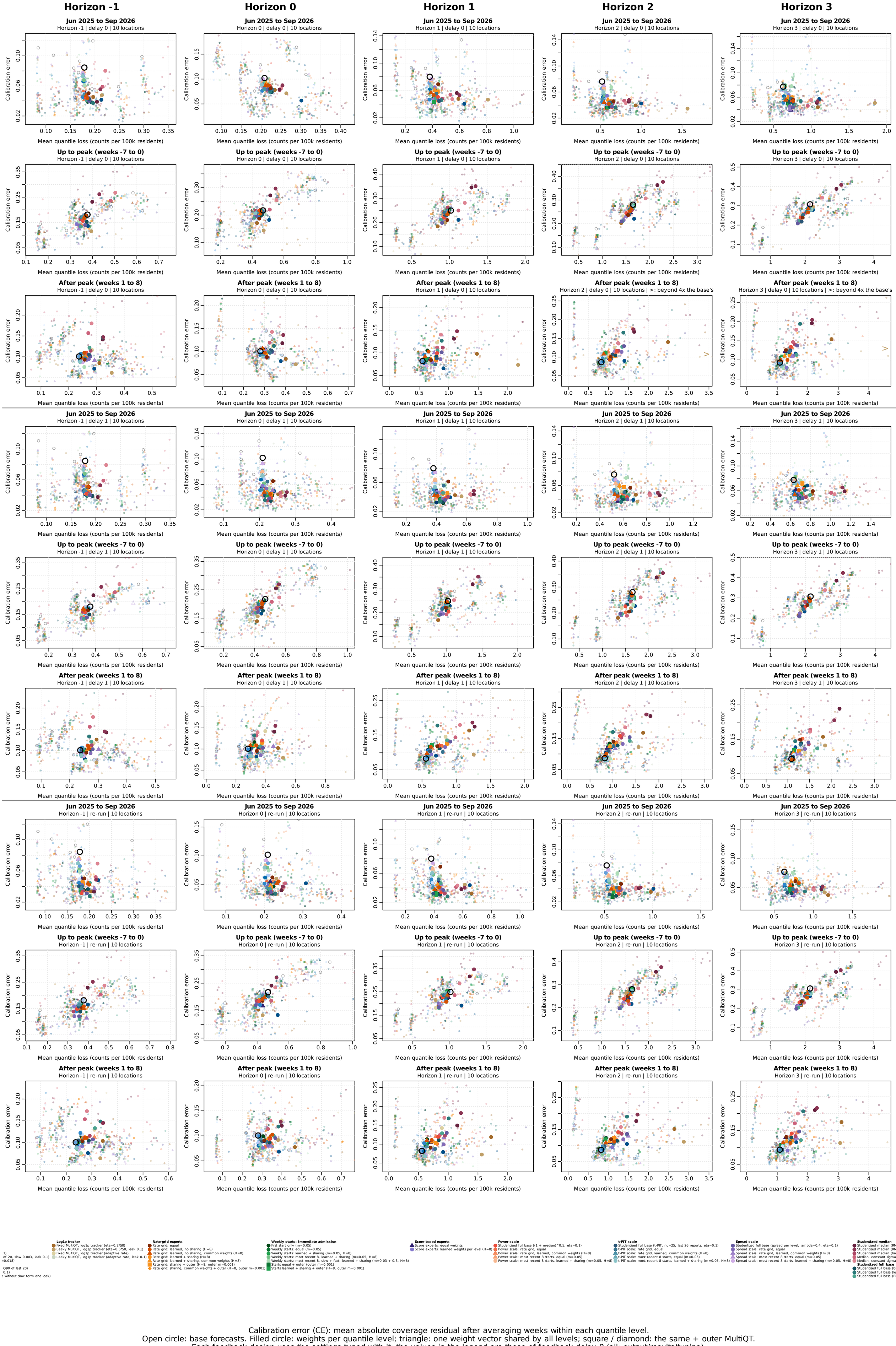
Rows: feedback design (delay d: learn once, d weeks after the first report, from the report current then; re-run: re-run every week on the reports known then) and window; columns: horizon.



Calibration error (CE): mean absolute coverage residual after averaging weeks within each quantile level.
Open circle: base forecasts. Filled circle: weights per quantile level; triangle: one weight vector shared by all levels; square / diamond: the same + outer MultiQT.
Each feedback design uses the settings tuned with it; the values in the legend are those of feedback delay 0 (all: output/results/tuning).

flu | the 10 largest locations without Florida | evaluation: Jun 2025 to Sep 2026 | raw-count updates, loss in counts per 100k residents

Small: single locations; large: means over locations; axes: 1st to 99th percentile of both.



Calibration error (CE): mean absolute coverage residual after averaging weeks within each quantile level.
Open circle: base forecasts. Filled circle: weights per quantile level; triangle: one weight vector shared by all levels; square / diamond: the same + outer MultiQT.
Each feedback design uses the settings tuned with it; the values in the legend are those of feedback delay 0 (all: output/results/tuning).

flu | the 10 largest locations without Florida | evaluation: Jun 2025 to Sep 2026 | raw-count updates, loss in counts per 100k residents

Means over locations only; axes fitted to the means.

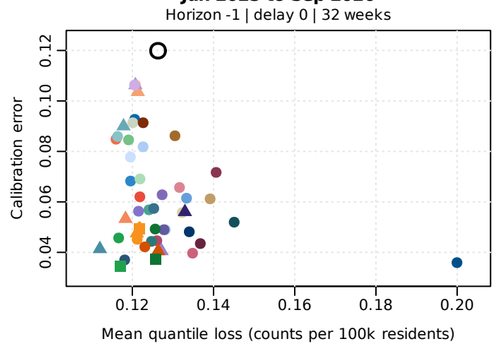
Calibration error (CE): mean absolute coverage residual after averaging weeks within each quantile level.
Open circle: base forecasts. Filled circle: weights per quantile level; triangle: one weight vector shared by all levels; square / diamond: the same + outer MultiQT.
Each feedback design uses the settings tuned with it; the values in the legend are those of feedback delay 0 (all: output/results/tuning).

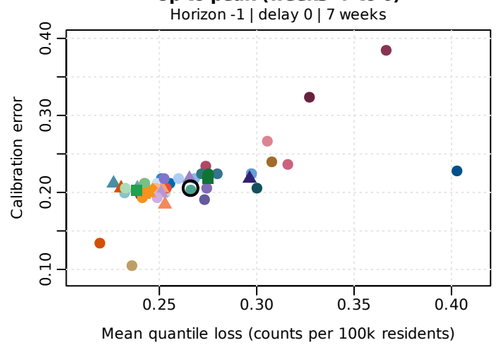
flu | United States (US) | evaluation: Jun 2025 to Sep 2026 | raw-count updates, loss in counts per 100k residents

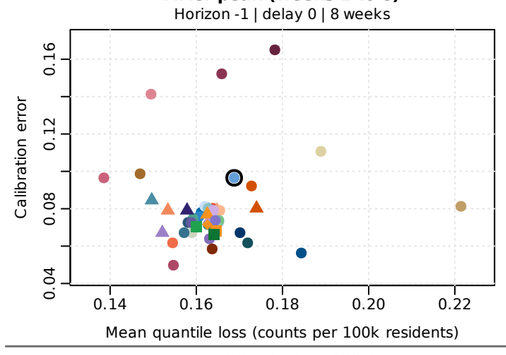
Rows: feedback design (delay d: learn once, d weeks after the first report, from the report current then; re-run: re-run every week on the reports known then) and window; columns: horizon.

Feedback delay 0

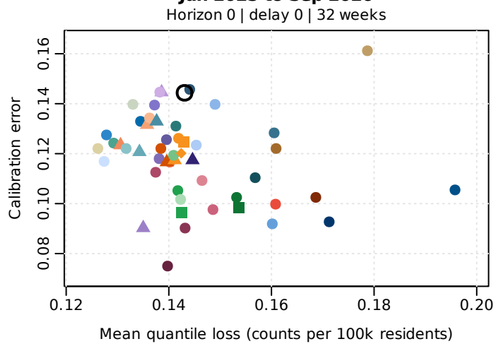
Horizon -1

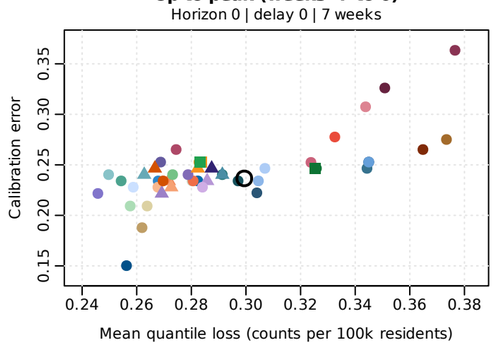


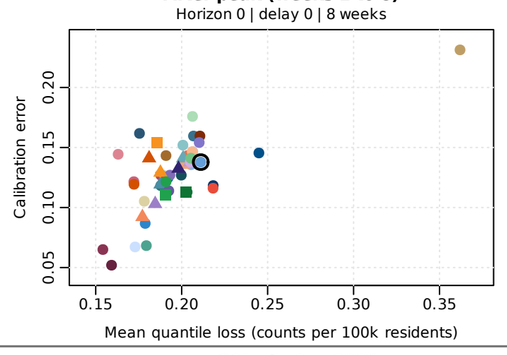




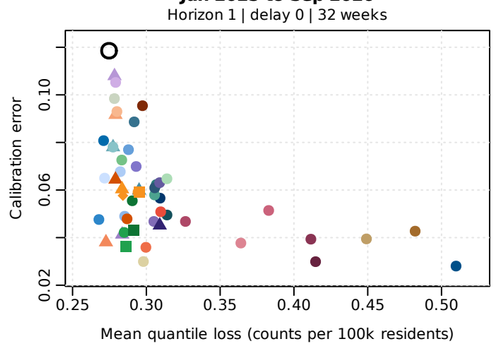
Horizon 0

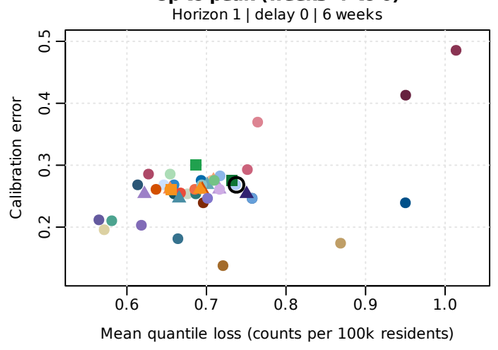


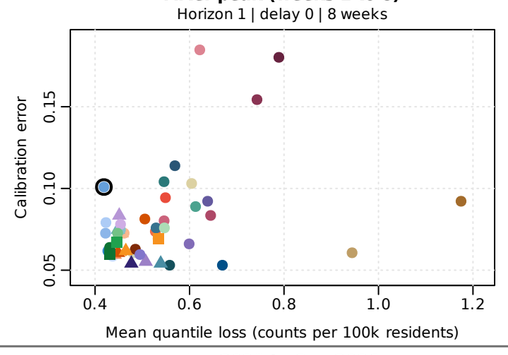




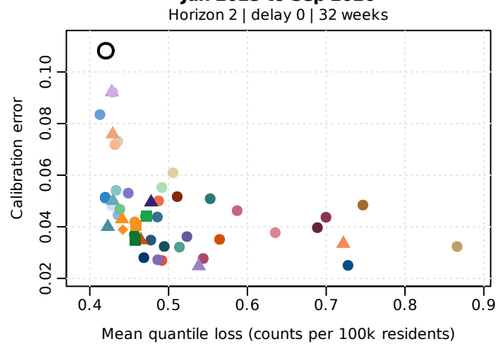
Horizon 1

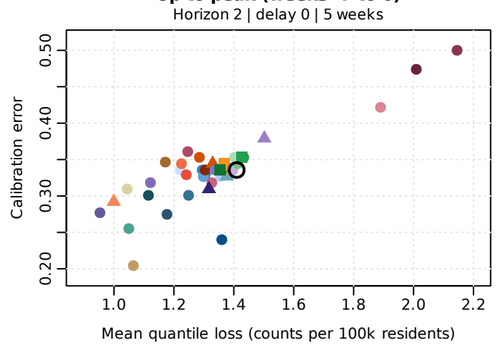


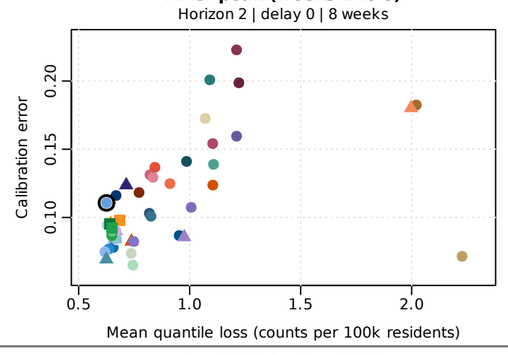




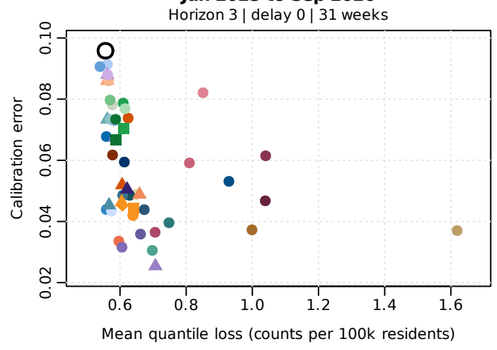
Horizon 2

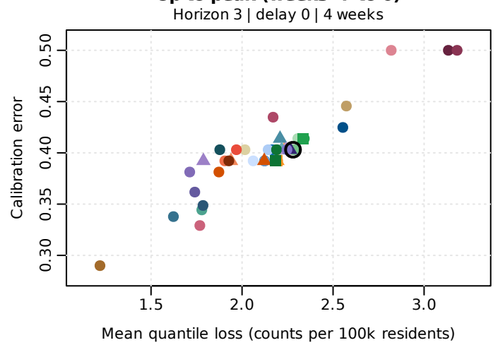


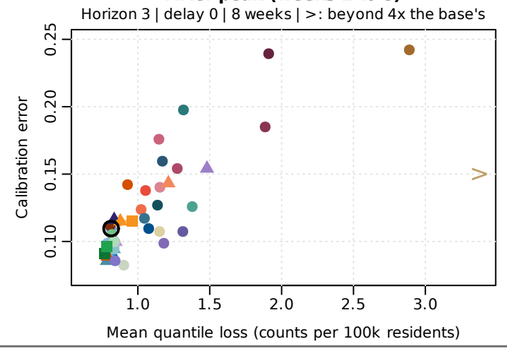




Horizon 3

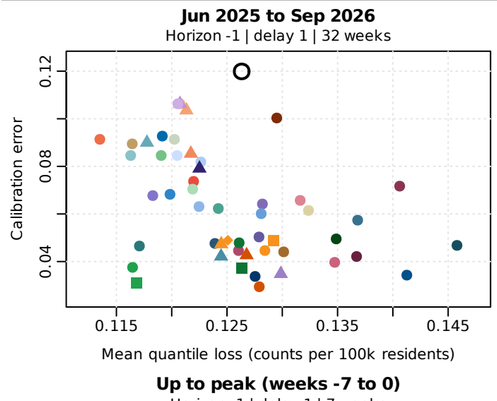


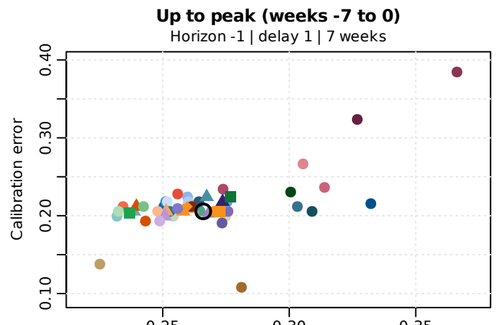


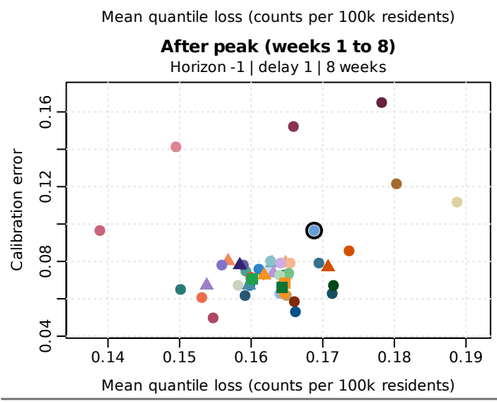


Feedback delay 1

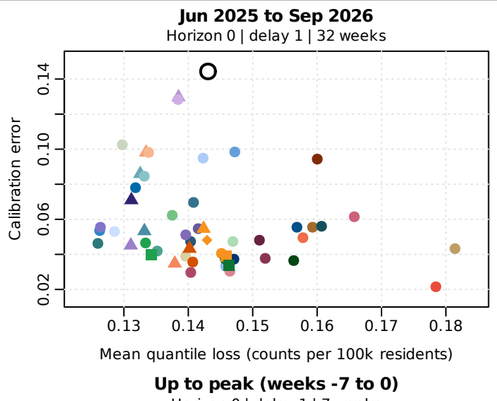
Horizon -1

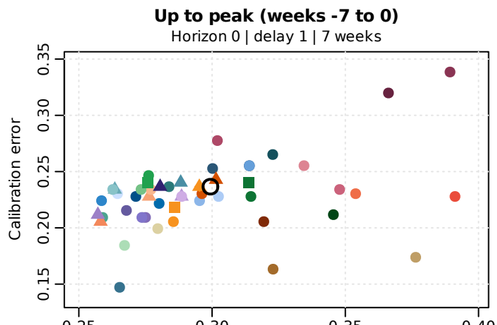


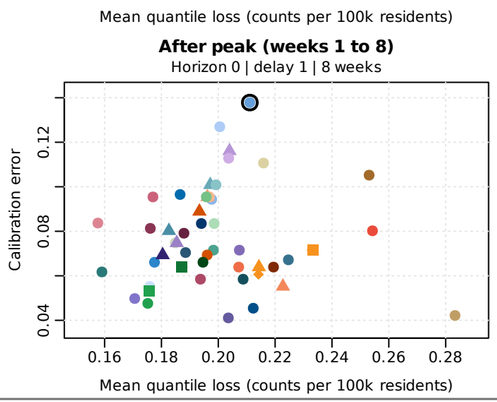




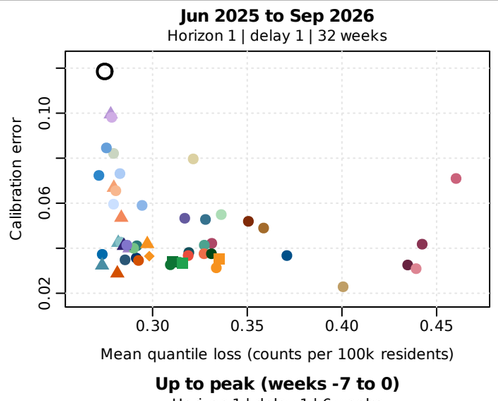
Horizon 0

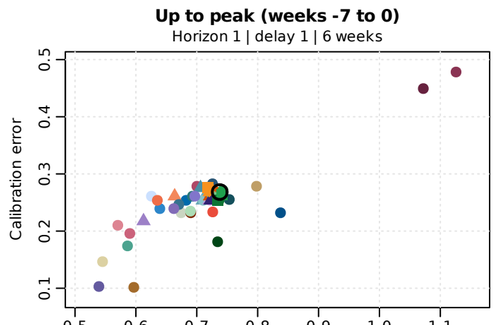


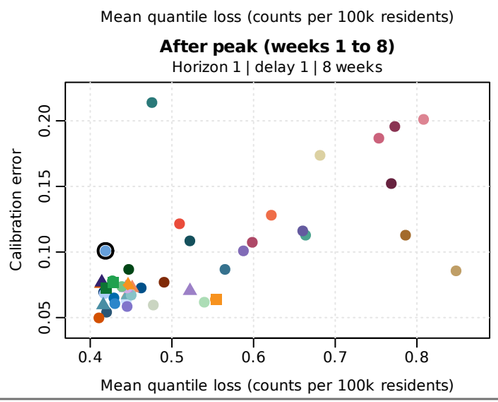




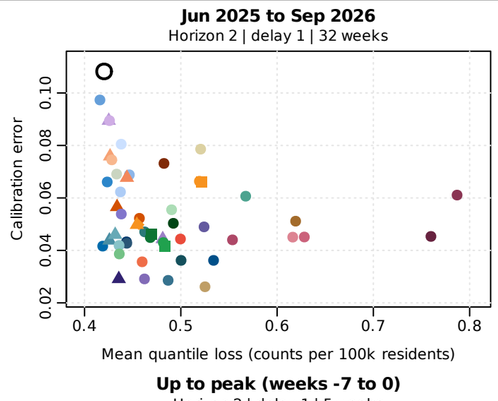
Horizon 1

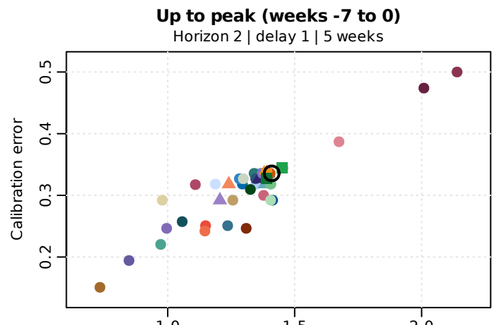


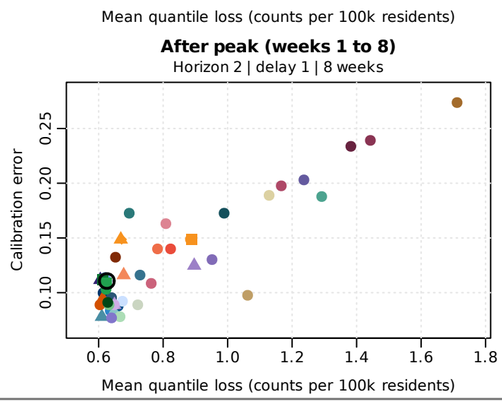




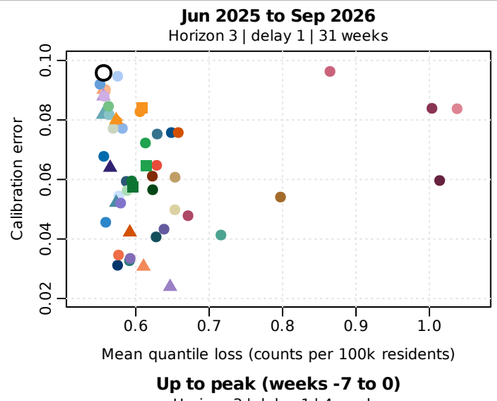
Horizon 2

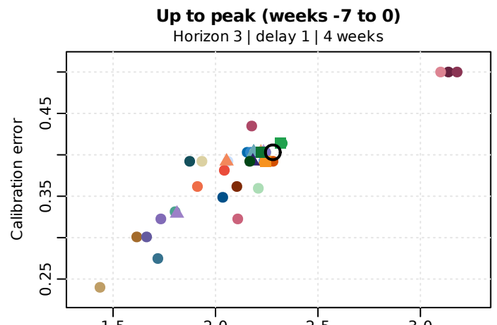


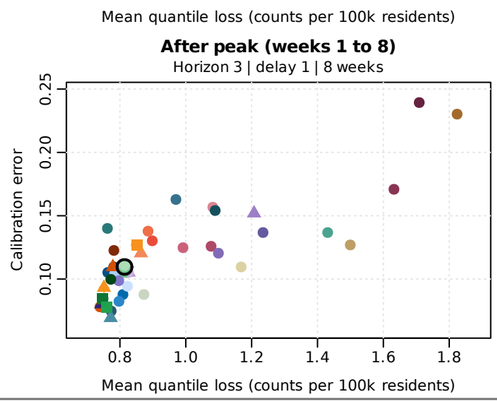




Horizon 3

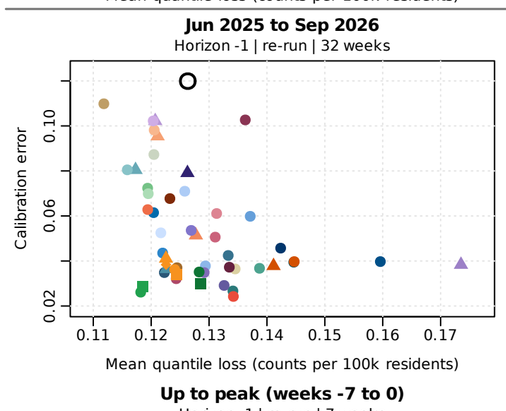


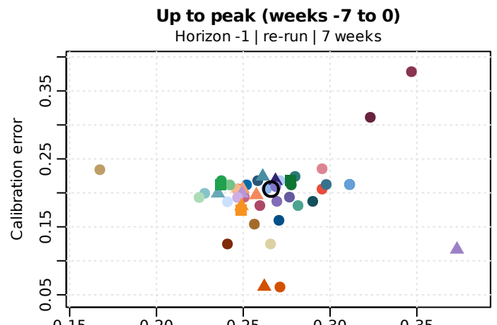


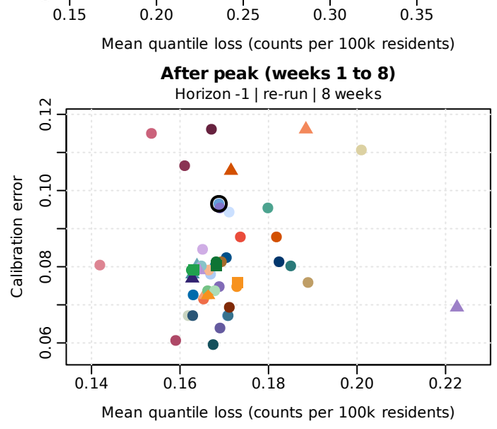


Feedback re-run

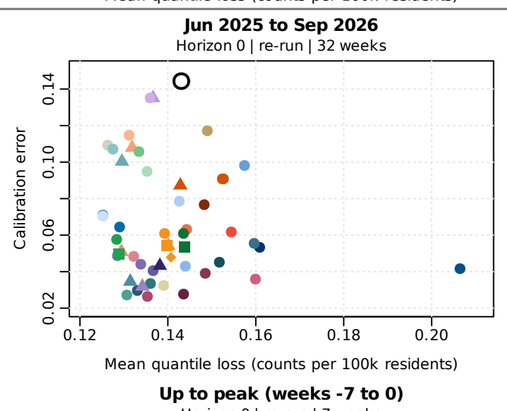
Horizon -1

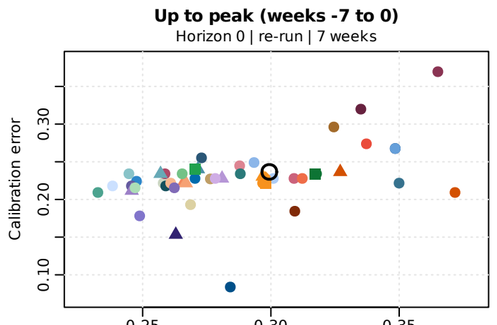


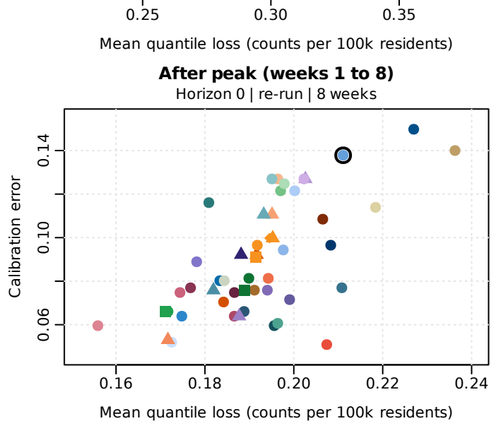




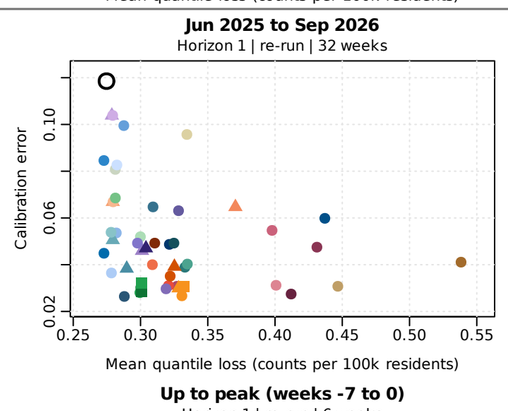
Horizon 0

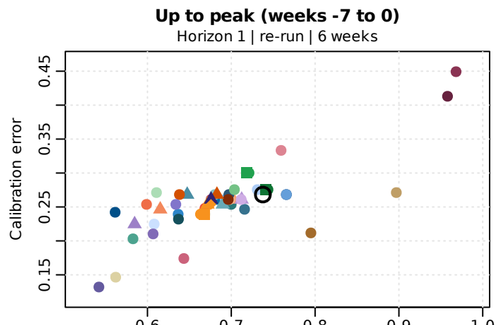


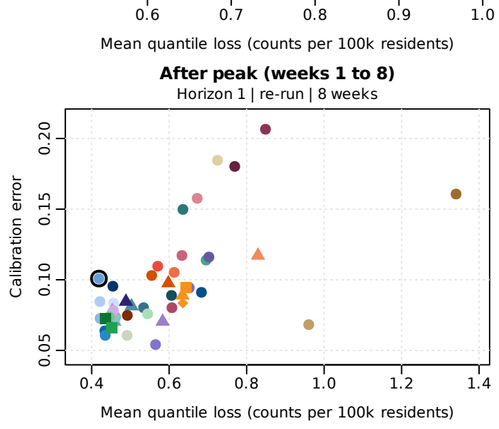




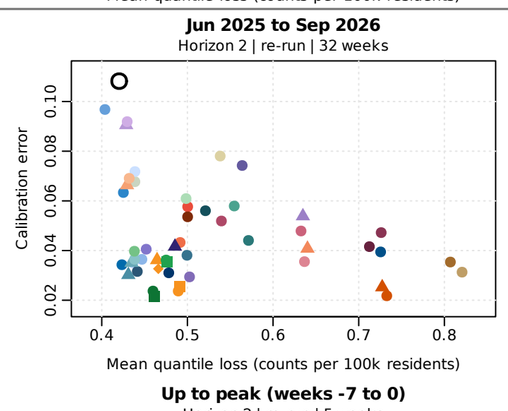
Horizon 1

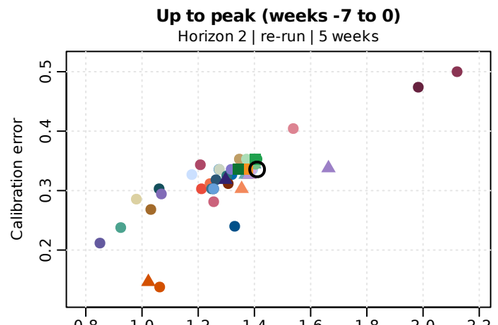


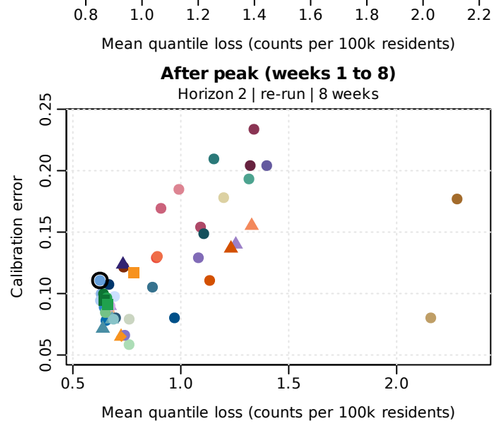




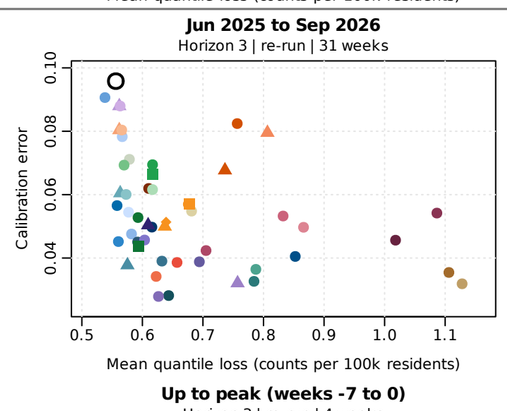
Horizon 2

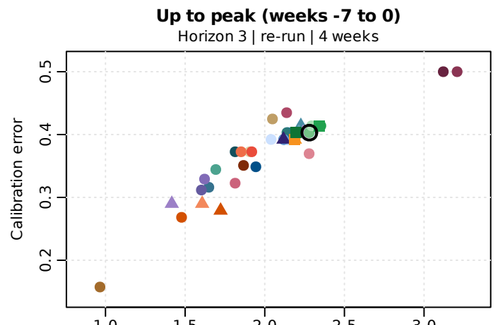


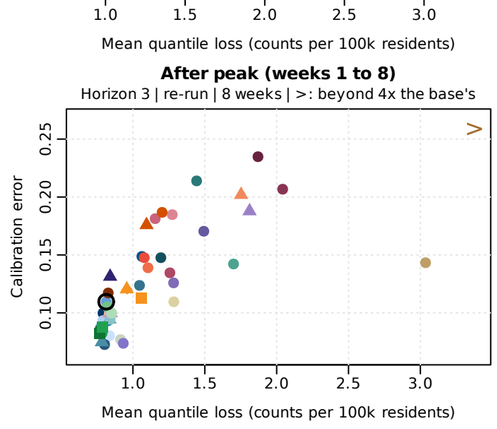


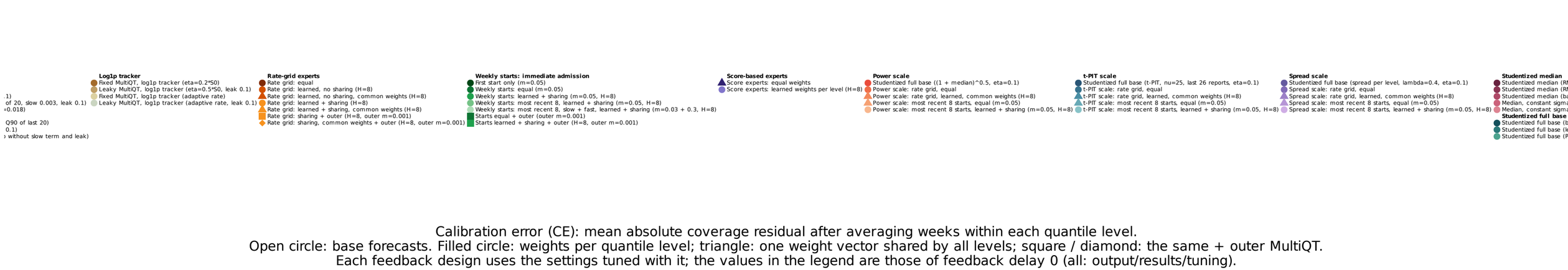


Horizon 3









Calibration error (CE): mean absolute coverage residual after averaging weeks within each quantile level.
Open circle: base forecasts. Filled circle: weights per quantile level; triangle: one weight vector shared by all levels; square / diamond: the same + outer MultiQT.
Each feedback design uses the settings tuned with it; the values in the legend are those of feedback delay 0 (all: output/results/tuning).

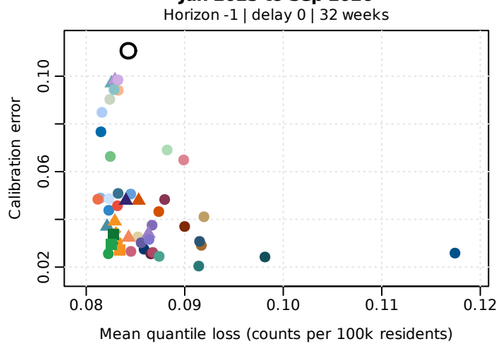
flu | California (06) | evaluation: Jun 2025 to Sep 2026 | raw-count updates, loss in counts per 100k residents

Rows: feedback design (delay d: learn once, d weeks after the first report, from the report current then; re-run: re-run every week on the reports known then) and window; columns: horizon.

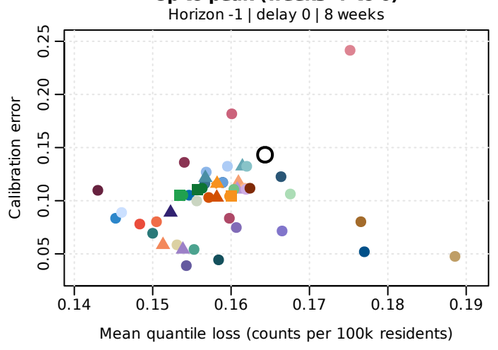
Horizon -1

Feedback delay 0

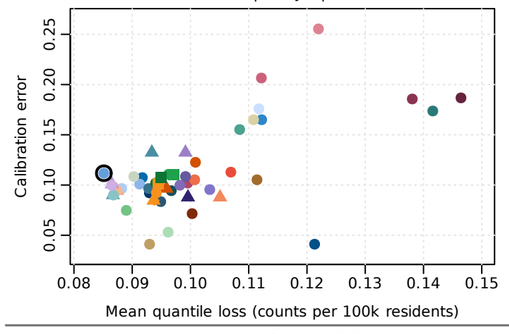
Jun 2025 to Sep 2026
Horizon -1 | delay 0 | 32 weeks



Up to peak (weeks -7 to 0)
Horizon -1 | delay 0 | 8 weeks

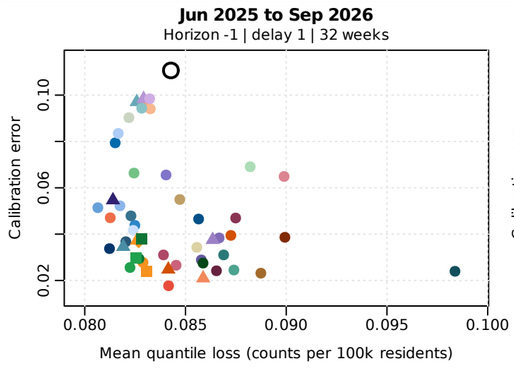


After peak (weeks 1 to 8)
Horizon -1 | delay 0 | 8 weeks



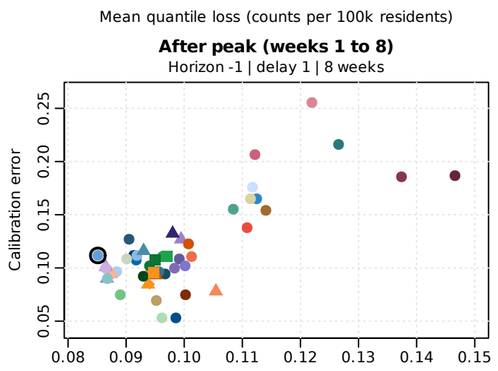
Feedback delay 1

Jun 2025 to Sep 2026
Horizon -1 | delay 1 | 32 weeks



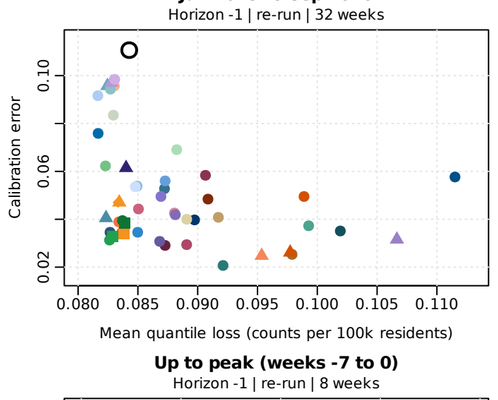
Up to peak (weeks -7 to 0)
Horizon -1 | delay 1 | 8 weeks

After peak (weeks 1 to 8)
Horizon -1 | delay 1 | 8 weeks



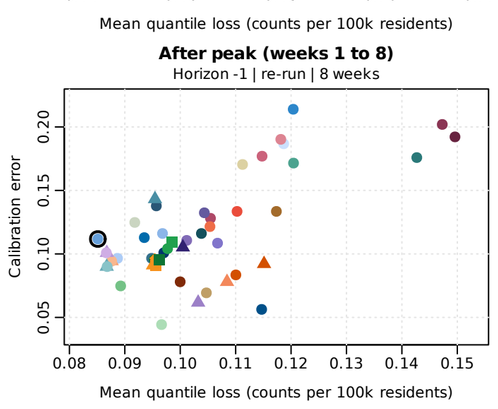
Feedback re-run

Jun 2025 to Sep 2026
Horizon -1 | re-run | 32 weeks



Up to peak (weeks -7 to 0)
Horizon -1 | re-run | 8 weeks

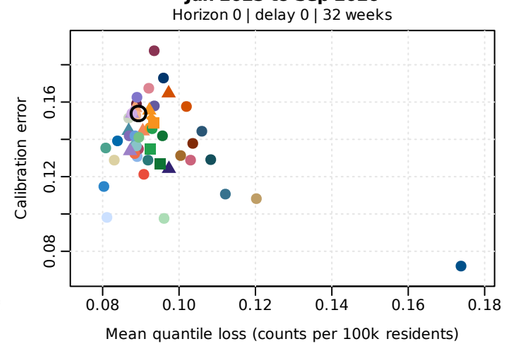
After peak (weeks 1 to 8)
Horizon -1 | re-run | 8 weeks



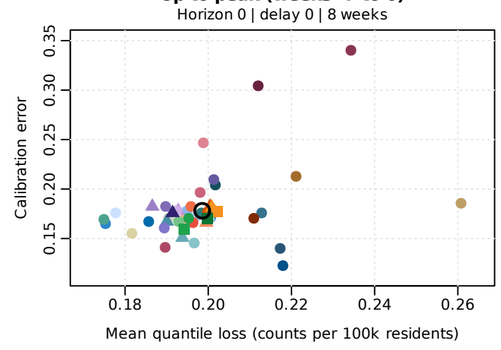
Horizon 0

Feedback delay 0

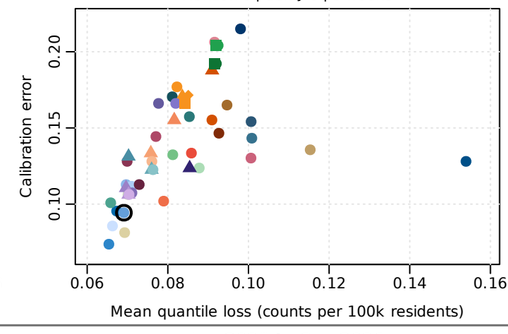
Jun 2025 to Sep 2026
Horizon 0 | delay 0 | 32 weeks



Up to peak (weeks -7 to 0)
Horizon 0 | delay 0 | 8 weeks

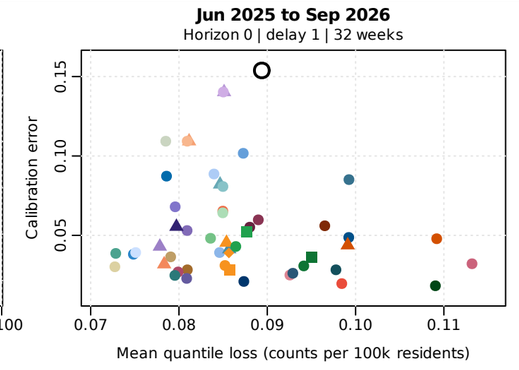


After peak (weeks 1 to 8)
Horizon 0 | delay 0 | 8 weeks



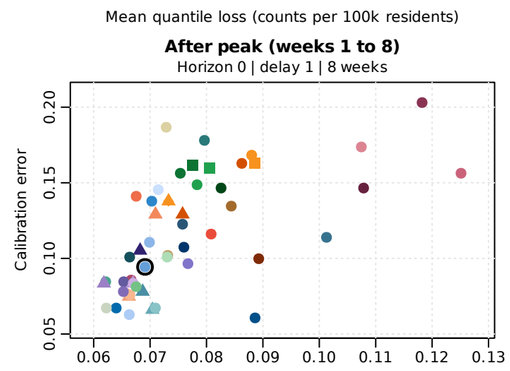
Feedback delay 1

Jun 2025 to Sep 2026
Horizon 0 | delay 1 | 32 weeks



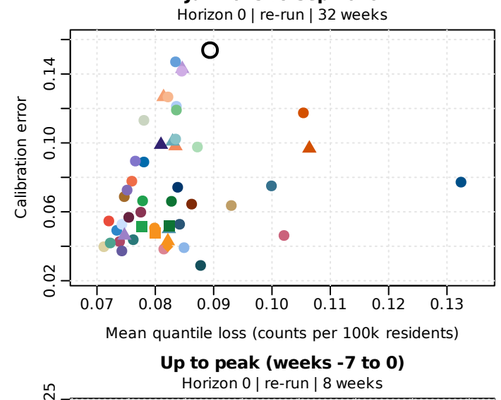
Up to peak (weeks -7 to 0)
Horizon 0 | delay 1 | 8 weeks

After peak (weeks 1 to 8)
Horizon 0 | delay 1 | 8 weeks



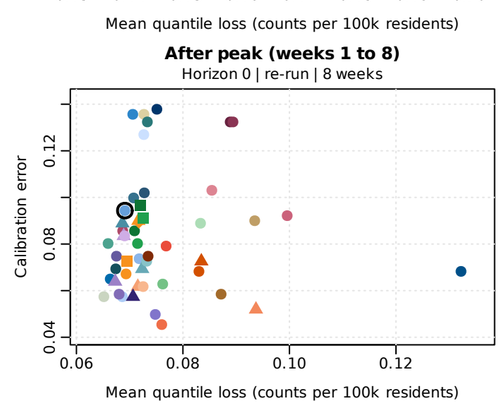
Feedback re-run

Jun 2025 to Sep 2026
Horizon 0 | re-run | 32 weeks



Up to peak (weeks -7 to 0)
Horizon 0 | re-run | 8 weeks

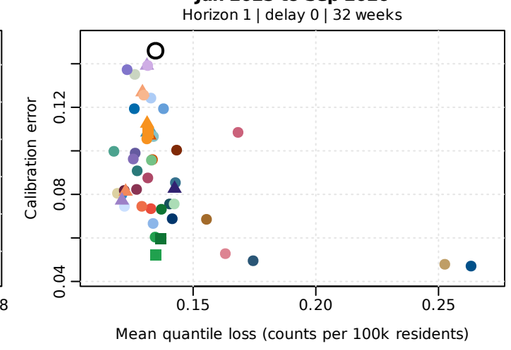
After peak (weeks 1 to 8)
Horizon 0 | re-run | 8 weeks



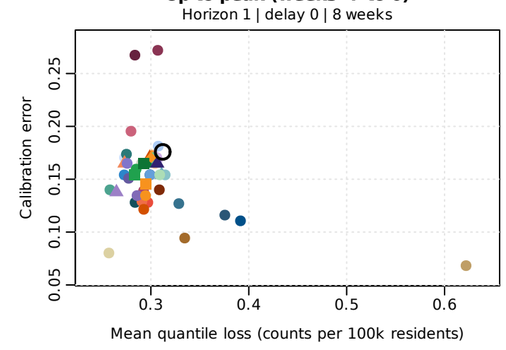
Horizon 1

Feedback delay 0

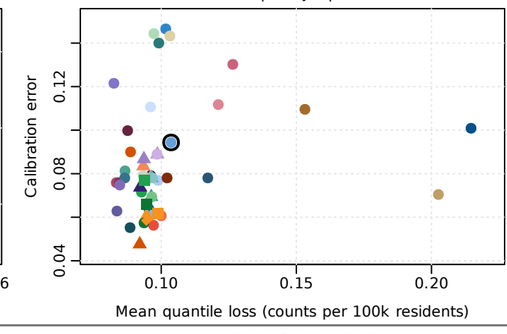
Jun 2025 to Sep 2026
Horizon 1 | delay 0 | 32 weeks



Up to peak (weeks -7 to 0)
Horizon 1 | delay 0 | 8 weeks

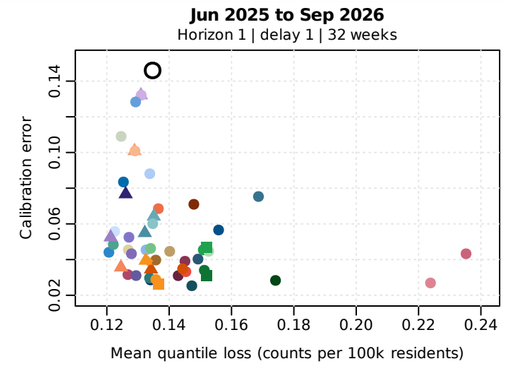


After peak (weeks 1 to 8)
Horizon 1 | delay 0 | 8 weeks



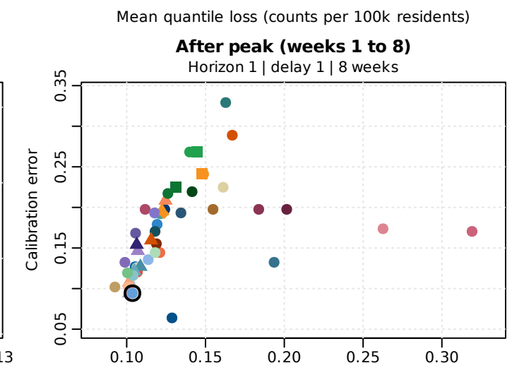
Feedback delay 1

Jun 2025 to Sep 2026
Horizon 1 | delay 1 | 32 weeks



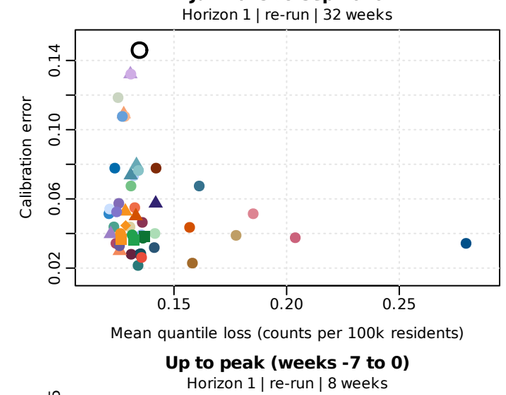
Up to peak (weeks -7 to 0)
Horizon 1 | delay 1 | 8 weeks

After peak (weeks 1 to 8)
Horizon 1 | delay 1 | 8 weeks



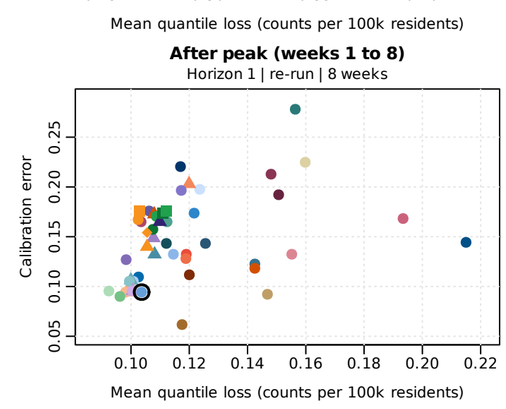
Feedback re-run

Jun 2025 to Sep 2026
Horizon 1 | re-run | 32 weeks



Up to peak (weeks -7 to 0)
Horizon 1 | re-run | 8 weeks

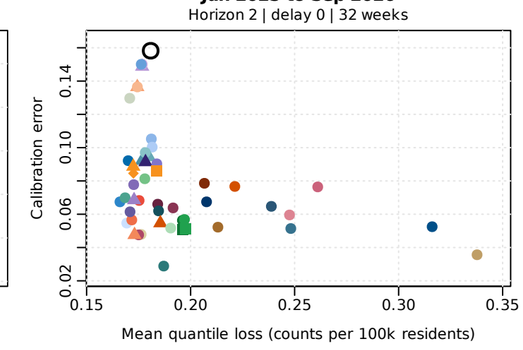
After peak (weeks 1 to 8)
Horizon 1 | re-run | 8 weeks



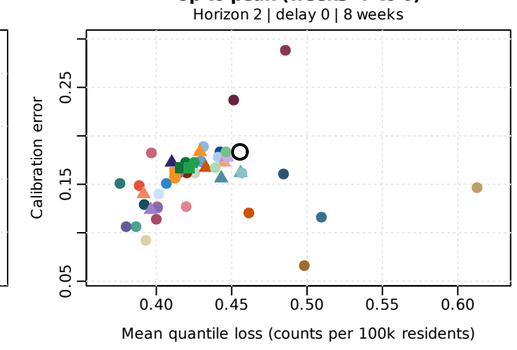
Horizon 2

Feedback delay 0

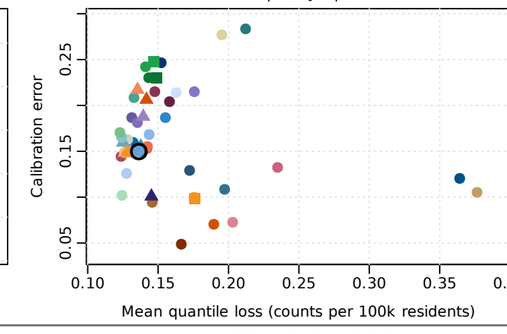
Jun 2025 to Sep 2026
Horizon 2 | delay 0 | 32 weeks



Up to peak (weeks -7 to 0)
Horizon 2 | delay 0 | 8 weeks

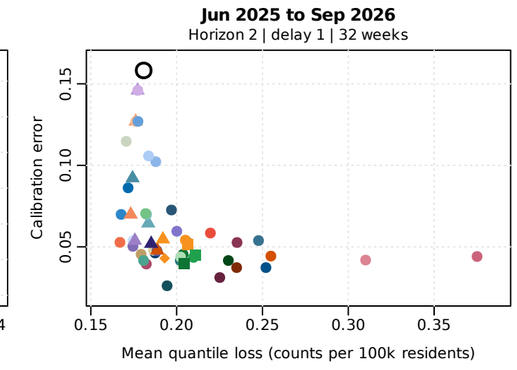


After peak (weeks 1 to 8)
Horizon 2 | delay 0 | 8 weeks



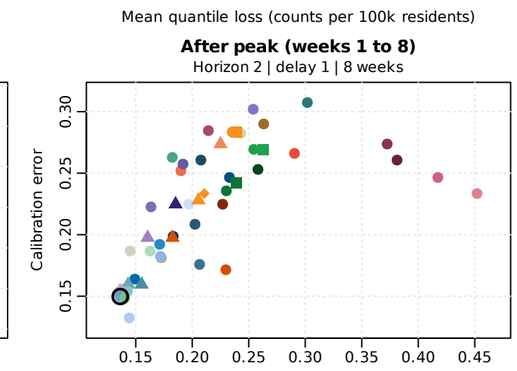
Feedback delay 1

Jun 2025 to Sep 2026
Horizon 2 | delay 1 | 32 weeks



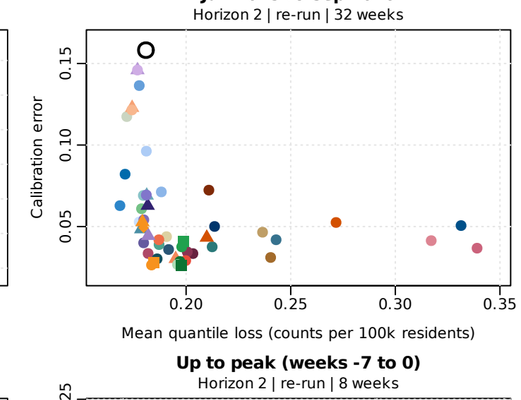
Up to peak (weeks -7 to 0)
Horizon 2 | delay 1 | 8 weeks

After peak (weeks 1 to 8)
Horizon 2 | delay 1 | 8 weeks



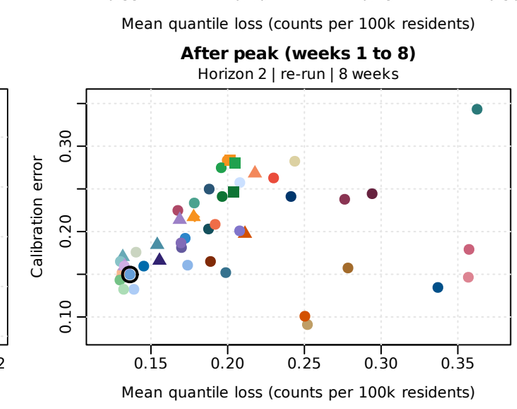
Feedback re-run

Jun 2025 to Sep 2026
Horizon 2 | re-run | 32 weeks



Up to peak (weeks -7 to 0)
Horizon 2 | re-run | 8 weeks

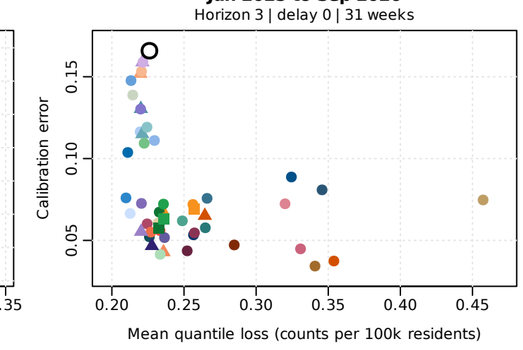
After peak (weeks 1 to 8)
Horizon 2 | re-run | 8 weeks



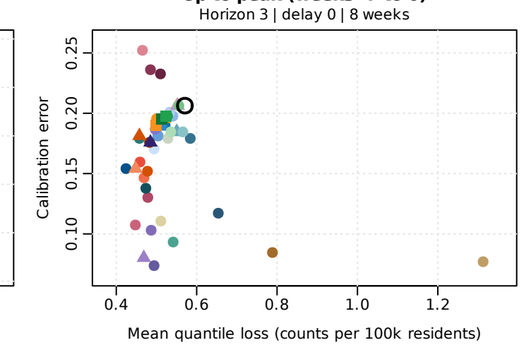
Horizon 3

Feedback delay 0

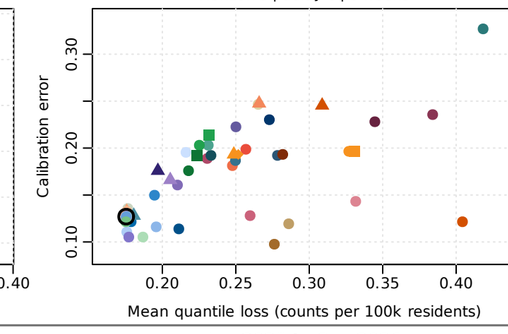
Jun 2025 to Sep 2026
Horizon 3 | delay 0 | 31 weeks



Up to peak (weeks -7 to 0)
Horizon 3 | delay 0 | 8 weeks

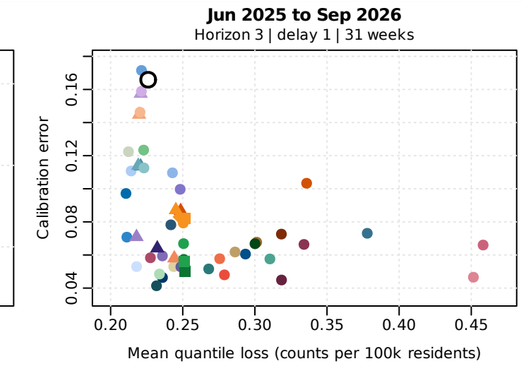


After peak (weeks 1 to 8)
Horizon 3 | delay 0 | 8 weeks



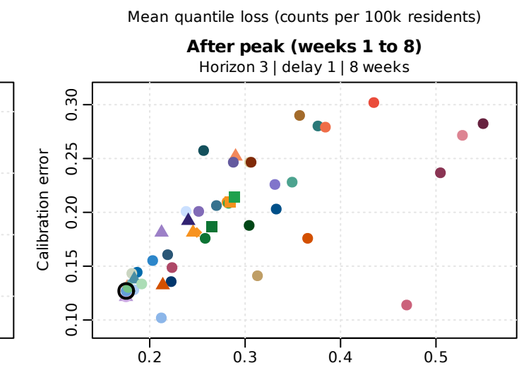
Feedback delay 1

Jun 2025 to Sep 2026
Horizon 3 | delay 1 | 31 weeks



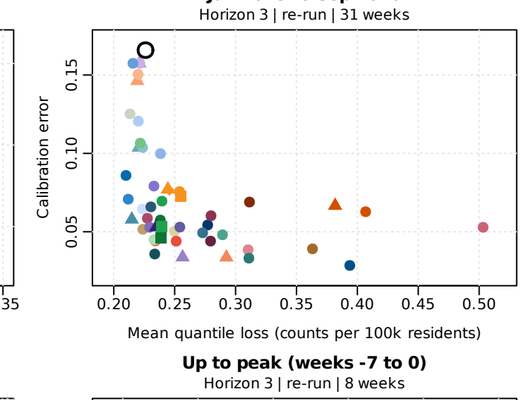
Up to peak (weeks -7 to 0)
Horizon 3 | delay 1 | 8 weeks

After peak (weeks 1 to 8)
Horizon 3 | delay 1 | 8 weeks



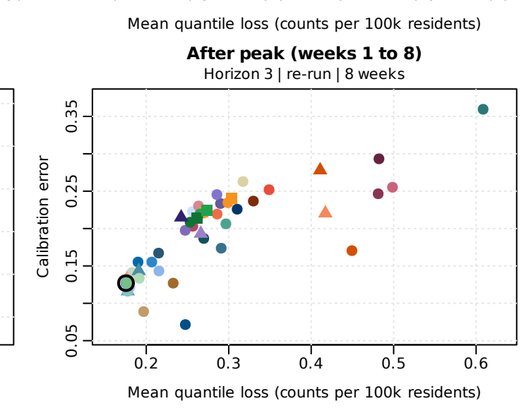
Feedback re-run

Jun 2025 to Sep 2026
Horizon 3 | re-run | 31 weeks



Up to peak (weeks -7 to 0)
Horizon 3 | re-run | 8 weeks

After peak (weeks 1 to 8)
Horizon 3 | re-run | 8 weeks





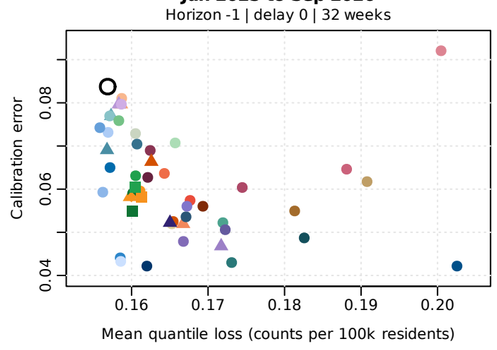
Calibration error (CE): mean absolute coverage residual after averaging weeks within each quantile level.
Open circle: base forecasts. Filled circle: weights per quantile level; triangle: one weight vector shared by all levels; square / diamond: the same + outer MultiQT.
Each feedback design uses the settings tuned with it; the values in the legend are those of feedback delay 0 (all: output/results/tuning).

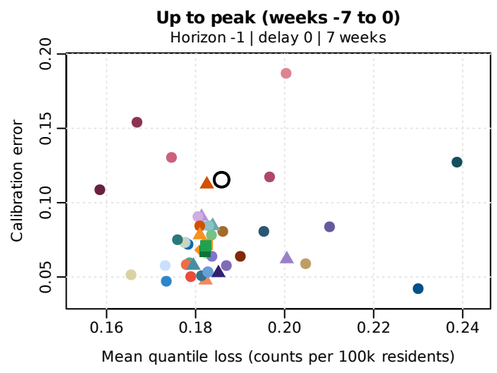
flu | Texas (48) | evaluation: Jun 2025 to Sep 2026 | raw-count updates, loss in counts per 100k residents

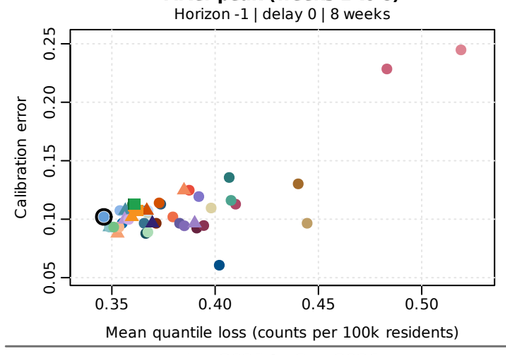
Rows: feedback design (delay d: learn once, d weeks after the first report, from the report current then; re-run: re-run every week on the reports known then) and window; columns: horizon.

Feedback delay 0

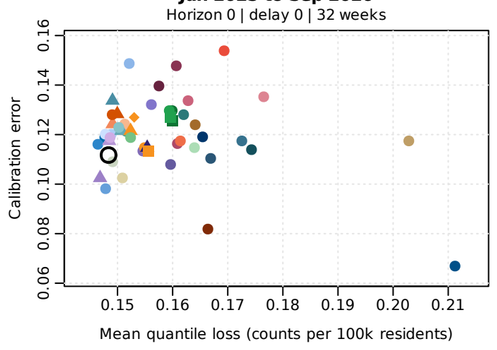
Horizon -1

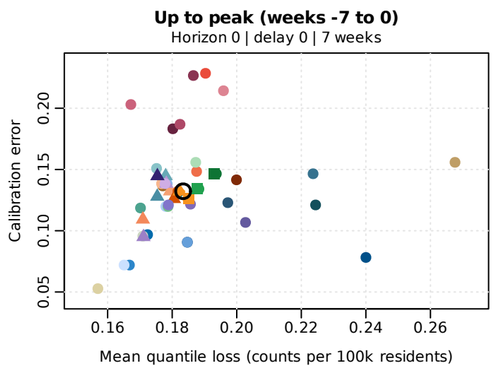


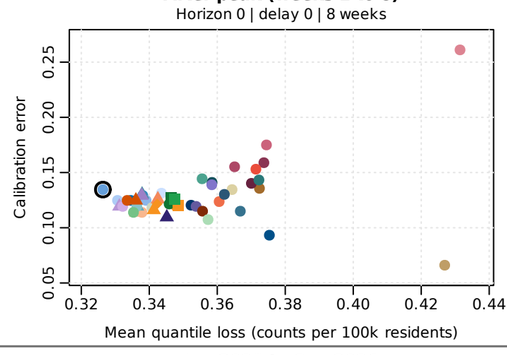




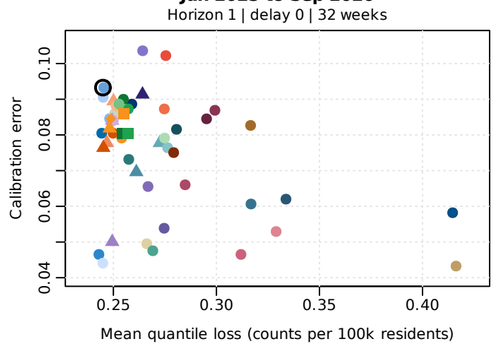
Horizon 0

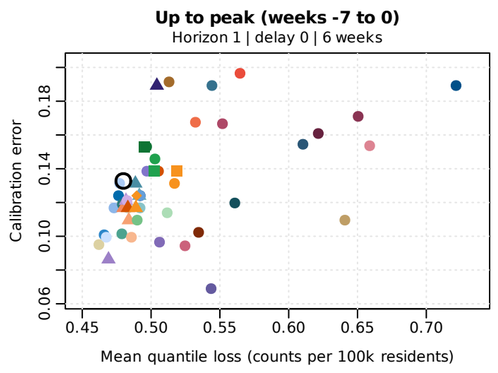


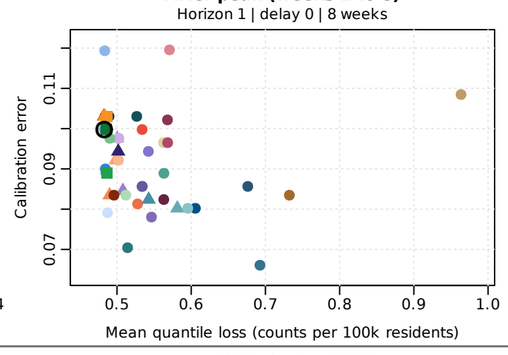




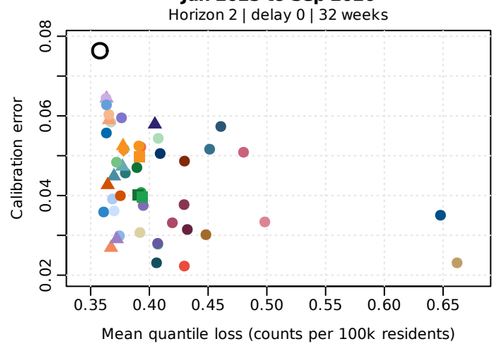
Horizon 1

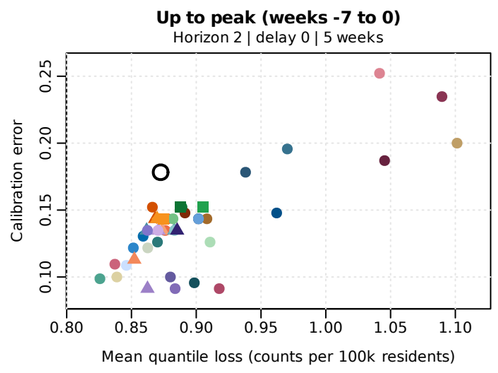


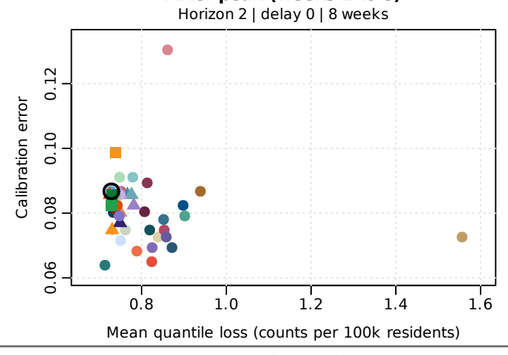




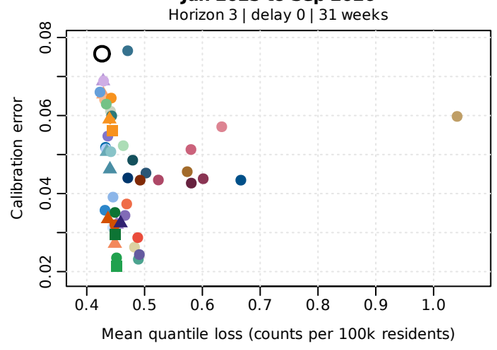
Horizon 2

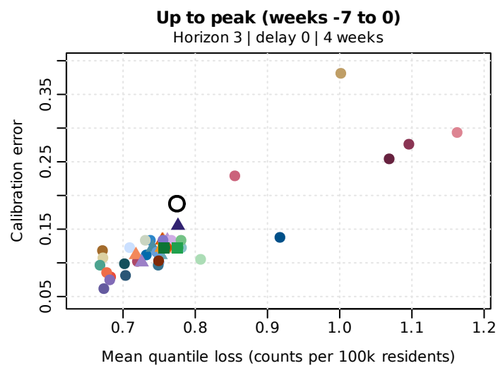


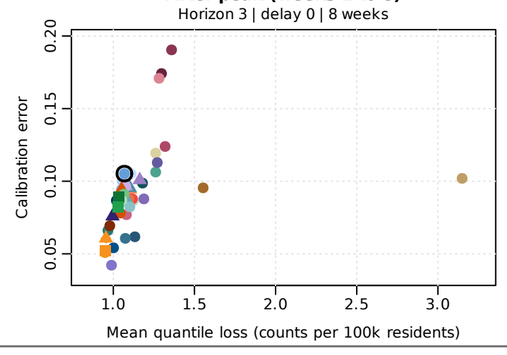




Horizon 3

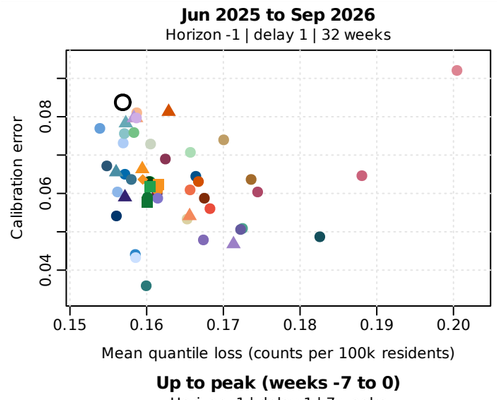


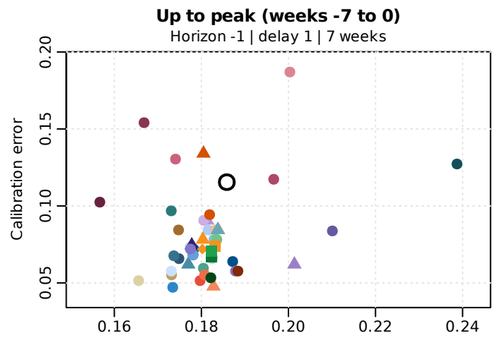


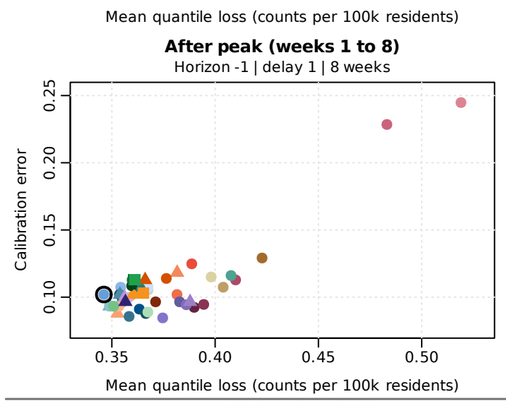


Feedback delay 1

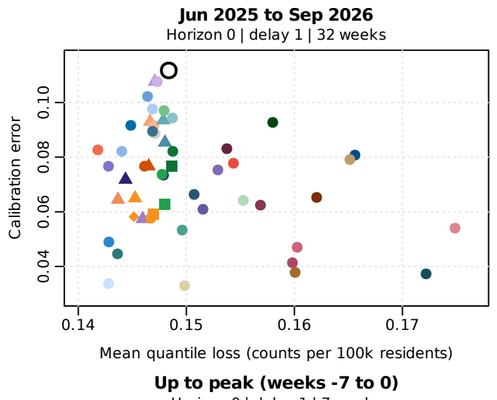
Horizon -1

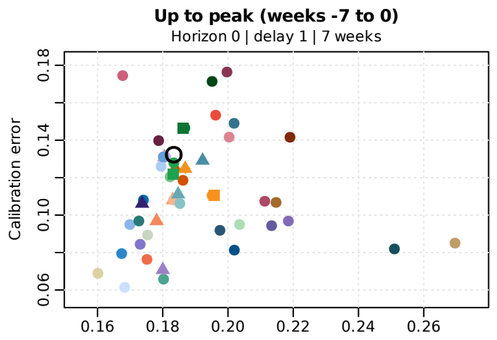


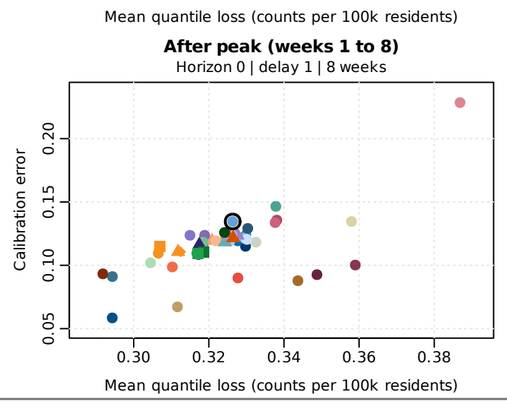




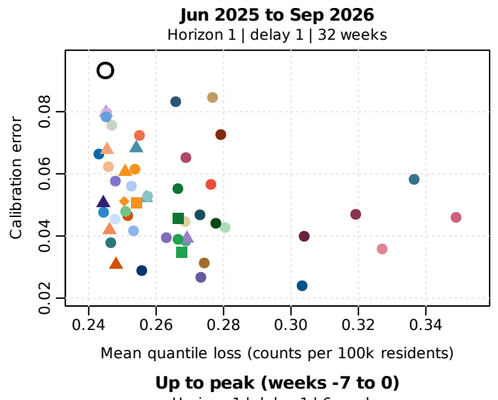
Horizon 0

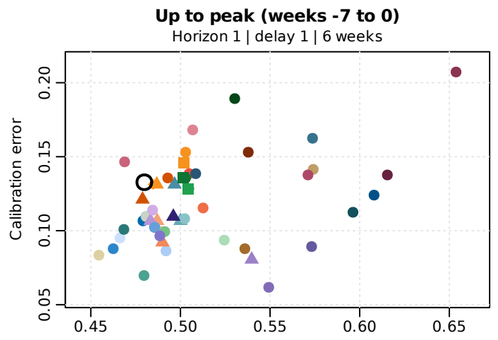


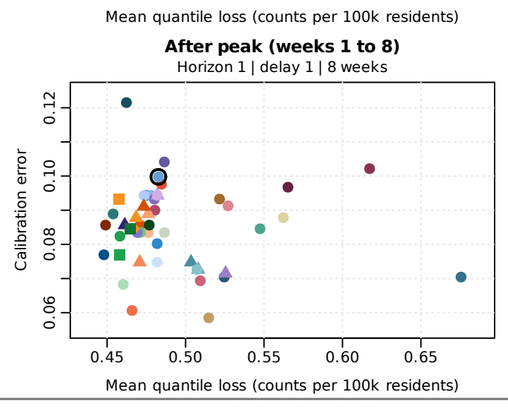




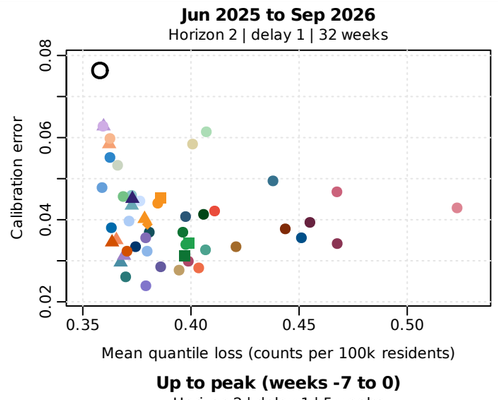
Horizon 1

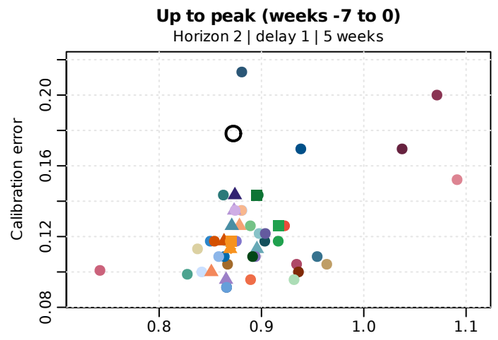


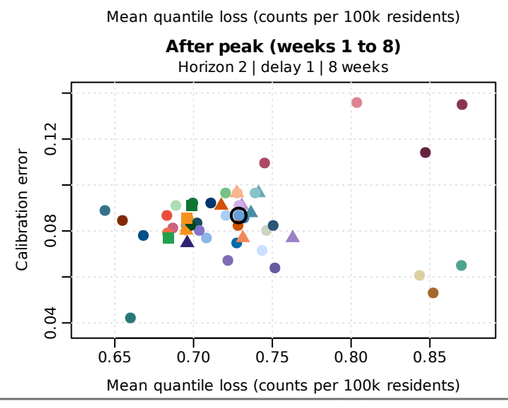




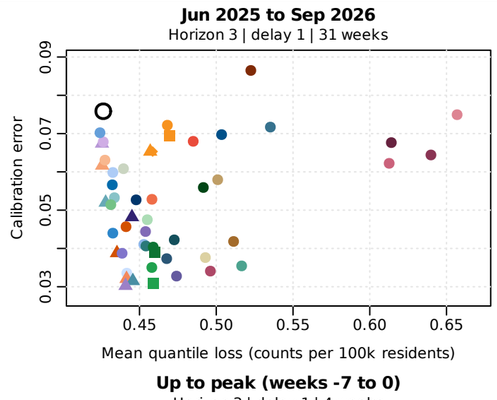
Horizon 2

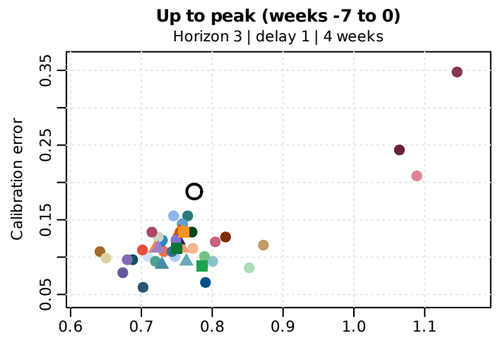


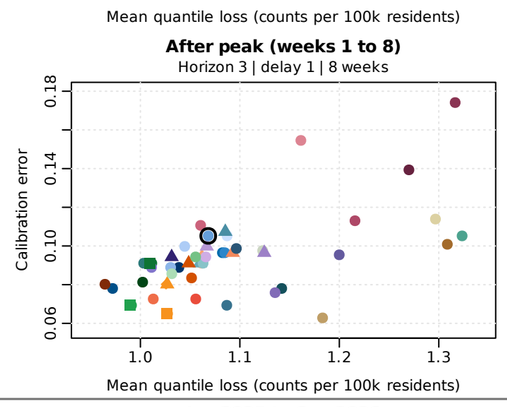




Horizon 3

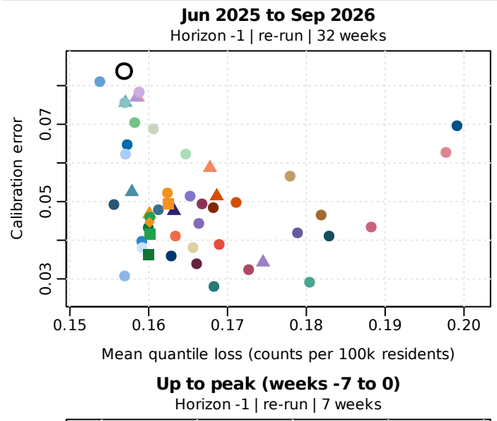


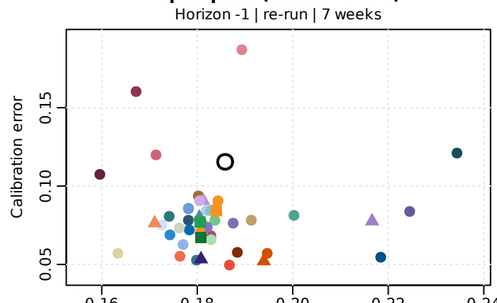


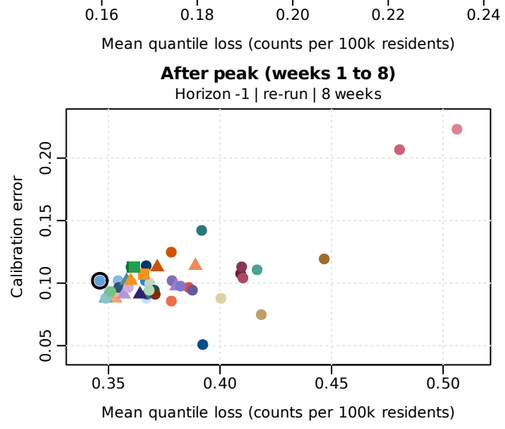


Feedback re-run

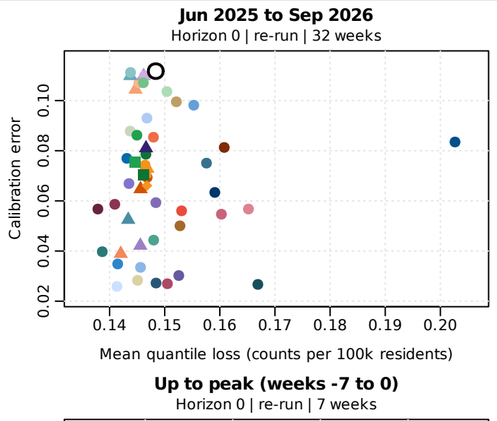
Horizon -1

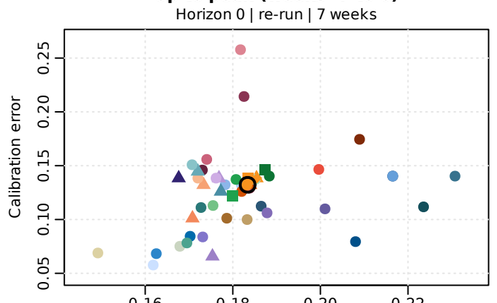


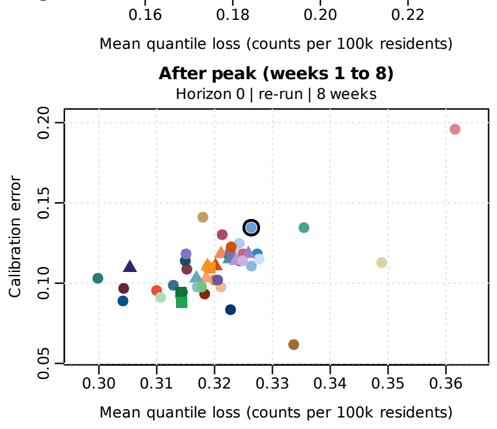




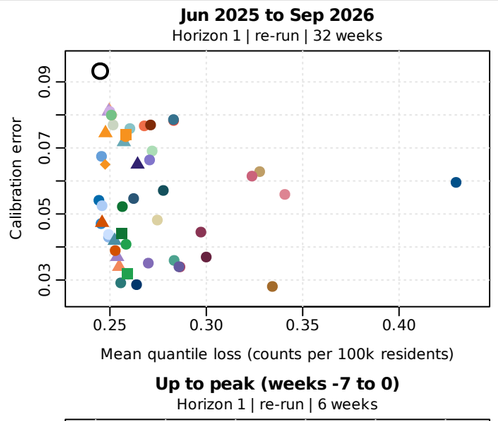
Horizon 0

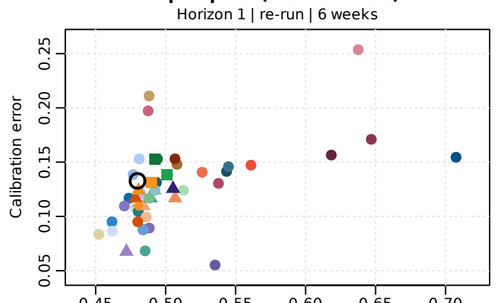


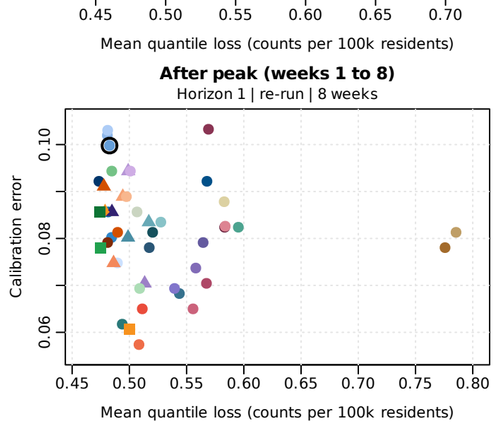




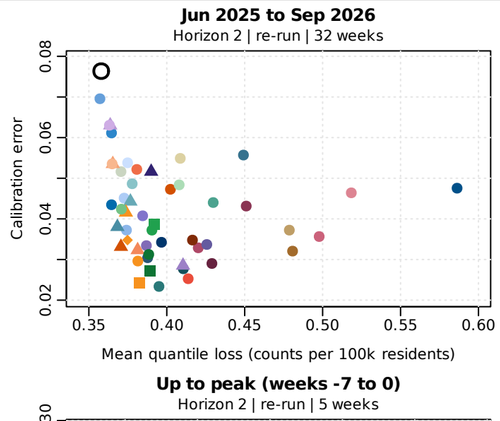
Horizon 1

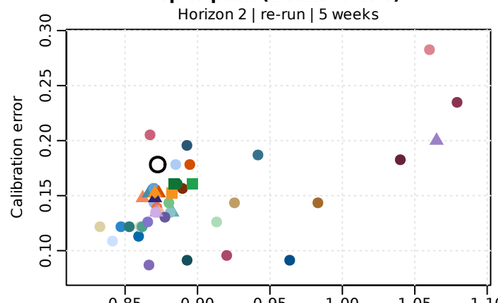


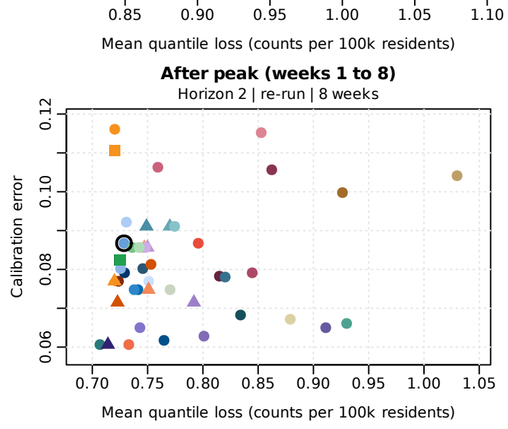




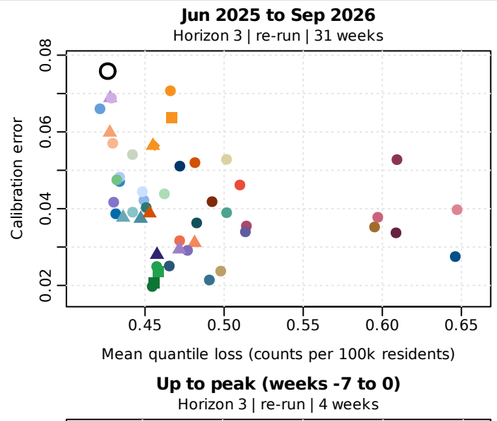
Horizon 2

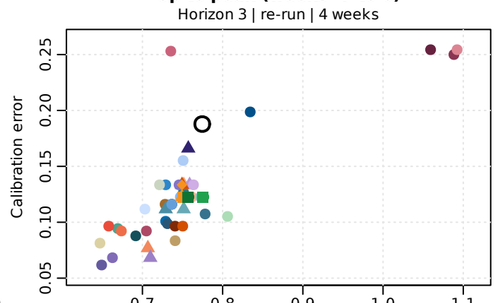


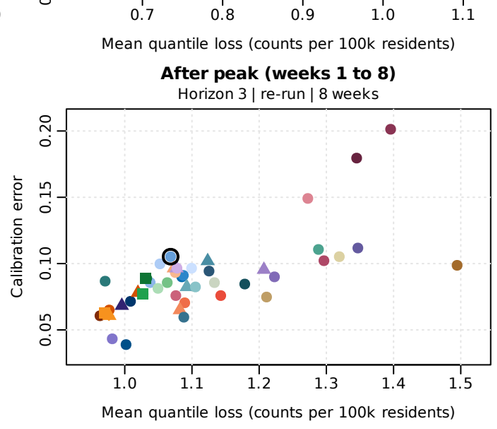


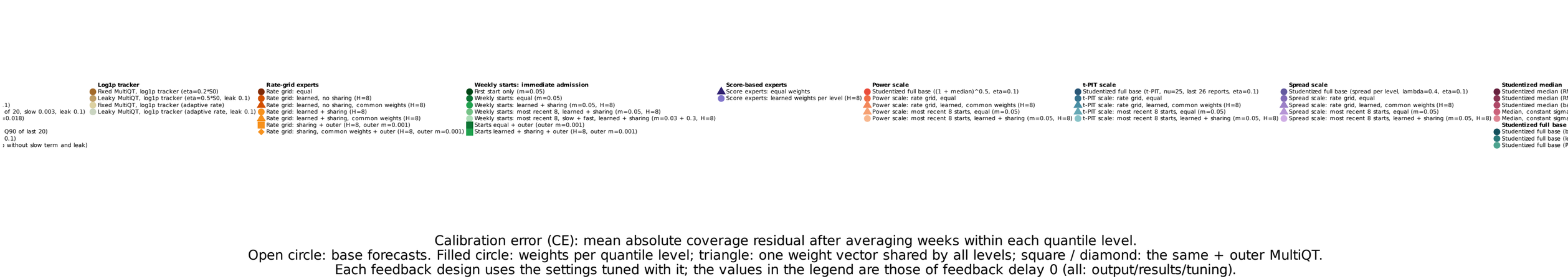


Horizon 3









Calibration error (CE): mean absolute coverage residual after averaging weeks within each quantile level.
Open circle: base forecasts. Filled circle: weights per quantile level; triangle: one weight vector shared by all levels; square / diamond: the same + outer MultiQT.
Each feedback design uses the settings tuned with it; the values in the legend are those of feedback delay 0 (all: output/results/tuning).

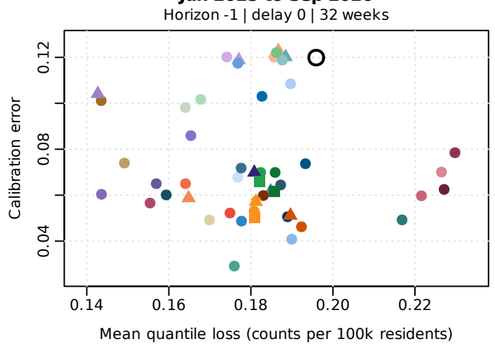
flu | New York (36) | evaluation: Jun 2025 to Sep 2026 | raw-count updates, loss in counts per 100k residents

Rows: feedback design (delay d: learn once, d weeks after the first report, from the report current then; re-run: re-run every week on the reports known then) and window; columns: horizon.

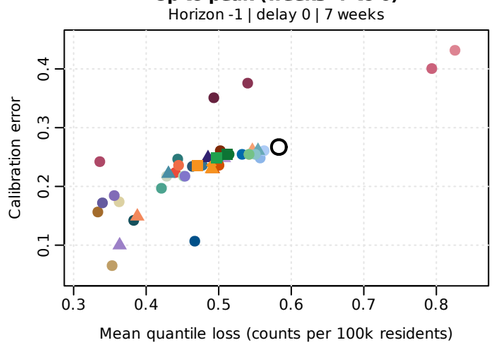
Feedback delay 0

Horizon -1

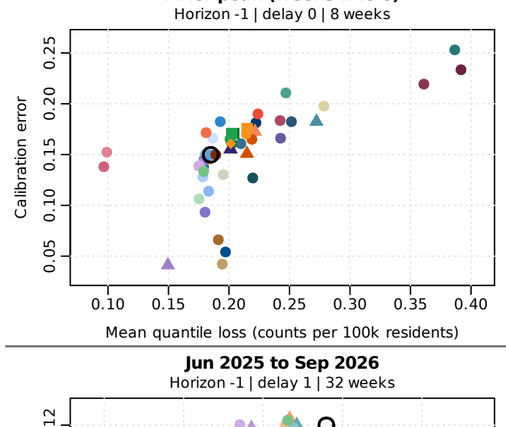
Jun 2025 to Sep 2026
Horizon -1 | delay 0 | 32 weeks



Up to peak (weeks -7 to 0)
Horizon -1 | delay 0 | 7 weeks

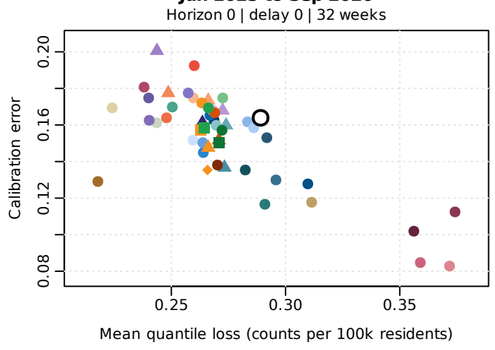


After peak (weeks 1 to 8)
Horizon -1 | delay 0 | 8 weeks

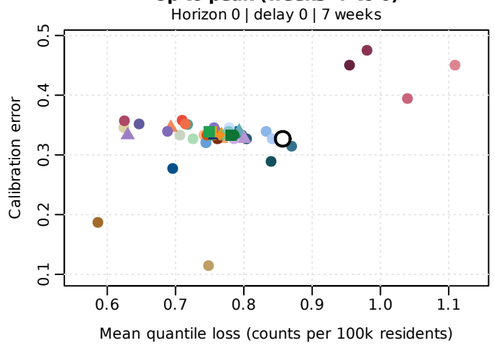


Horizon 0

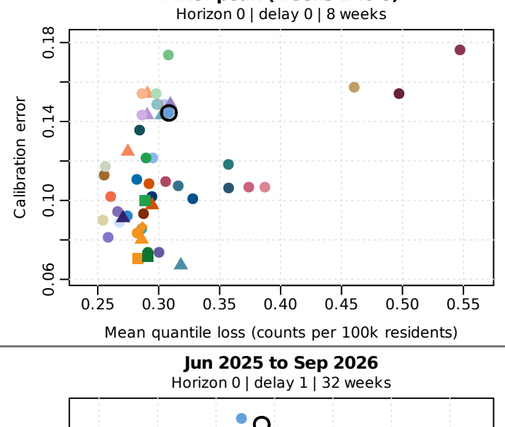
Jun 2025 to Sep 2026
Horizon 0 | delay 0 | 32 weeks



Up to peak (weeks -7 to 0)
Horizon 0 | delay 0 | 7 weeks

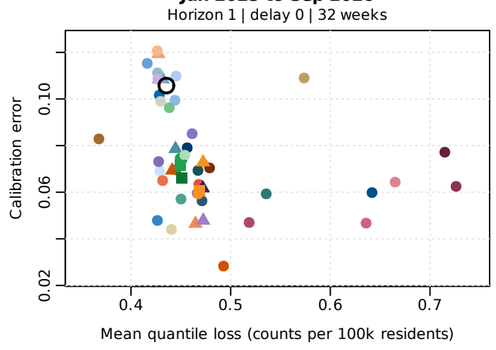


After peak (weeks 1 to 8)
Horizon 0 | delay 0 | 8 weeks

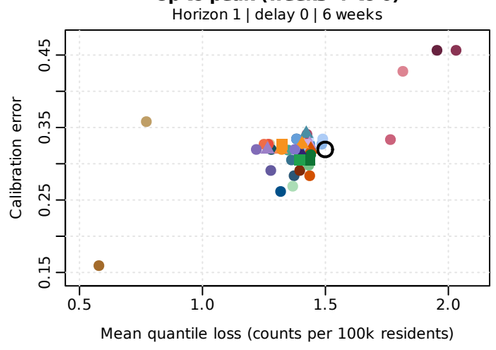


Horizon 1

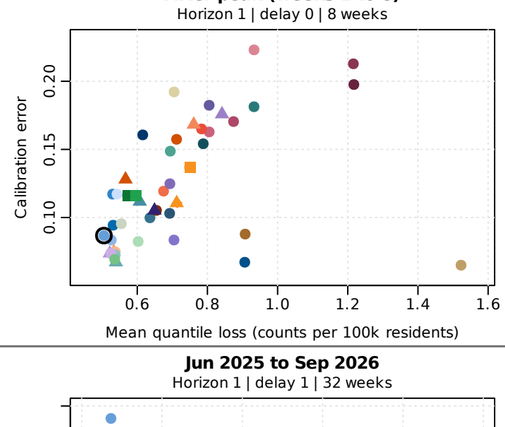
Jun 2025 to Sep 2026
Horizon 1 | delay 0 | 32 weeks



Up to peak (weeks -7 to 0)
Horizon 1 | delay 0 | 6 weeks

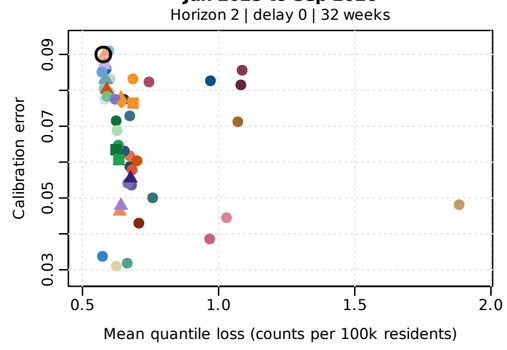


After peak (weeks 1 to 8)
Horizon 1 | delay 0 | 8 weeks

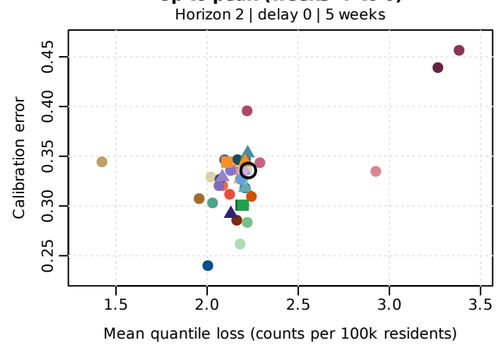


Horizon 2

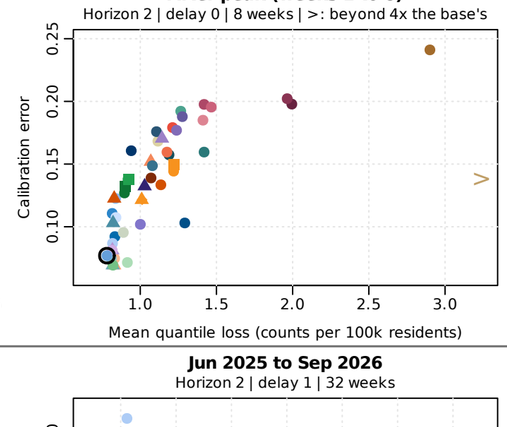
Jun 2025 to Sep 2026
Horizon 2 | delay 0 | 32 weeks



Up to peak (weeks -7 to 0)
Horizon 2 | delay 0 | 5 weeks

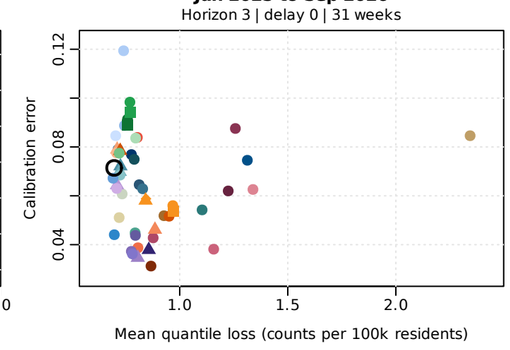


After peak (weeks 1 to 8)
Horizon 2 | delay 0 | 8 weeks | >: beyond 4x the base's

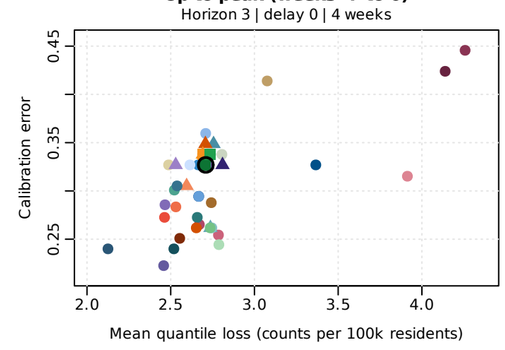


Horizon 3

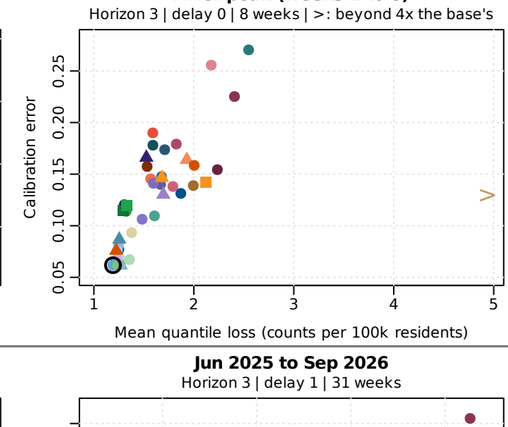
Jun 2025 to Sep 2026
Horizon 3 | delay 0 | 31 weeks



Up to peak (weeks -7 to 0)
Horizon 3 | delay 0 | 4 weeks



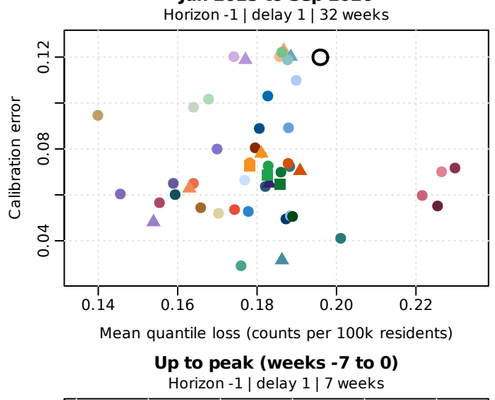
After peak (weeks 1 to 8)
Horizon 3 | delay 0 | 8 weeks | >: beyond 4x the base's



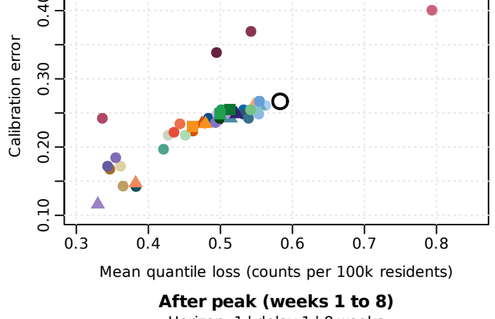
Feedback delay 1

Horizon -1

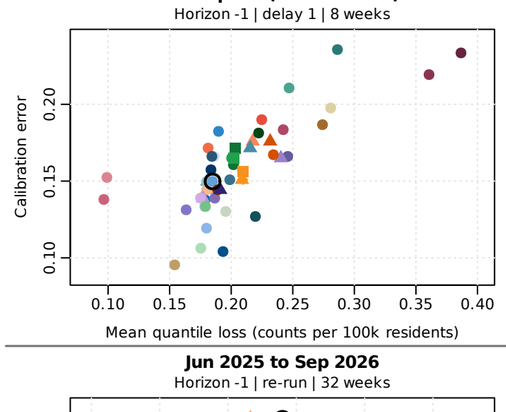
Jun 2025 to Sep 2026
Horizon -1 | delay 1 | 32 weeks



Up to peak (weeks -7 to 0)
Horizon -1 | delay 1 | 7 weeks

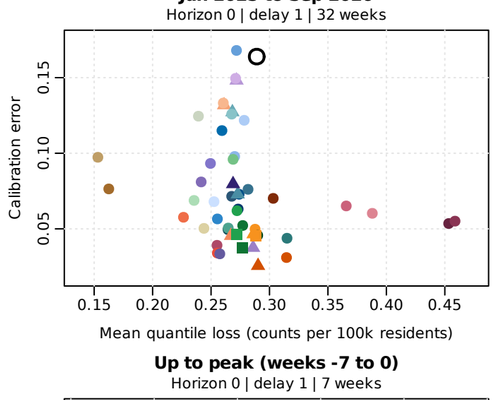


After peak (weeks 1 to 8)
Horizon -1 | delay 1 | 8 weeks

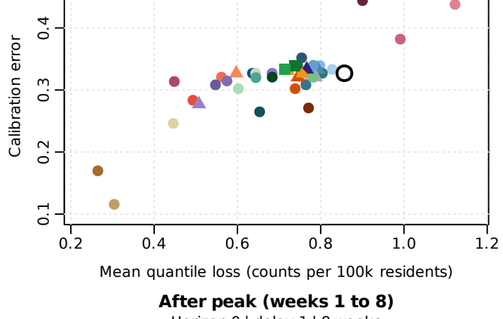


Horizon 0

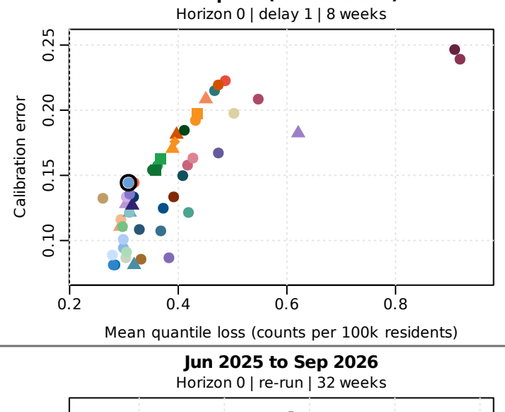
Jun 2025 to Sep 2026
Horizon 0 | delay 1 | 32 weeks



Up to peak (weeks -7 to 0)
Horizon 0 | delay 1 | 7 weeks

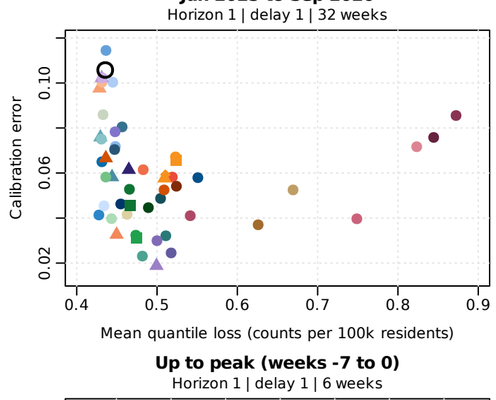


After peak (weeks 1 to 8)
Horizon 0 | delay 1 | 8 weeks

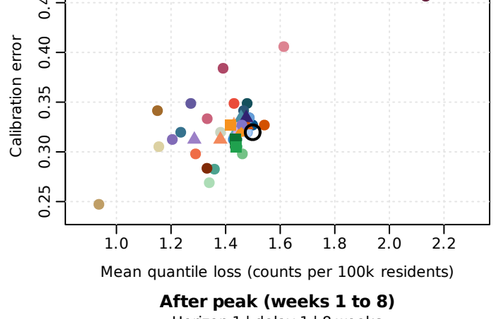


Horizon 1

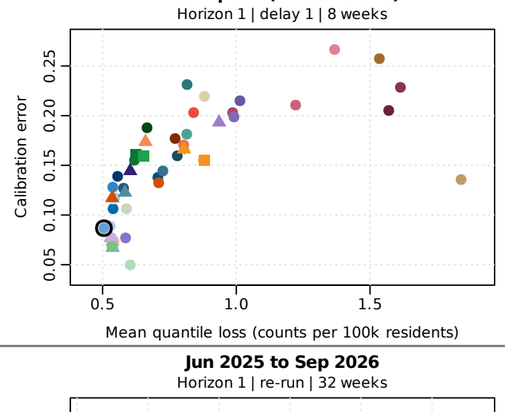
Jun 2025 to Sep 2026
Horizon 1 | delay 1 | 32 weeks



Up to peak (weeks -7 to 0)
Horizon 1 | delay 1 | 6 weeks

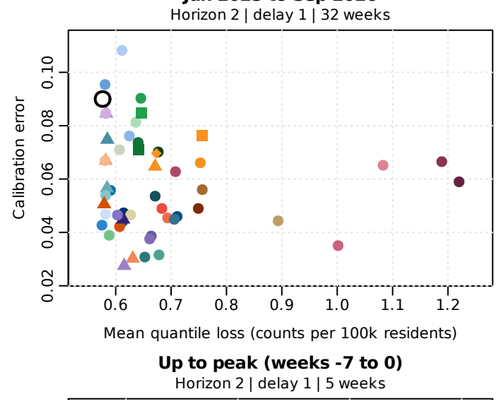


After peak (weeks 1 to 8)
Horizon 1 | delay 1 | 8 weeks

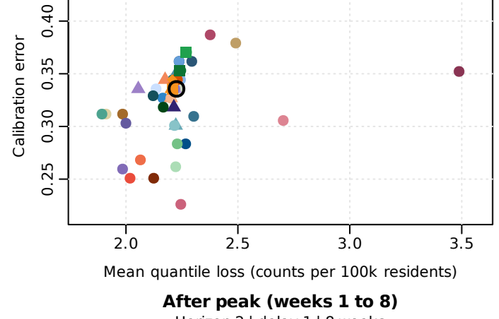


Horizon 2

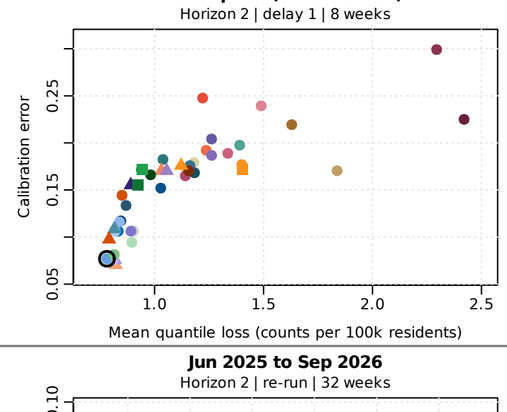
Jun 2025 to Sep 2026
Horizon 2 | delay 1 | 32 weeks



Up to peak (weeks -7 to 0)
Horizon 2 | delay 1 | 5 weeks

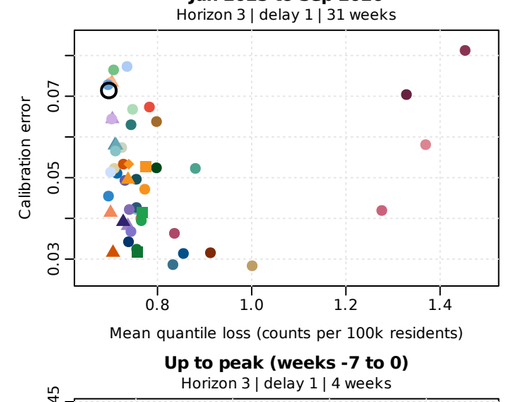


After peak (weeks 1 to 8)
Horizon 2 | delay 1 | 8 weeks

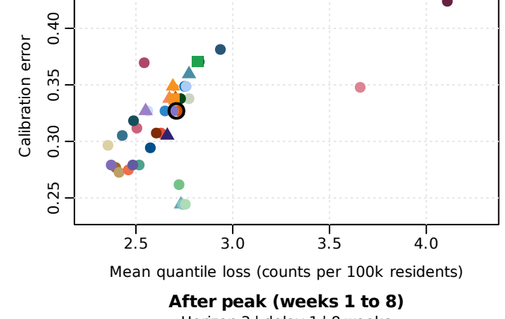


Horizon 3

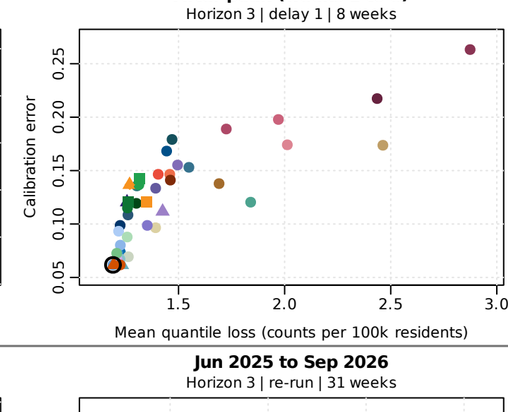
Jun 2025 to Sep 2026
Horizon 3 | delay 1 | 31 weeks



Up to peak (weeks -7 to 0)
Horizon 3 | delay 1 | 4 weeks



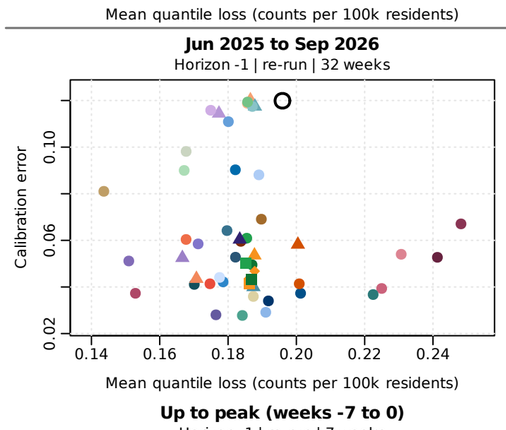
After peak (weeks 1 to 8)
Horizon 3 | delay 1 | 8 weeks



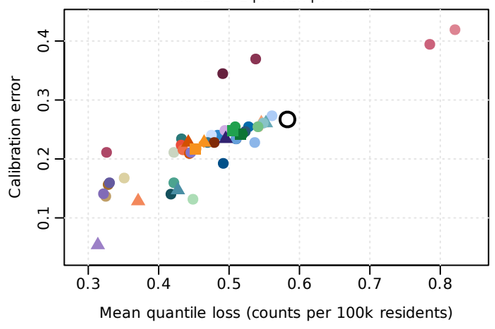
Feedback re-run

Horizon -1

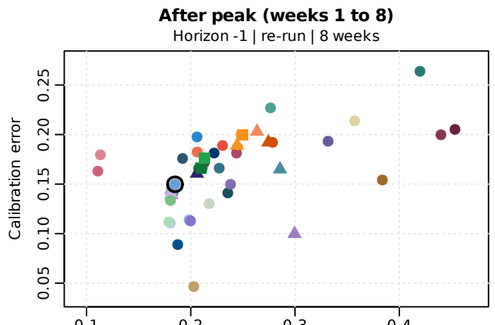
Jun 2025 to Sep 2026
Horizon -1 | re-run | 32 weeks



Up to peak (weeks -7 to 0)
Horizon -1 | re-run | 7 weeks

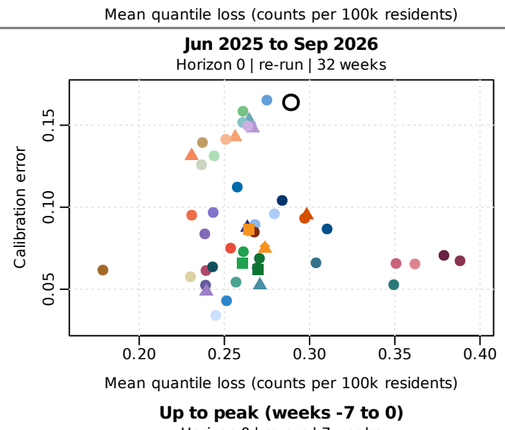


After peak (weeks 1 to 8)
Horizon -1 | re-run | 8 weeks

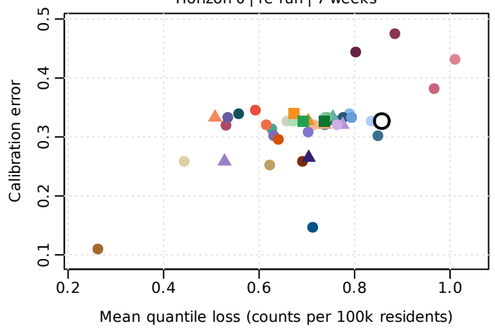


Horizon 0

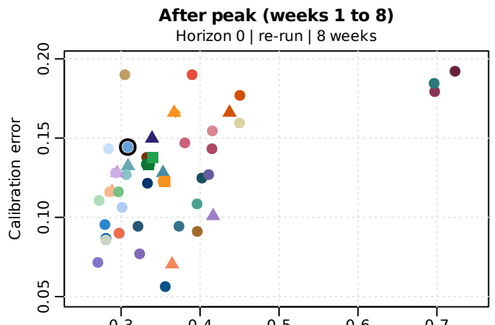
Jun 2025 to Sep 2026
Horizon 0 | re-run | 32 weeks



Up to peak (weeks -7 to 0)
Horizon 0 | re-run | 7 weeks

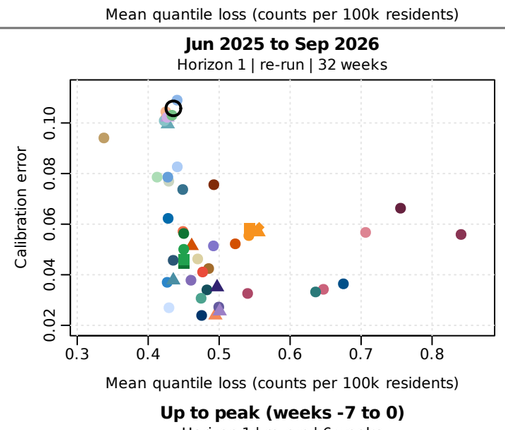


After peak (weeks 1 to 8)
Horizon 0 | re-run | 8 weeks

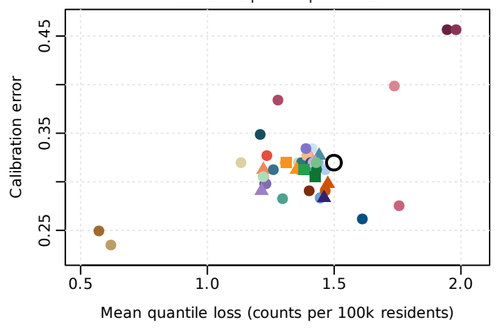


Horizon 1

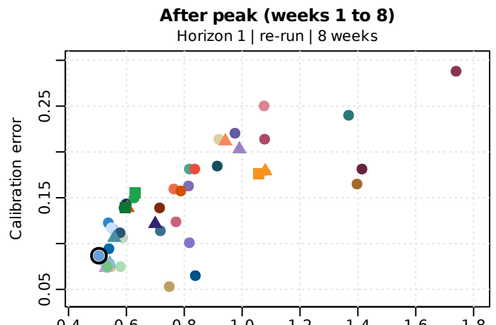
Jun 2025 to Sep 2026
Horizon 1 | re-run | 32 weeks



Up to peak (weeks -7 to 0)
Horizon 1 | re-run | 6 weeks

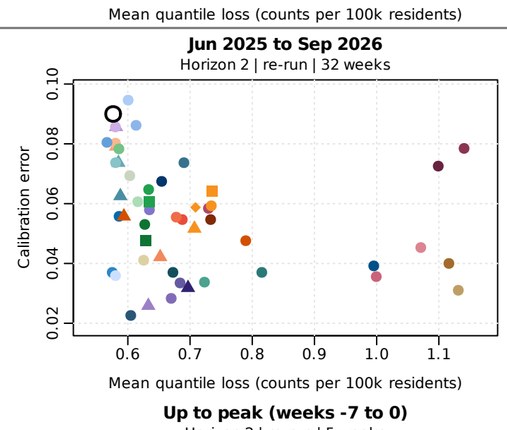


After peak (weeks 1 to 8)
Horizon 1 | re-run | 8 weeks

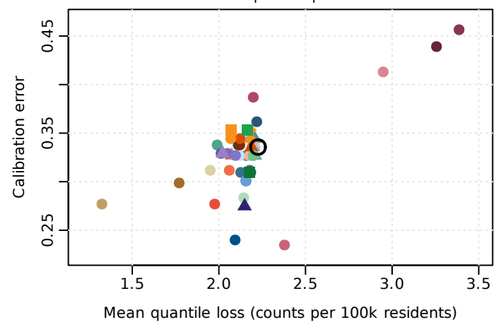


Horizon 2

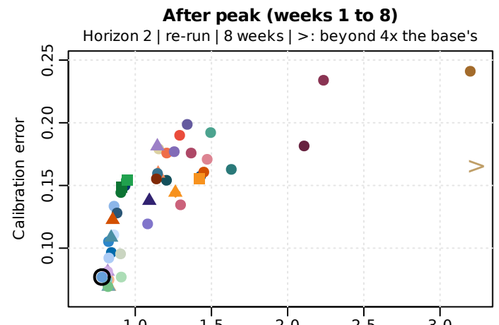
Jun 2025 to Sep 2026
Horizon 2 | re-run | 32 weeks



Up to peak (weeks -7 to 0)
Horizon 2 | re-run | 5 weeks

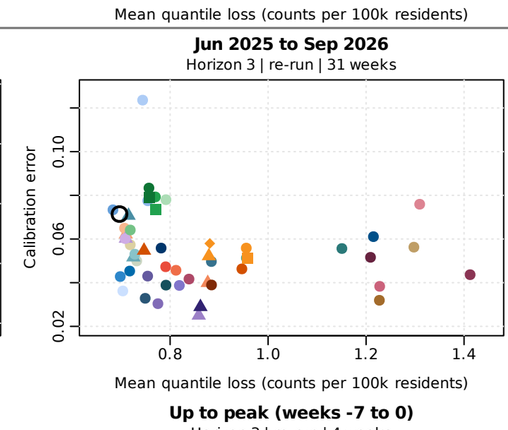


After peak (weeks 1 to 8)
Horizon 2 | re-run | 8 weeks | >: beyond 4x the base's

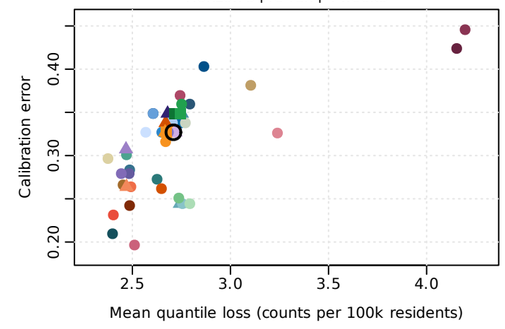


Horizon 3

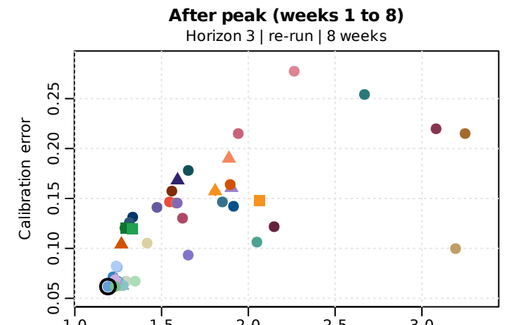
Jun 2025 to Sep 2026
Horizon 3 | re-run | 31 weeks

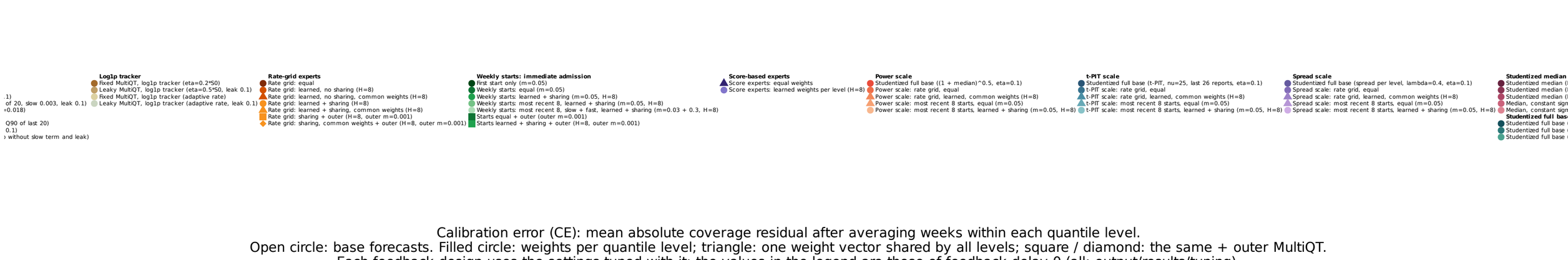


Up to peak (weeks -7 to 0)
Horizon 3 | re-run | 4 weeks



After peak (weeks 1 to 8)
Horizon 3 | re-run | 8 weeks





Calibration error (CE): mean absolute coverage residual after averaging weeks within each quantile level.
Open circle: base forecasts. Filled circle: weights per quantile level; triangle: one weight vector shared by all levels; square / diamond: the same + outer MultiQT.
Each feedback design uses the settings tuned with it; the values in the legend are those of feedback delay 0 (all: output/results/tuning).

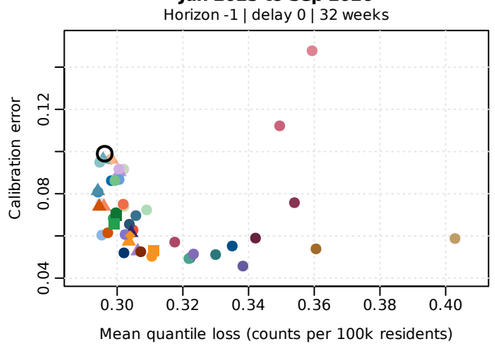
flu | Pennsylvania (42) | evaluation: Jun 2025 to Sep 2026 | raw-count updates, loss in counts per 100k residents

Rows: feedback design (delay d: learn once, d weeks after the first report, from the report current then; re-run: re-run every week on the reports known then) and window; columns: horizon.

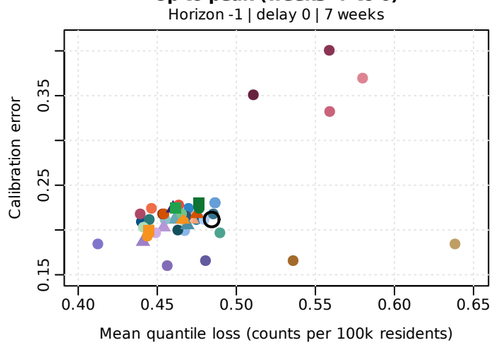
Horizon -1

Feedback delay 0

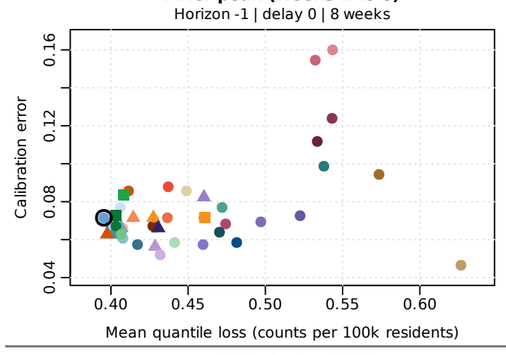
Jun 2025 to Sep 2026
Horizon -1 | delay 0 | 32 weeks



Up to peak (weeks -7 to 0)
Horizon -1 | delay 0 | 7 weeks

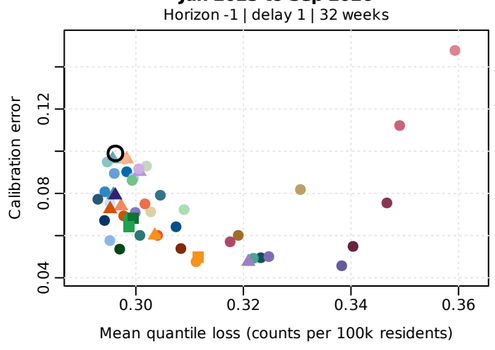


After peak (weeks 1 to 8)
Horizon -1 | delay 0 | 8 weeks

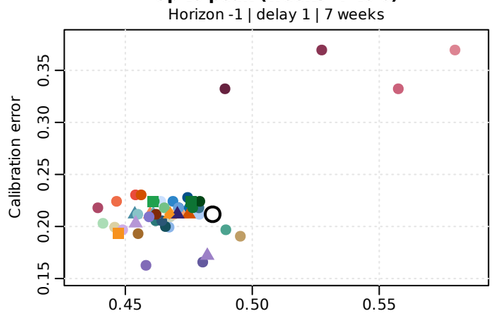


Feedback delay 1

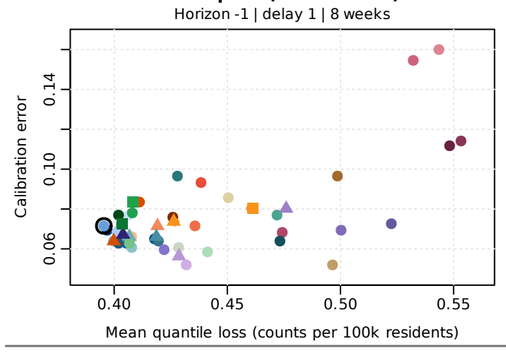
Jun 2025 to Sep 2026
Horizon -1 | delay 1 | 32 weeks



Up to peak (weeks -7 to 0)
Horizon -1 | delay 1 | 7 weeks

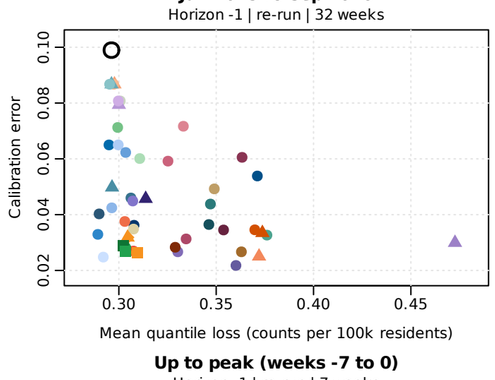


After peak (weeks 1 to 8)
Horizon -1 | delay 1 | 8 weeks

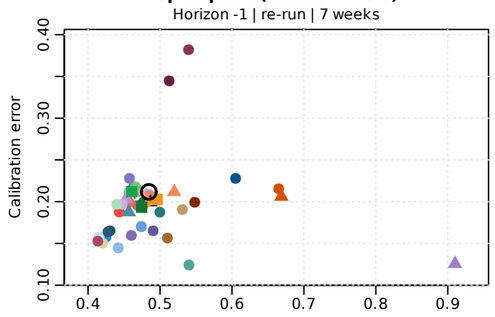


Feedback re-run

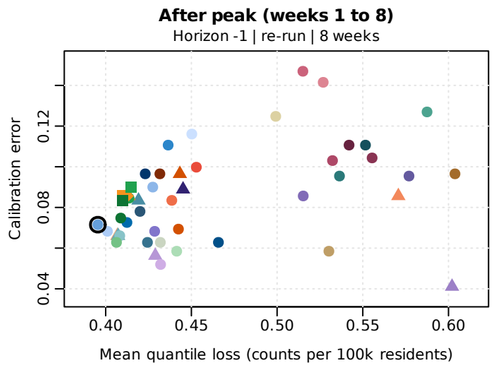
Jun 2025 to Sep 2026
Horizon -1 | re-run | 32 weeks



Up to peak (weeks -7 to 0)
Horizon -1 | re-run | 7 weeks



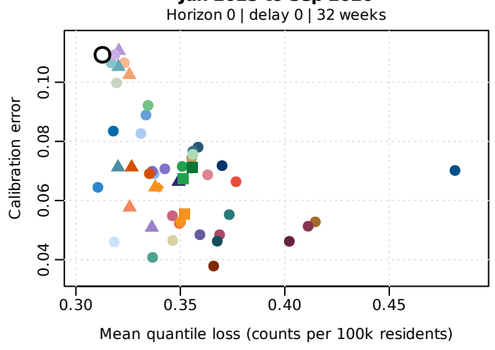
After peak (weeks 1 to 8)
Horizon -1 | re-run | 8 weeks



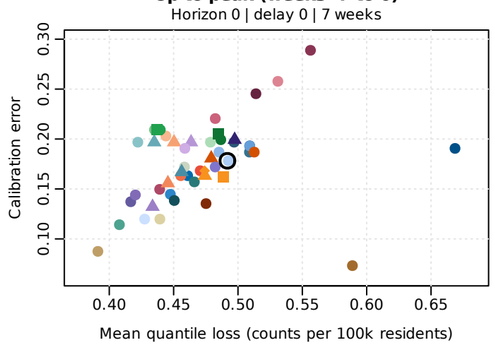
Horizon 0

Feedback delay 0

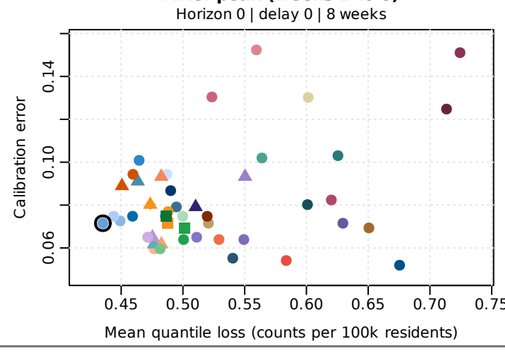
Jun 2025 to Sep 2026
Horizon 0 | delay 0 | 32 weeks



Up to peak (weeks -7 to 0)
Horizon 0 | delay 0 | 7 weeks

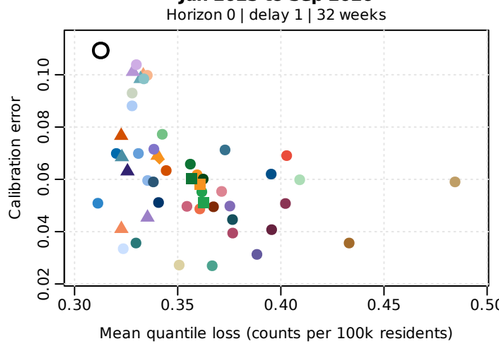


After peak (weeks 1 to 8)
Horizon 0 | delay 0 | 8 weeks

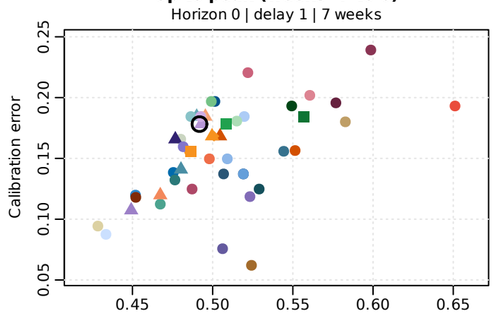


Feedback delay 1

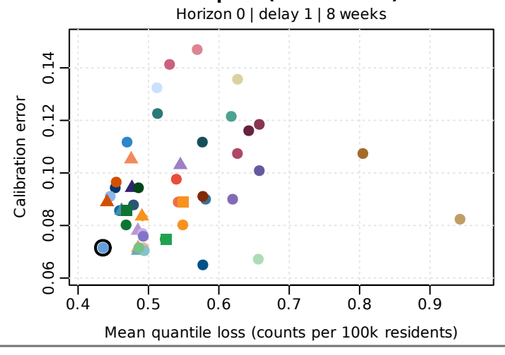
Jun 2025 to Sep 2026
Horizon 0 | delay 1 | 32 weeks



Up to peak (weeks -7 to 0)
Horizon 0 | delay 1 | 7 weeks

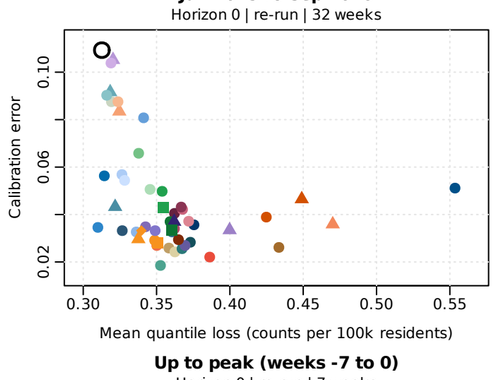


After peak (weeks 1 to 8)
Horizon 0 | delay 1 | 8 weeks

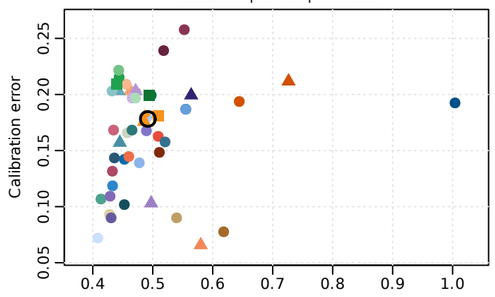


Feedback re-run

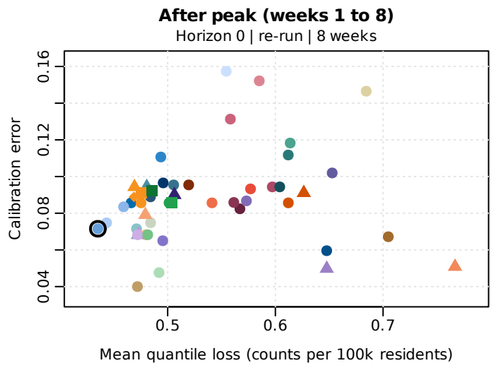
Jun 2025 to Sep 2026
Horizon 0 | re-run | 32 weeks



Up to peak (weeks -7 to 0)
Horizon 0 | re-run | 7 weeks



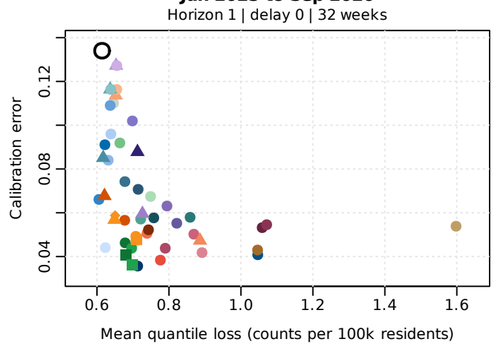
After peak (weeks 1 to 8)
Horizon 0 | re-run | 8 weeks



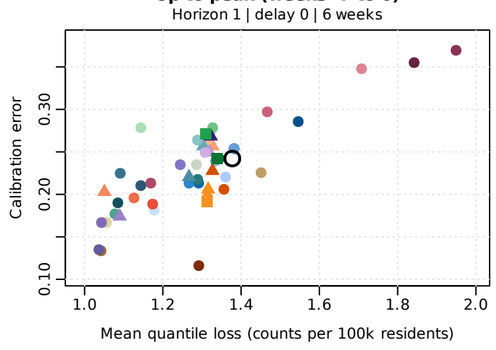
Horizon 1

Feedback delay 0

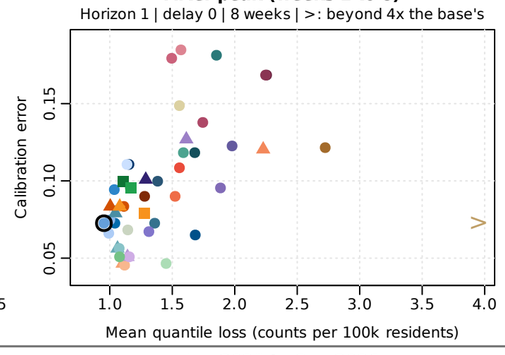
Jun 2025 to Sep 2026
Horizon 1 | delay 0 | 32 weeks



Up to peak (weeks -7 to 0)
Horizon 1 | delay 0 | 6 weeks

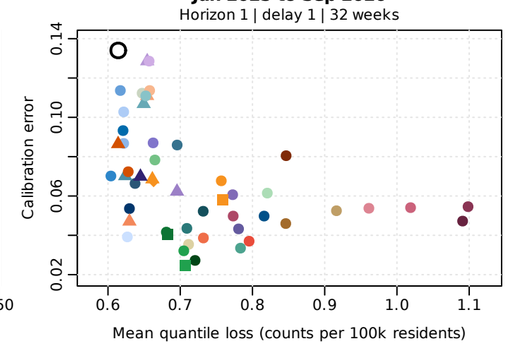


After peak (weeks 1 to 8)
Horizon 1 | delay 0 | 8 weeks | >: beyond 4x the base's

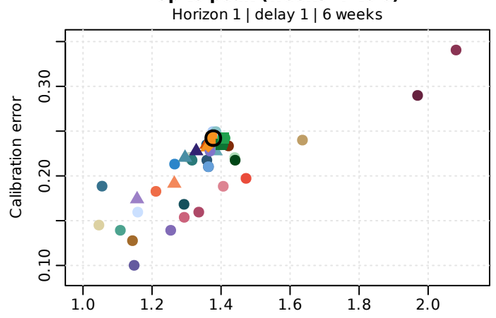


Feedback delay 1

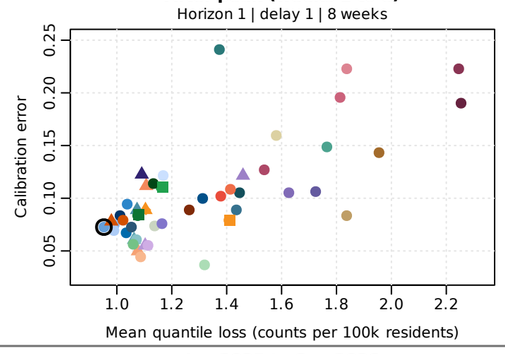
Jun 2025 to Sep 2026
Horizon 1 | delay 1 | 32 weeks



Up to peak (weeks -7 to 0)
Horizon 1 | delay 1 | 6 weeks

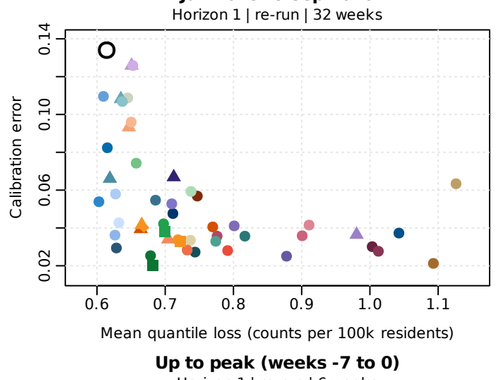


After peak (weeks 1 to 8)
Horizon 1 | delay 1 | 8 weeks

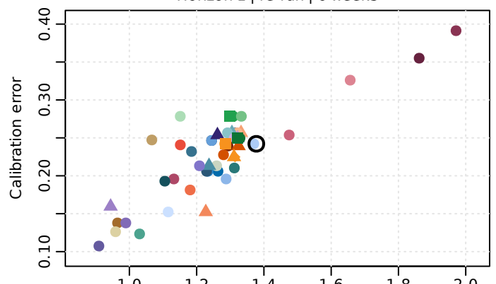


Feedback re-run

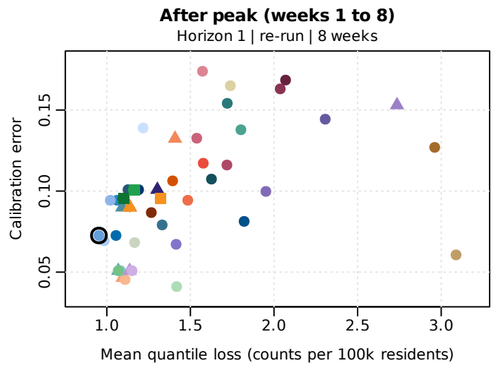
Jun 2025 to Sep 2026
Horizon 1 | re-run | 32 weeks



Up to peak (weeks -7 to 0)
Horizon 1 | re-run | 6 weeks



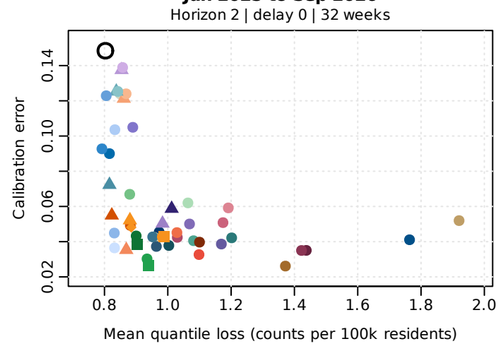
After peak (weeks 1 to 8)
Horizon 1 | re-run | 8 weeks



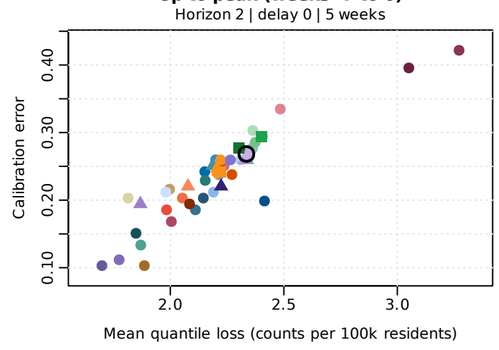
Horizon 2

Feedback delay 0

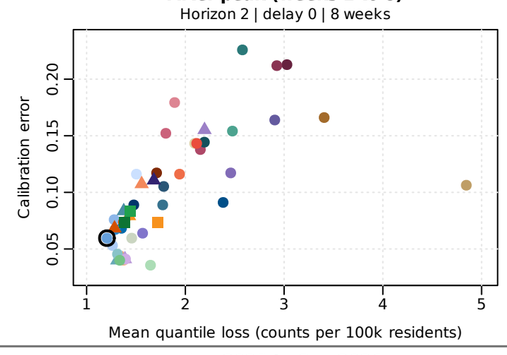
Jun 2025 to Sep 2026
Horizon 2 | delay 0 | 32 weeks



Up to peak (weeks -7 to 0)
Horizon 2 | delay 0 | 5 weeks

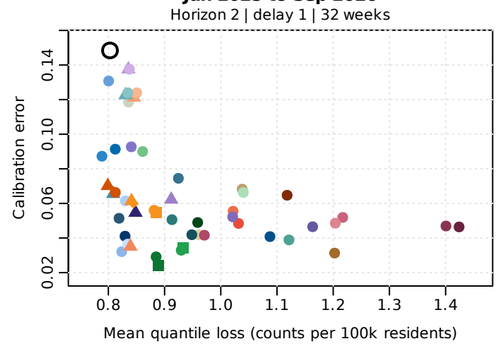


After peak (weeks 1 to 8)
Horizon 2 | delay 0 | 8 weeks

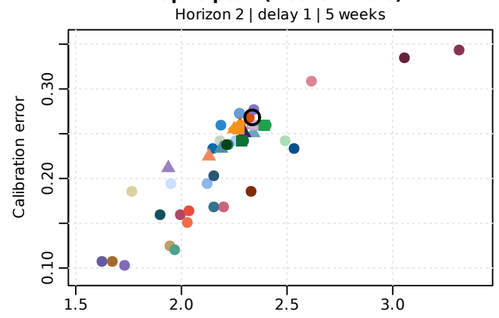


Feedback delay 1

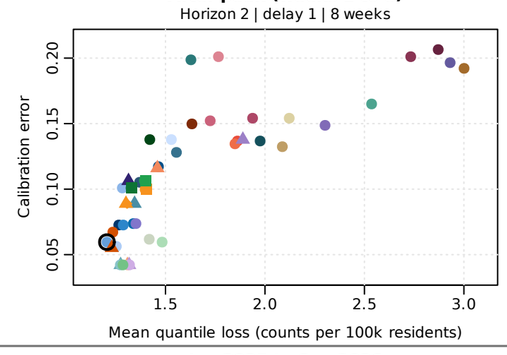
Jun 2025 to Sep 2026
Horizon 2 | delay 1 | 32 weeks



Up to peak (weeks -7 to 0)
Horizon 2 | delay 1 | 5 weeks

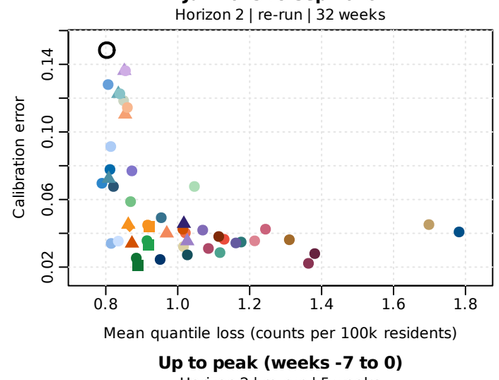


After peak (weeks 1 to 8)
Horizon 2 | delay 1 | 8 weeks

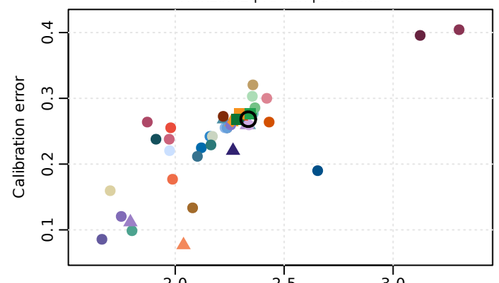


Feedback re-run

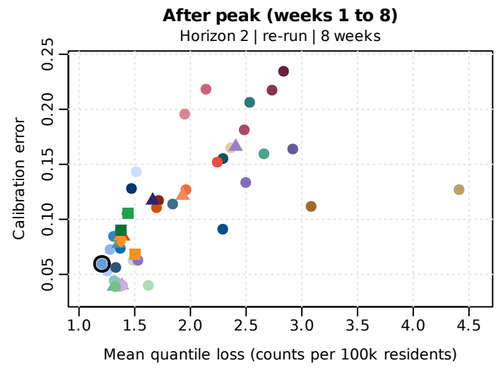
Jun 2025 to Sep 2026
Horizon 2 | re-run | 32 weeks



Up to peak (weeks -7 to 0)
Horizon 2 | re-run | 5 weeks



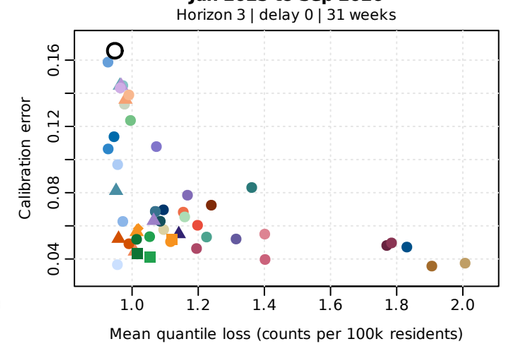
After peak (weeks 1 to 8)
Horizon 2 | re-run | 8 weeks



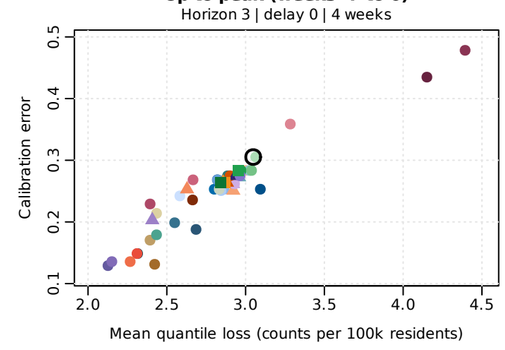
Horizon 3

Feedback delay 0

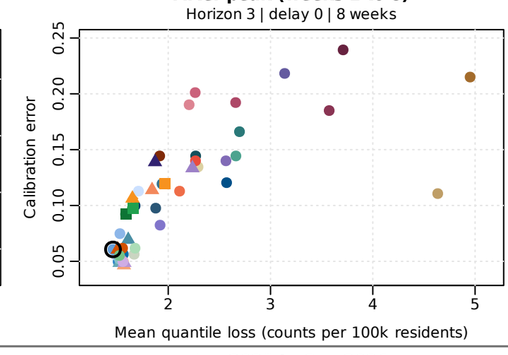
Jun 2025 to Sep 2026
Horizon 3 | delay 0 | 31 weeks



Up to peak (weeks -7 to 0)
Horizon 3 | delay 0 | 4 weeks

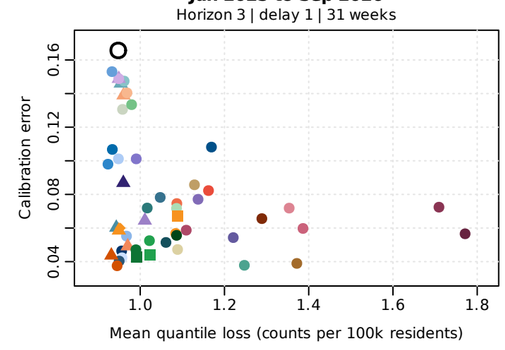


After peak (weeks 1 to 8)
Horizon 3 | delay 0 | 8 weeks

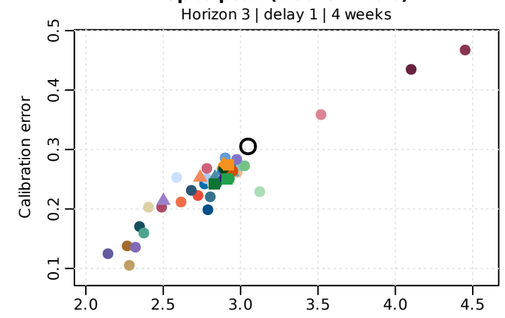


Feedback delay 1

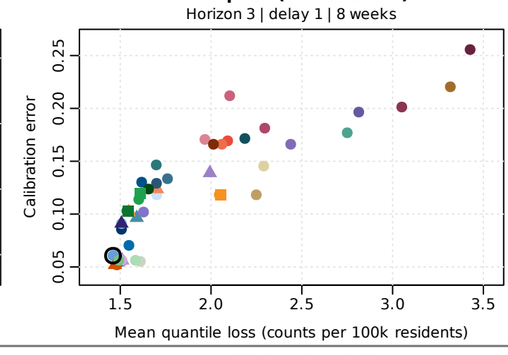
Jun 2025 to Sep 2026
Horizon 3 | delay 1 | 31 weeks



Up to peak (weeks -7 to 0)
Horizon 3 | delay 1 | 4 weeks

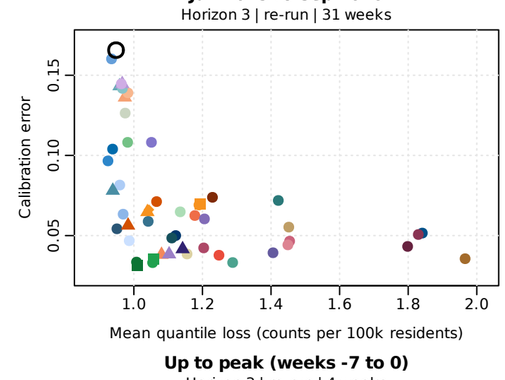


After peak (weeks 1 to 8)
Horizon 3 | delay 1 | 8 weeks

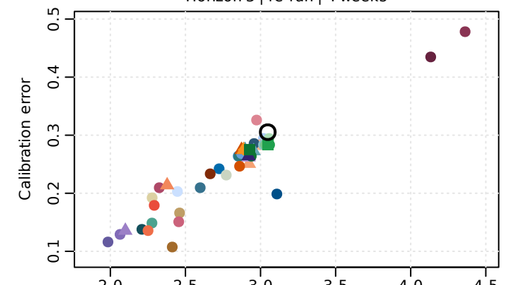


Feedback re-run

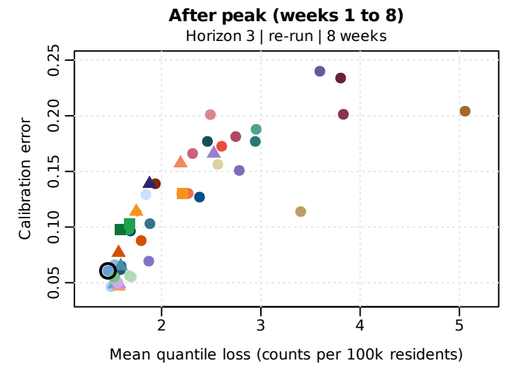
Jun 2025 to Sep 2026
Horizon 3 | re-run | 31 weeks

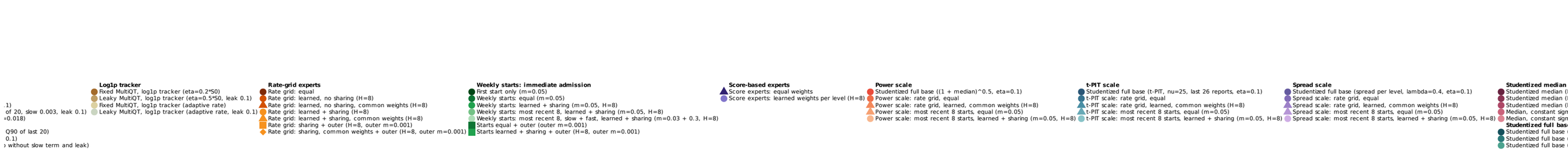


Up to peak (weeks -7 to 0)
Horizon 3 | re-run | 4 weeks



After peak (weeks 1 to 8)
Horizon 3 | re-run | 8 weeks





Calibration error (CE): mean absolute coverage residual after averaging weeks within each quantile level.
Open circle: base forecasts. Filled circle: weights per quantile level; triangle: one weight vector shared by all levels; square / diamond: the same + outer MultiQT.
Each feedback design uses the settings tuned with it; the values in the legend are those of feedback delay 0 (all: output/results/tuning).